

ISC Class XII Mathematics

Chapter 6 Applications of Derivatives

80 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Sections Covered

- I. Rate of Change of Quantities
- II. Tangents, Normals, and Angle of Intersection
- III. Approximations Using Differentials
- IV. Increasing and Decreasing Functions
- V. Maxima and Minima – Direct Method
- VI. Local Maxima and Minima – Derivative Tests
- VII. Practical Problems on Maxima and Minima (Word Problems)

Note: Exercise 6.3 (Approximations) was not available in the source scan supplied for this chapter; its questions in this document are sourced from the standard ML Aggarwal ISC Class 12 exercise set for this topic rather than from the scanned pages used for every other section.

Contents

1	I. Rate of Change of Quantities	8
2	II. Tangents, Normals, and Angle of Intersection	22
3	III. Approximations Using Differentials	45
4	IV. Increasing and Decreasing Functions	52
5	V. Maxima and Minima – Direct Method	68
6	VI. Local Maxima and Minima – Derivative Tests	75
7	VII. Practical Problems on Maxima and Minima (Word Problems)	81

Index of Types

Type	Method / Pattern	★	Source(s)
Section I: Rate of Change of Quantities			
1	Differentiating a standard geometric formula and evaluating the rate at a point	Yes	Ex 6.1 Q6 (NCERT); Ex 6.1 Q13 (NCERT) (alt.)
2	Marginal revenue / marginal cost (economics rate of change)	–	Ex 6.1 Q10(i)(ii) (NCERT)
3	Proving a general proportionality between two rates	–	Ex 6.1 Q15
4	Related rates on a curve: point where two rates satisfy a stated condition	–	Ex 6.1 Q16; Ex 6.1 Q20 (alt.)
5	Two independently-changing quantities combined via sum/product rule	–	Ex 6.1 Q18
6	A single given rate converted through a shared variable to a different quantity's rate	–	Ex 6.1 Q19
7	Rate problem needing a derived geometric relation first (e.g. median-side)	–	Ex 6.1 Q17
8	Similar-triangles "shadow" word problems	Yes	Ex 6.1 Q21; Ex 6.1 Q22 (alt.)

Type	Method / Pattern	★	Source(s)
9	Pythagorean-relation word problems (kite/string)	–	Ex 6.1 Q23
10	Cone volume via height alone (fixed radius-height ratio)	Yes	Ex 6.1 Q24; Ex 6.1 Q25 (alt.)
11	Cone volume via slant height, or a fixed-vessel similar-triangle setup	–	Ex 6.1 Q27; Ex 6.1 Q26 (alt.)
12	Rate of change from an abstract real-world growth/decay model	Yes	Ex 6.1 Q28; Text Example 28 (alt.)
Section II: Tangents, Normals, and Angle of Intersection			
13	Slope of the tangent or normal at a given point (explicit/implicit curve)	Yes	Ex 6.2 Q1(ii); Ex 6.2 Q1(iii)+Q2(i) (alt.)
14	Point(s) where the tangent has a specified (non-zero) slope	–	Ex 6.2 Q3; Ex 6.2 Q5 (alt.)
15	Point(s) where the tangent is horizontal (or vertical)	Yes	Ex 6.2 Q6(i); Ex 6.2 Q16 (alt.)
16	Point(s) where the tangent is parallel to a given line	Yes	Ex 6.2 Q17; Ex 6.2 Q18 (alt.)
17	Point(s) where the tangent is perpendicular to a given line	–	Ex 6.2 Q13
18	Proving tangents at two different points are parallel to each other	–	Ex 6.2 Q11 (NCERT)
19	Finding unknown constants from a tangent condition at a point	–	Ex 6.2 Q12
20	Slopes of tangents at multiple points found by first solving for intersections	–	Ex 6.2 Q10
21	Equations of BOTH tangent and normal at a given point	Yes	Ex 6.2 Q19(ii); Ex 6.2 Q19(v) (alt.)
22	Tangent/normal equation at a general parametric point	–	Ex 6.2 Q20
23	Tangent equation for an implicit curve needing implicit differentiation	Yes	Ex 6.2 Q21(ii) (NCERT); Ex 6.2 Q21(i) (alt.)
24	Tangent equation for a parametric curve at a numeric parameter value	Yes	Ex 6.2 Q22(i) (NCERT); Ex 6.2 Q22(ii) (alt.)
25	Tangent(s) with a given slope, including a "no such tangent" case	–	Ex 6.2 Q23(i) (NCERT)
26	Tangent equation parallel or perpendicular to a given line	Yes	Ex 6.2 Q24(ii) (NCERT); Ex 6.2 Q25(i) (alt.)
27	Normal equation parallel or perpendicular to a given line	Yes	Ex 6.2 Q27; Ex 6.2 Q28 (alt.)
28	Normal equation through an external point not on the curve	–	Ex 6.2 Q29
29	Angle of intersection between two curves	Yes	Ex 6.2 Q30(i); Ex 6.2 Q31 (alt.)
30	Proving two curves intersect orthogonally (at a point, or as a general condition)	Yes	Ex 6.2 Q32; Ex 6.2 Q33 (alt.)

Type	Method / Pattern	★	Source(s)
31	Proving two curves touch each other	–	Ex 6.2 Q34 (Competency Focused)
Section III: Approximations Using Differentials			
32	Approximate value of an n^{th} root using differentials	Yes	Ex 6.3 Q1(ii) (NCERT); Ex 6.3 Q4(i) (NCERT) (alt.)
33	Approximate value of a power/polynomial function at a nearby point	Yes	Ex 6.3 Q7(i) (NCERT); Ex 6.3 Q6 (NCERT Exemplar) (alt.)
34	Approximate change in area/volume/surface area from a small dimension change	Yes	Ex 6.3 Q8; Ex 6.3 Q9 (alt.)
35	Symbolic percentage change in volume/surface area of a solid	–	Ex 6.3 Q10(i)(ii)
36	Error propagation (absolute/percentage error in a derived quantity)	Yes	Ex 6.3 Q11(i) (NCERT); Ex 6.3 Q11(ii) (alt.)
Section IV: Increasing and Decreasing Functions			
37	Proving strictly increasing/decreasing on $\mathbb{R}$ (constant-sign derivative)	Yes	Ex 6.4 Q1(ii) (NCERT); Ex 6.4 Q4(i)(ii) (alt.)
38	Proving inc/dec where the derivative's sign needs algebraic simplification	–	Ex 6.4 Q3(ii); Ex 6.4 Q4(iii) (NCERT) (alt.)
39	Finding the range of a parameter for which a function is increasing	–	Ex 6.4 Q5 (ISC Specimen 2026); Ex 6.4 Q17 (alt.)
40	Finding interval(s) of increase/decrease via $f'(x) = 0$ and sign-testing	Yes	Ex 6.4 Q24(ii) (NCERT); Ex 6.4 Q6(iii) (alt.)
41	Proving increasing/decreasing via an inequality-style argument	–	Ex 6.4 Q7
42	Finding an unknown constant for which a function is decreasing	–	Ex 6.4 Q8
43	General/definitional proofs (iff statements, greatest-integer function)	–	Ex 6.4 Q9(i); Ex 6.4 Q9(ii) (alt.)
44	Inc/dec on a restricted sub-interval using trig identities	Yes	Ex 6.4 Q18; Ex 6.4 Q13(ii) (alt.)
45	Inc/dec using a log-based function and a known inequality	Yes	Ex 6.4 Q14(i) (NCERT); Ex 6.4 Q14(ii) (alt.)
46	Quick recognition of inc/dec at stated x -values/sub-domains (VSA style)	–	Ex 6.4 Q15; Ex 6.4 Q16 (NCERT) (alt.)
47	Proving inc on one half and dec on the other of a periodic function's domain	Yes	Ex 6.4 Q19 (NCERT); Ex 6.4 Q20 (NCERT) (alt.)
48	Proving a function is neither increasing nor decreasing on an interval	–	Ex 6.4 Q21 (NCERT)
49	Interval of inc/dec for an exponential/log-combined function	–	Ex 6.4 Q25(i) (NCERT); Ex 6.4 Q25(ii) (alt.)
Section V: Maxima and Minima – Direct Method			
50	Max/min of a quadratic via completing the square (no calculus)	–	Ex 6.5 Q1(ii) (NCERT); Ex 6.5 Q2(ii) (alt.)

Type	Method / Pattern	★	Source(s)
51	Max/min of a modulus-based function using $ \cdot \geq 0$	Yes	Ex 6.5 Q4(i); Ex 6.5 Q6(ii) (NCERT) (alt.)
52	Max/min of trig functions using known bounds (incl. auxiliary angle)	Yes	Ex 6.5 Q9; Ex 6.5 Q8(ii) (alt.)
53	Recognising when a function has no max/min	–	Ex 6.5 Q10; Ex 6.5 Q11 (alt.)
54	Absolute max/min on a closed interval (candidates test)	Yes	Ex 6.5 Q12(ii) (NCERT); Ex 6.5 Q13 (NCERT) (alt.)
Section VI: Local Maxima and Minima – Derivative Tests			
55	Turning points via the second-derivative test, incl. point of inflexion	–	Ex 6.6 Q1(i); Ex 6.6 Q1(ii) (alt.)
56	Local max/min for standard functions (mechanical practice cluster)	–	Ex 6.6 Q3(vii)(viii); Ex 6.6 Q2 (alt.)
57	Finding unknown constants from local max/min location conditions	Yes	Ex 6.6 Q4; Ex 6.6 Q5 (Competency Focused) (alt.)
58	Applied/real-world stationary-point questions	–	Ex 6.6 Q6 (Competency Focused); Ex 6.6 Q7 (alt.)
Section VII: Practical Problems on Maxima and Minima (Word Problems)			
59	Two-number problems (fixed sum/product, optimize a combination)	Yes	Ex 6.7 Q4 (NCERT); Ex 6.7 Q1 (NCERT) (alt.)
60	Three-number / ratio-constrained number problems	–	Ex 6.7 Q7
61	Economic optimization (profit/cost functions)	Yes	Ex 6.7 Q8 (NCERT); Ex 6.7 Q9 (alt.)
62	Proving a rectangle-square optimality theorem (fixed area/perimeter)	Yes	Ex 6.7 Q10; Ex 6.7 Q12 (alt.)
63	Rectangle with a numeric perimeter/area constraint – find the actual max/min	–	Ex 6.7 Q11
64	Right triangle optimality theorems (isosceles is optimal)	–	Ex 6.7 Q13; Ex 6.7 Q15 (alt.)
65	Triangle with two fixed sides and a variable included angle	–	Ex 6.7 Q14
66	Triangle with sides as algebraic expressions – maximize area (Heron's formula)	–	Ex 6.7 Q16; Ex 6.7 Q17 (alt.)
67	Rectangle inscribed in a fixed circle – prove square is optimal	–	Ex 6.7 Q19
68	Printing/margin problem – minimize total paper for a fixed printed area	–	Ex 6.7 Q18
69	Circle + square/triangle, fixed combined perimeter – minimize combined area	Yes	Ex 6.7 Q20(i) (NCERT); Ex 6.7 Q21 (alt.)
70	Norman-window style composite shape – maximize area/light	–	Ex 6.7 Q23

Type	Method / Pattern	★	Source(s)
71	Fenced area with different costs per side – minimize total cost	–	Ex 6.7 Q24
72	Nearest point on a curve to a given external point	Yes	Ex 6.7 Q25(i); Ex 6.7 Q25(ii) (alt.)
73	Running-track composite-perimeter word problem – maximize enclosed area	–	Ex 6.7 Q26 (ISC 2023)
74	Triangle/polygon inscribed in an ellipse – maximize area	–	Ex 6.7 Q27
75	Box/cuboid optimality theorems (prove cube or a fixed ratio is optimal)	Yes	Ex 6.7 Q28 (ISC 2017); Ex 6.7 Q31 (alt.)
76	Numeric box optimization with cost or corner-cutting twist	Yes	Ex 6.7 Q33; Ex 6.7 Q30 (ISC 2019) (alt.)
77	Cone optimization (semi-vertical angle, given slant height)	–	Ex 6.7 Q34
78	Cylinder optimization – given volume minimize area, or given area maximize volume	Yes	Ex 6.7 Q37 (ISC 2020); Ex 6.7 Q36(i) (alt.)
79	Solid inscribed in a sphere – maximize volume or curved surface area	Yes	Ex 6.7 Q39 (ISC 2018); Ex 6.7 Q38 (alt.)
80	Solid inscribed in a cone / composite-solid optimization	Yes	Ex 6.7 Q41 (Competency Focused); Ex 6.7 Q43 (alt.)

Formula Sheet

Section I – Rate of Change

$$\frac{dQ}{dt} = \frac{dQ}{dx} \cdot \frac{dx}{dt}$$

The chain rule linking a quantity's rate of change to the rate of its underlying variable

Section I – Standard Geometric Formulas

$$A_{\text{circle}} = \pi r^2, \quad V_{\text{sphere}} = \frac{4}{3}\pi r^3, \quad S_{\text{sphere}} = 4\pi r^2, \quad V_{\text{cube}} = x^3, \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h$$

The formulas most frequently differentiated in rate-of-change problems

Section II – Tangents and Normals

$$\text{Slope of tangent at } (x_1, y_1) = \left. \frac{dy}{dx} \right|_{(x_1, y_1)}; \quad \text{Slope of normal} = \frac{-1}{dy/dx}$$

The tangent's slope is the derivative itself; the normal is perpendicular to it

Section II – Line Equations

$$y - y_1 = m(x - x_1)$$

Point-slope form, used to write both tangent (m = slope of tangent) and normal (m = slope of normal) equations

Section II – Angle of Intersection

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

Angle between two curves at their point of intersection, using each curve's tangent slope there

Section II – Orthogonality

$$\text{Curves intersect orthogonally} \iff m_1 m_2 = -1$$

Special case of the angle formula when $\theta = 90^\circ$

Section III – Approximations

$$\Delta y \approx dy = f'(x) \cdot \Delta x, \quad \text{so } f(x + \Delta x) \approx f(x) + f'(x)\Delta x$$

The core linear-approximation formula: the actual change Δy is approximated by the differential dy

Section III – Percentage Error

$$\% \text{ error in } y \approx \frac{dy}{y} \times 100 = \left(\frac{f'(x)}{f(x)} \right) \Delta x \times 100$$

Relates percentage error in x to percentage error in a derived quantity $y = f(x)$

Section IV – Increasing/Decreasing Functions

$$f \text{ strictly increasing on } I \iff f'(x) > 0 \forall x \in I;$$

$$f \text{ strictly decreasing on } I \iff f'(x) < 0 \forall x \in I$$

The fundamental derivative-sign test for monotonicity

Section V – Direct Method for Max/Min

$$-1 \leq \sin \theta, \cos \theta \leq 1; \quad |u| \geq 0 \forall u; \quad a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$$

Bounds used to read off max/min values directly, without calculus

Section VI – Second Derivative Test

$$f'(c) = 0 \text{ and } f''(c) < 0 \Rightarrow \text{local maximum at } c; \quad f'(c) = 0 \text{ and } f''(c) > 0 \Rightarrow \text{local minimum at } c$$

The standard test to classify a stationary point; if $f''(c) = 0$ too, fall back to the first-derivative sign test (point of inflexion is possible)

Section VII – Optimization Method

Express the quantity to optimize as a function of ONE variable using the g

$$0$$

The universal three-step method: (1) set up $f(x)$ using the constraint to eliminate extra variables, (2) solve $f'(x) = 0$ for critical points, (3) confirm max/min using $f''(x)$ or a sign check

Section VII – Common Solid Formulas

$$V_{\text{cylinder}} = \pi r^2 h, \quad S_{\text{cylinder, closed}} = 2\pi r^2 + 2\pi r h, \quad V_{\text{cone}} = \frac{1}{3}\pi r^2 h$$

Needed repeatedly across the box/cylinder/cone optimization word problems

I. Rate of Change of Quantities

Type 1

Differentiating a standard geometric formula (area, volume, surface area) with respect to its defining variable, then evaluating the resulting rate at a given instant using the chain rule with a supplied rate for that variable.

★ **Board Favorite** – This is the foundational rate-of-change skill, tested constantly across ISC papers – the book tags multiple instances directly to NCERT.

Formula Used: $\frac{dQ}{dt} = \frac{dQ}{dr} \cdot \frac{dr}{dt}$

Source: Exercise 6.1, Q6 (NCERT)

Given: A spherical balloon has a variable radius. Find the rate at which its volume is increasing w.

Solution:

Step 1: Write the volume formula: $V = \frac{4}{3}\pi r^3$.

Step 2: Differentiate with respect to r : $\frac{dV}{dr} = 4\pi r^2$ – this IS the rate asked for, since the question wants the rate w.r.t. radius directly (not w.r.t. time).

Step 3: Substitute $r = 10$: $\frac{dV}{dr} = 4\pi(10)^2 = 400\pi$.

Final Answer

$$\frac{dV}{dr} = 400\pi \text{ cm}^3/\text{cm}$$

Common Mistake: Assuming every rate-of-change question needs dt somewhere – when the question explicitly asks for the rate "with respect to radius," it wants $\frac{dV}{dr}$ directly, with no time variable involved at all.

Alternate Board-Style Example (rate w.r.t. time, via a given dr/dt)

Source: Exercise 6.1, Q13 (NCERT)

Given: A stone dropped into a quiet lake creates circular waves at 5 cm/sec. When the radius is 8 cm,

Solution:

Step 1: Write the area formula: $A = \pi r^2$.

Step 2: Differentiate both sides with respect to TIME t (using the chain rule, since r itself depends on t): $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$.

Step 3: Substitute the given rate $\frac{dr}{dt} = 5$ cm/sec and the instant $r = 8$ cm: $\frac{dA}{dt} = 2\pi(8)(5) = 80\pi$.

Final Answer

$$\frac{dA}{dt} = 80\pi \text{ cm}^2/\text{sec}$$

Quick Recall: Write the standard formula, differentiate with the chain rule (bringing in $\frac{dr}{dt}$ whenever the rate is with respect to time), then substitute the given instant – this single method handles circles, squares, cubes, spheres, cylinders, soap bubbles, and tanks alike (see Ex 6.1 Q1–Q9, Q14 for further practice with the exact same technique).

Type 2

Marginal revenue or marginal cost – an economics application where the "rate of change" is simply the derivative of a revenue or cost function with respect to the number of units produced/sold.

Formula Used: Marginal Revenue = $\frac{d}{dx}R(x)$; Marginal Cost = $\frac{d}{dx}C(x)$

Source: Exercise 6.1, Q10(i) (NCERT)

Given: $R(x) = 3x^2 + 36x + 5$ (total revenue in rupees from selling x units). Find the marginal revenue at $x = 5$.

Solution:

Step 1: "Marginal revenue" is just standard terminology for $\frac{dR}{dx}$ – the instantaneous rate of change of revenue with respect to the number of units.

Step 2: Differentiate: $\frac{dR}{dx} = 6x + 36$.

Step 3: Substitute $x = 5$: $\frac{dR}{dx} = 6(5) + 36 = 66$.

Final Answer

Marginal revenue = Rs. 66

Common Mistake: Confusing marginal revenue (the DERIVATIVE, i.e. the rate at which revenue changes per additional unit) with the average revenue per unit ($R(x)/x$) – these are different quantities entirely.

Quick Recall: "Marginal" always means "take the derivative" in economics contexts – differentiate the given revenue or cost function and substitute the specified x .

Type 3

Proving a general PROPORTIONALITY relationship between two rates of change, without evaluating any specific number – a symbolic derivation rather than a numeric answer.

Source: Exercise 6.1, Q15

Given: If the circumference of a circle is increasing at a constant rate, prove that the rate of change

Solution:

Step 1: Let the circumference be $C = 2\pi r$, and let $\frac{dC}{dt} = k$ (a constant, as given) – introducing a symbol for the constant rate.

Step 2: Differentiate $C = 2\pi r$ with respect to t : $\frac{dC}{dt} = 2\pi \frac{dr}{dt} \Rightarrow k = 2\pi \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{k}{2\pi}$ – this shows $\frac{dr}{dt}$ is ALSO constant (a key intermediate fact).

Step 3: Now consider the area $A = \pi r^2$. Differentiate with respect to t : $\frac{dA}{dt} = 2\pi r \frac{dr}{dt}$.

Step 4: Substitute $\frac{dr}{dt} = \frac{k}{2\pi}$ from Step 2: $\frac{dA}{dt} = 2\pi r \cdot \frac{k}{2\pi} = kr$.

Step 5: Since k is a fixed constant, $\frac{dA}{dt} = kr$ shows $\frac{dA}{dt}$ is directly proportional to r , with constant of proportionality k .

Final Answer

$$\frac{dA}{dt} = kr, \text{ so } \frac{dA}{dt} \propto r$$

Common Mistake: Trying to prove the statement using specific numeric values instead of keeping the constant rate as a general symbol k throughout – a proportionality proof must hold for the GENERAL constant, not just illustrate one case.

Quick Recall: "Prove a proportionality" questions want you to introduce a symbol for the given constant rate, substitute it through the chain-rule relationship, and show the result is (constant) $\times$ (the claimed proportional variable).

Type 4

A particle or point moves along a given curve; find the specific point on the curve at which two rates ($\frac{dx}{dt}$ and $\frac{dy}{dt}$) satisfy some stated numeric relationship (e.g. equal, or one is a multiple of the other).

Source: Exercise 6.1, Q16

Given: Find the point on the curve $y^2 = 8x$ for which the abscissa and ordinate change at the same

Solution:

Step 1: "Abscissa and ordinate change at the same rate" means $\frac{dx}{dt} = \frac{dy}{dt}$ – translating the words into an equation between the two rates.

Step 2: Differentiate the curve $y^2 = 8x$ implicitly with respect to t : $2y\frac{dy}{dt} = 8\frac{dx}{dt}$.

Step 3: Substitute the condition $\frac{dx}{dt} = \frac{dy}{dt}$ (call this common rate k , though it will cancel):
 $2yk = 8k \Rightarrow 2y = 8 \Rightarrow y = 4$ (dividing by k , valid since the rates are nonzero for genuine motion).

Step 4: Substitute $y = 4$ back into the curve equation to find x : $16 = 8x \Rightarrow x = 2$.

Final Answer

The point is $(2, 4)$

Common Mistake: Trying to solve for x and y independently without first using the curve equation itself to link them – the point must lie ON the curve, so the curve equation is essential for finding the second coordinate.

Alternate Board-Style Example

Source: Exercise 6.1, Q20

Given: A particle moves along the curve $y = \frac{2}{3}x^3 + 1$.
 1. Find the points where the y -coordinate is changing twice as fast as the x -coordinate.

Solution:

Step 1: Translate the condition: $\frac{dy}{dt} = 2\frac{dx}{dt}$.

Step 2: Differentiate $y = \frac{2}{3}x^3 + 1$ implicitly with respect to t : $\frac{dy}{dt} = 2x^2 \frac{dx}{dt}$.

Step 3: Substitute the condition: $2\frac{dx}{dt} = 2x^2 \frac{dx}{dt} \Rightarrow 2 = 2x^2$ (dividing by $\frac{dx}{dt}$, nonzero for real motion) $\Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.

Step 4: Find y for each: at $x = 1$, $y = \frac{2}{3} + 1 = \frac{5}{3}$; at $x = -1$, $y = -\frac{2}{3} + 1 = \frac{1}{3}$.

Final Answer

The points are $\left(1, \frac{5}{3}\right)$ and $\left(-1, \frac{1}{3}\right)$

Quick Recall: Translate the words into an equation relating $\frac{dx}{dt}$ and $\frac{dy}{dt}$ FIRST, differentiate the curve implicitly with respect to t , substitute the condition (the common rate factor cancels), then use the curve equation to find the missing coordinate.

Type 5

Two quantities (like a rectangle's length and width) each change at their OWN independently-given rate – find the rate of change of a combined quantity (perimeter, area) using the sum or product rule, substituting BOTH given rates.

Source: Exercise 6.1, Q18

Given: A rectangle's length x decreases at 5 cm/min and width y increases at 4 cm/min. When $x = 8$, $y = 6$, find the rate of change of (i) the perimeter (ii) the area.

Solution:

Case: (i) Perimeter

$P = 2(x + y)$. Differentiate w.r.t. t : $\frac{dP}{dt} = 2\left(\frac{dx}{dt} + \frac{dy}{dt}\right)$. Substitute $\frac{dx}{dt} = -5$ (decreasing) and $\frac{dy}{dt} = 4$ (increasing): $\frac{dP}{dt} = 2(-5 + 4) = -2$.

Case: (ii) Area

$A = xy$. Differentiate using the PRODUCT rule w.r.t. t : $\frac{dA}{dt} = x\frac{dy}{dt} + y\frac{dx}{dt}$. Substitute $x = 8$, $y = 6$, $\frac{dx}{dt} = -5$, $\frac{dy}{dt} = 4$: $\frac{dA}{dt} = 8(4) + 6(-5) = 32 - 30 = 2$.

Final Answer

$\frac{dP}{dt} = -2$ cm/min (decreasing); $\frac{dA}{dt} = 2$ cm²/min (increasing)

Common Mistake: Forgetting the crucial NEGATIVE sign on a "decreasing" rate – a quantity described as decreasing at 5 cm/min must be substituted as $\frac{dx}{dt} = -5$, not +5.

Quick Recall: When two variables each have their own given rate, differentiate the combined formula using the ordinary sum/product rule (treating both x and y as functions of t), then substitute ALL given values (including correct signs for decreasing quantities) at once.

Type 6

A single given rate for ONE quantity (e.g. volume) must be converted, through a shared underlying variable (e.g. edge length), into the rate of a DIFFERENT quantity (e.g. surface area) – requires first back-solving for the hidden variable's own rate.

Source: Exercise 6.1, Q19

Given: The volume of a cube increases at $9 \text{ cm}^3/\text{sec}$. How fast is the surface area increasing when $t = 10$?

Solution:

Step 1: Both volume and surface area depend on the SAME underlying variable, the edge length x – so first relate $\frac{dV}{dt}$ to $\frac{dx}{dt}$, then use $\frac{dx}{dt}$ to find $\frac{dS}{dt}$.

Step 2: $V = x^3 \Rightarrow \frac{dV}{dt} = 3x^2 \frac{dx}{dt}$. Substitute the given $\frac{dV}{dt} = 9$ and $x = 10$: $9 = 3(100) \frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{9}{300} = \frac{3}{100}$ – this intermediate rate is the KEY link between the two quantities.

Step 3: Now use $S = 6x^2 \Rightarrow \frac{dS}{dt} = 12x \frac{dx}{dt}$. Substitute $x = 10$ and the just-found $\frac{dx}{dt} = \frac{3}{100}$:

$$\frac{dS}{dt} = 12(10) \left(\frac{3}{100} \right) = \frac{360}{100} = 3.6.$$

Final Answer

$$\frac{dS}{dt} = 3.6 \text{ cm}^2/\text{sec}$$

Common Mistake: Trying to relate $\frac{dV}{dt}$ directly to $\frac{dS}{dt}$ without going through the shared variable x – volume and surface area are not directly proportional to each other, so the edge length MUST be used as the connecting link.

Quick Recall: When a rate is given for one quantity but asked for a DIFFERENT quantity, identify the shared underlying variable, back-solve for its rate first using the given information, then substitute that rate into the second quantity's own formula.

Type 7

A rate-of-change problem where the needed geometric formula isn't standard – a relationship (like median in terms of side) must be DERIVED or recalled first, before differentiating.

Source: Exercise 6.1, Q17

Given: The median of an equilateral triangle is increasing at $2\sqrt{3}$ cm/sec. Find the rate at which the side length is increasing.

Solution:

Step 1: Recall (or derive via Pythagoras on half the equilateral triangle) the relationship between the median m and side a of an equilateral triangle: $m = \frac{\sqrt{3}}{2}a$.

Step 2: Differentiate with respect to t : $\frac{dm}{dt} = \frac{\sqrt{3}}{2} \frac{da}{dt}$.

Step 3: Substitute the given $\frac{dm}{dt} = 2\sqrt{3}$: $2\sqrt{3} = \frac{\sqrt{3}}{2} \frac{da}{dt} \Rightarrow \frac{da}{dt} = \frac{2\sqrt{3} \times 2}{\sqrt{3}} = 4$.

Final Answer

$$\frac{da}{dt} = 4 \text{ cm/sec}$$

Common Mistake: Not knowing (or mis-recalling) the median-side relationship for an equilateral triangle – since all three medians of an equilateral triangle are equal and also serve as the altitude, $m = \frac{\sqrt{3}}{2}a$ is worth memorizing directly.

Quick Recall: When the problem's geometric quantity isn't a standard area/volume, first recall or quickly re-derive the formula linking it to a more familiar variable (like side length), then apply the same differentiate-and-substitute method as any other rate problem.

Type 8

Classic "shadow" word problems: a person of fixed height walks away from a lamp post at a given speed – use similar triangles to relate the shadow's length (or its tip's distance from the post) to the person's distance from the post, then differentiate.

★ **Board Favorite** – Shadow/lamppost problems are a perennial favorite in board exams for related rates, testing the ability to set up a similar-triangles relationship before any differentiation happens.

Formula	Used:	Similar triangles:	$\frac{\text{lamp height}}{\text{person height}} =$
$\frac{\text{person's distance} + \text{shadow length}}{\text{shadow length}}$			

Source: Exercise 6.1, Q21

Given: A man of height 2 m walks at a uniform speed of 5 km/h away from a lamp post 6 m high.

Solution:

Step 1: Set up the picture: let the man be a distance x from the lamp post, and let s be the length of his shadow. The lamp post (height 6 m) and the man (height 2 m) are two parallel vertical segments, with the sun's ray forming similar triangles from the tip of the lamp post to the tip of the shadow.

Step 2: By similar triangles, comparing the BIG triangle (lamp post height 6, base $x + s$) to the SMALL triangle (man's height 2, base s): $\frac{6}{x + s} = \frac{2}{s}$.

Step 3: Cross-multiply and simplify: $6s = 2(x + s) \Rightarrow 6s = 2x + 2s \Rightarrow 4s = 2x \Rightarrow s = \frac{x}{2}$ – a clean linear relationship between the shadow length and the man's distance.

Step 4: Differentiate with respect to t : $\frac{ds}{dt} = \frac{1}{2} \frac{dx}{dt}$.

Step 5: Substitute the man's walking speed $\frac{dx}{dt} = 5$ km/h: $\frac{ds}{dt} = \frac{1}{2}(5) = 2.5$.

Final Answer

The shadow length increases at 2.5 km/h

Common Mistake: Setting up the similar-triangles ratio upside down (e.g. comparing height to base inconsistently between the big and small triangle) – always match "lamp height corresponds to the FULL base" with "man's height corresponds to the shadow-only base" consistently.

Alternate Board-Style Example

Source: Exercise 6.1, Q22

Given: A girl of height 1.6 m walks away from a lamp post 4 m high at 50 metres per minute. How

Solution:

Step 1: By similar triangles as before: $\frac{4}{x + s} = \frac{1.6}{s} \Rightarrow 4s = 1.6(x + s) \Rightarrow 4s - 1.6s = 1.6x \Rightarrow$

$$2.4s = 1.6x \Rightarrow s = \frac{1.6}{2.4}x = \frac{2}{3}x.$$

Step 2: Differentiate: $\frac{ds}{dt} = \frac{2}{3} \frac{dx}{dt}$.

Step 3: Substitute $\frac{dx}{dt} = 50$: $\frac{ds}{dt} = \frac{2}{3}(50) = \frac{100}{3}$.

Final Answer

The shadow lengthens at $\frac{100}{3}$ metres per minute

Quick Recall: Draw the similar-triangles picture first (lamp post and person as two parallel heights, sun's ray as the hypotenuse), derive the linear relationship $s = kx$ between shadow length and walking distance, THEN differentiate – the algebra is always easier after simplifying to $s = kx$ first.

Type 9

Pythagorean-relation word problems: two quantities related by $x^2 + y^2 = z^2$ (like the horizontal distance and height of a kite, connected by the string length) – differentiate the Pythagorean relation implicitly.

Source: Exercise 6.1, Q23

Given: A kite is 120 m high and 130 m of string is out. If the kite moves horizontally at 5.2 m/sec,

Solution:

Step 1: Let x be the horizontal distance of the kite and l be the length of string out. Since the height (120 m, constant) is fixed, the string, height, and horizontal distance form a right triangle: $x^2 + (120)^2 = l^2$.

Step 2: At the given instant, use the given string length to find x : $x^2 = 130^2 - 120^2 = 16900 - 14400 = 2500 \Rightarrow x = 50$.

Step 3: Differentiate the relation $x^2 + 14400 = l^2$ with respect to t (the constant 14400 vanishes): $2x \frac{dx}{dt} = 2l \frac{dl}{dt} \Rightarrow x \frac{dx}{dt} = l \frac{dl}{dt}$.

Step 4: Substitute $x = 50$, $\frac{dx}{dt} = 5.2$, $l = 130$: $50(5.2) = 130 \frac{dl}{dt} \Rightarrow \frac{dl}{dt} = \frac{260}{130} = 2$.

Final Answer

String is being paid out at 2 m/sec

Common Mistake: Forgetting that the height is FIXED (constant) in this problem, so it contributes zero to the differentiated equation – only the horizontal distance x and string length l are actually changing with time.

Quick Recall: Identify which side of the right triangle is fixed (contributes 0 when differentiated) and which two are changing – differentiate the Pythagorean relation implicitly with respect to t , then substitute the instant's values (finding any missing side via plain Pythagoras first if needed).

Type 10

A cone (funnel, sand-heap, or similar) has volume that can be expressed purely in terms of its height h alone, using a FIXED ratio between radius and height (from a given semi-vertical angle or a stated proportion) – substitute this ratio into the volume formula before differentiating.

★ **Board Favorite** – Cone-shaped container/heap problems are a classic, frequently-tested related-rates pattern precisely because they require the extra "eliminate r using the fixed ratio" step before the calculus even begins.

Formula Used: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$

Source: Exercise 6.1, Q24

Given: A circular cone, semi-vertical angle 45° , is fixed vertex-down. Water is poured in at $2 \text{ cm}^3/\text{s}$

Solution:

Step 1: For a cone with semi-vertical angle 45° , the radius and height of the water cone at any instant are related by $\tan 45^\circ = \frac{r}{h} \Rightarrow r = h$ (since $\tan 45^\circ = 1$) – this is the key geometric simplification.

Step 2: Substitute $r = h$ into the volume formula: $V = \frac{1}{3}\pi h^2 \cdot h = \frac{1}{3}\pi h^3$ – now V is a function of h ALONE.

Step 3: Differentiate with respect to t : $\frac{dV}{dt} = \pi h^2 \frac{dh}{dt}$.

Step 4: Substitute the given $\frac{dV}{dt} = 2$ and $h = 4$: $2 = \pi(16) \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{2}{16\pi} = \frac{1}{8\pi}$.

Final Answer

$$\frac{dh}{dt} = \frac{1}{8\pi} \text{ cm/sec}$$

Common Mistake: Trying to differentiate $V = \frac{1}{3}\pi r^2 h$ directly with BOTH r and h as separate changing variables – since r and h are tied together by the fixed angle, r must be eliminated FIRST, reducing the formula to a single-variable function of h .

Alternate Board-Style Example

Source: Exercise 6.1, Q25

Given: Sand poured from a pipe at $12 \text{ cm}^3/\text{sec}$ forms a cone whose radius is always one-sixth of its

Solution:

Step 1: Given ratio: $r = \frac{h}{6}$.

Step 2: Substitute into $V = \frac{1}{3}\pi r^2 h$: $V = \frac{1}{3}\pi \left(\frac{h}{6}\right)^2 h = \frac{1}{3}\pi \cdot \frac{h^2}{36} \cdot h = \frac{\pi h^3}{108}$.

Step 3: Differentiate: $\frac{dV}{dt} = \frac{\pi}{108} \cdot 3h^2 \frac{dh}{dt} = \frac{\pi h^2}{36} \frac{dh}{dt}$.

Step 4: Substitute $\frac{dV}{dt} = 12$ and $h = 4$: $12 = \frac{\pi(16)}{36} \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{12 \times 36}{16\pi} = \frac{27}{\pi}$.

Final Answer

$$\frac{dh}{dt} = \frac{27}{\pi} \text{ cm/sec}$$

Quick Recall: Whenever a cone problem gives a fixed angle or a fixed ratio between radius and height, substitute that ratio into $V = \frac{1}{3}\pi r^2 h$ FIRST to reduce it to a function of one variable, and only then differentiate with respect to time.

Type 11

A cone problem where either (a) the volume is more naturally expressed via the SLANT HEIGHT rather than the vertical height, or (b) a FIXED container (with its own given, fixed radius and height) constrains the water/sand's radius-to-height ratio via similar triangles, and this ratio must be derived from the container's own dimensions before substituting.

Formula Used: Slant height parametrization: $r = l \sin \alpha$, $h = l \cos \alpha \Rightarrow V = \frac{1}{3}\pi l^3 \sin^2 \alpha \cos \alpha$

Source: Exercise 6.1, Q27

Given: Water drips at a steady rate of $1 \text{ cm}^3/\text{sec}$ through a tiny hole at the vertex of a conical vessel ($\frac{1}{6}$). When the slant height of the water is 4 cm, find the rate of decrease of the slant height.

Solution:

Step 1: Here it's more convenient to express the water cone's dimensions via its SLANT HEIGHT l rather than h : with semi-vertical angle α , $r = l \sin \alpha$ and $h = l \cos \alpha$ (standard right-triangle decomposition of the cone's cross-section).

Step 2: Substitute into $V = \frac{1}{3}\pi r^2 h$: $V = \frac{1}{3}\pi(l \sin \alpha)^2(l \cos \alpha) = \frac{1}{3}\pi l^3 \sin^2 \alpha \cos \alpha$ – now V depends on l alone (since α is a FIXED constant angle).

Step 3: Differentiate with respect to t : $\frac{dV}{dt} = \pi l^2 \sin^2 \alpha \cos \alpha \frac{dl}{dt}$.

Step 4: Since water is DRIPPING OUT, the volume is decreasing: $\frac{dV}{dt} = -1$. Substitute $l = 4$ and the given α (with $\sin^2 \alpha \cos \alpha$ computed from $\tan \alpha = \frac{1}{6}$, giving a specific numeric coefficient), then solve for $\frac{dl}{dt}$.

Final Answer

The slant height decreases at a rate found by solving $-\quad 1 \quad = \pi(16) \sin^2 \alpha \cos \alpha \frac{dl}{dt}$ for $\frac{dl}{dt}$ using the given α

Common Mistake: Using the vertical height h formula out of habit when the question specifically asks about SLANT height – the slant-height parametrization ($r = l \sin \alpha$, $h = l \cos \alpha$) must be used instead of the usual r, h substitution.

Alternate Board-Style Example (fixed-vessel similar-triangle setup)

Source: Exercise 6.1, Q26

Given: A conical vessel with base radius 4 m and height 2 m is being filled with sand at $0.75 \text{ m}^3/\text{m}$

Solution:

Step 1: The FIXED container has base radius 4 and height 2, giving a fixed ratio $\frac{r}{h} = \frac{4}{2} = 2$ – by similar triangles, the sand's own radius r and height h at ANY instant must maintain this SAME ratio: $r = 2h$.

Step 2: Substitute into $V = \frac{1}{3}\pi r^2 h$: $V = \frac{1}{3}\pi(2h)^2 h = \frac{4}{3}\pi h^3$.

Step 3: Differentiate: $\frac{dV}{dt} = 4\pi h^2 \frac{dh}{dt}$.

Step 4: Substitute $\frac{dV}{dt} = 0.75$ and $h = 1.5$: $0.75 = 4\pi(1.5)^2 \frac{dh}{dt} = 4\pi(2.25) \frac{dh}{dt} = 9\pi \frac{dh}{dt} \Rightarrow \frac{dh}{dt} = \frac{0.75}{9\pi} = \frac{1}{12\pi}$.

Final Answer

$\frac{dh}{dt} = \frac{1}{12\pi} \text{ m/min}$

Quick Recall: For a FIXED container, the ratio $\frac{r}{h}$ of the growing sand/water cone always matches the container's OWN fixed $\frac{r}{h}$ ratio (by similar triangles) – derive that ratio from the container's given dimensions first; for slant-height questions, switch to the $r = l \sin \alpha$, $h = l \cos \alpha$ parametrization instead.

Type 12

A rate-of-change question set in a real-world (non-geometric) context – population growth, pollution levels, user growth on an app, etc. – modelled by an explicit function of time; differentiate and evaluate exactly as with any other rate problem, then interpret the sign (increasing or decreasing).

★ **Board Favorite** – Real-world modelling questions (population, growth curves) are an increasingly common modern ISC pattern, reflecting the applied, competency-based direction of the current syllabus.

Source: Exercise 6.1, Q28

Given: The population of rabbits in a forest is modelled by $P(t) = \frac{2000}{1 + e^{-0.5t}}$, where t is in years. Determine whether the rabbit population is increasing or not, and justify.

Solution:

Step 1: To determine increasing/decreasing, find $P'(t)$ and check its sign – write $P(t) = 2000(1 + e^{-0.5t})^{-1}$ to differentiate via the chain rule.

Step 2: $P'(t) = 2000 \times (-1)(1 + e^{-0.5t})^{-2} \times (-0.5e^{-0.5t}) = \frac{1000e^{-0.5t}}{(1 + e^{-0.5t})^2}$ – the two negative signs (from the -1 power and the -0.5 inside the exponential) cancel to give a positive coefficient.

Step 3: Analyse the sign: $e^{-0.5t} > 0$ for every real t , and $(1 + e^{-0.5t})^2 > 0$ always too – so $P'(t)$ is a ratio of two POSITIVE quantities, hence $P'(t) > 0$ for all t .

Step 4: Since $P'(t) > 0$ everywhere, the population is always increasing.

Final Answer

Yes, the rabbit population is always increasing, since $P'(t) > 0$ for all t (a ratio of positive quantities)

Common Mistake: Trying to evaluate $P'(t)$ at specific numeric values of t to "check a few points" instead of arguing the sign of $P'(t)$ in general – since $e^{-0.5t}$ is positive for EVERY real t , a single general sign argument settles the question for all time, not just isolated instants.

Companion Worked Example (from the textbook)

Source: Text Example 28

Given: The number of active users on a platform is modelled by $N(t) = 1000e^{0.1t}$, where t is in minutes. Find how fast the number of users is changing at $t = 10$.

Solution:

Step 1: Differentiate using the chain rule: $\frac{d}{dt}N(t) = 1000e^{0.1t} \times 0.1 = 100e^{0.1t}$.

Step 2: Substitute $t = 10$: $\left[\frac{d}{dt}N(t)\right]_{t=10} = 100e^{0.1 \times 10} = 100e^1 = 100e$.

Step 3: Since this value is positive, the number of active users is INCREASING at this instant.

Final Answer

Users are increasing at a rate of $100e$ persons per minute

Quick Recall: For real-world "modelled by" functions, the method never changes – differentiate the given function of t (using whatever chain/product rules the formula needs), substitute the specified instant if one is given, and read the SIGN of the result to answer any "increasing or decreasing" question.

II. Tangents, Normals, and Angle of Intersection

Type 13

Finding the slope of the tangent (or normal) to a curve at a given point or x -value – differentiate (implicitly if needed), substitute, and for the normal take the negative reciprocal.

★ **Board Favorite** – This is the foundational skill for the entire tangent/normal topic – every other type in this section builds on being able to find the slope quickly and correctly.

Formula Used: Slope of tangent = $\left. \frac{dy}{dx} \right|_{\text{point}}$; Slope of normal = $\frac{-1}{\text{slope of tangent}}$

Source: Exercise 6.2, Q1(ii) (NCERT)

Given: Find the slope of the tangent to $y = 3x^4 - 4x$ at $x = 4$.

Solution:

Step 1: Differentiate: $\frac{dy}{dx} = 12x^3 - 4$.

Step 2: Substitute $x = 4$: $\frac{dy}{dx} = 12(64) - 4 = 768 - 4 = 764$.

Final Answer

Slope of tangent = 764

Common Mistake: For curves given explicitly as $y = f(x)$, differentiating directly is enough – but forgetting to switch to IMPLICIT differentiation when x and y are mixed together (see the alternate example) is a common slip.

Alternate Board-Style Example (implicit curve + normal slope)

Source: Exercise 6.2, Q1(iii) and Q2(i)

Given: (i) Slope of tangent to $x^2 + 3y + y^2 = 5$ at $(1, 1)$. (ii) Slope of normal to $y = x^3 - 5x^2 - x + 1$ at $(1, -4)$.

Solution:

Case: (i) Implicit curve

Differentiate $x^2 + 3y + y^2 = 5$ implicitly: $2x + 3\frac{dy}{dx} + 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx}(3 + 2y) = -2x \Rightarrow \frac{dy}{dx} = \frac{-2x}{3 + 2y}$. Substitute $(1, 1)$: $\frac{dy}{dx} = \frac{-2}{5}$.

Case: (ii) Normal slope

Differentiate: $\frac{dy}{dx} = 3x^2 - 10x - 1$. At $(1, -4)$: $\frac{dy}{dx} = 3 - 10 - 1 = -8$. The normal's slope is the negative reciprocal: $\frac{-1}{-8} = \frac{1}{8}$.

Final Answer

(i) Slope of tangent = $-\frac{2}{5}$ (ii) Slope of normal = $\frac{1}{8}$

Quick Recall: Explicit curve $\Rightarrow$ differentiate directly; curve with x and y mixed $\Rightarrow$ differentiate implicitly, collecting $\frac{dy}{dx}$ terms; "slope of NORMAL" always means take the negative reciprocal of the tangent's slope at the end.

Type 14

Finding the point(s) on a curve where the tangent has a SPECIFIED (non-zero, non-special) slope – set $\frac{dy}{dx}$ equal to the given slope and solve for x , then find y .

Source: Exercise 6.2, Q3

Given: Find the point on $y = 3x^2 - 2x + 1$ at which the slope of the tangent is 4.

Solution:

Step 1: Differentiate: $\frac{dy}{dx} = 6x - 2$.

Step 2: Set equal to the given slope: $6x - 2 = 4 \Rightarrow 6x = 6 \Rightarrow x = 1$.

Step 3: Substitute $x = 1$ into the curve to find y : $y = 3(1) - 2(1) + 1 = 2$.

Final Answer

The point is $(1, 2)$

Common Mistake: Forgetting to substitute the found x -value back into the ORIGINAL curve equation to get y – the question asks for a point, not just an x -value.

Alternate Board-Style Example (special 45° slope)

Source: Exercise 6.2, Q5

Given: At what point on the curve $y = x^2$ does the tangent make an angle of 45° with the x -axis?

Solution:

Step 1: An angle of 45° with the x -axis means the slope is $\tan 45^\circ = 1$ – converting the angle condition into a slope condition first.

Step 2: Differentiate: $\frac{dy}{dx} = 2x$. Set equal to 1: $2x = 1 \Rightarrow x = \frac{1}{2}$.

Step 3: Substitute into the curve: $y = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Final Answer

The point is $\left(\frac{1}{2}, \frac{1}{4}\right)$

Quick Recall: Whenever an ANGLE is given instead of a slope, convert first using $\text{slope} = \tan(\text{angle})$ – then the method is identical to any other “find the point with a given slope” question.

Type 15

Finding the point(s) where the tangent is HORIZONTAL (parallel to the x -axis, slope = 0) – or, as the mirror case, VERTICAL (parallel to the y -axis, i.e. $\frac{dx}{dy} = 0$).

★ **Board Favorite** – Horizontal-tangent questions are extremely common since “parallel to the x -axis” appears constantly across VSA and SA questions – the book tags an instance directly to the ISC Specimen Paper 2024.

Formula Used: Tangent parallel to x -axis $\iff \frac{dy}{dx} = 0$

Source: Exercise 6.2, Q6(i) (NCERT)

Given: Find the point on $y = x^2 - 2x + 3$ at which the tangent is parallel to the x -axis.

Solution:

Step 1: “Parallel to the x -axis” means the slope is exactly 0.

Step 2: Differentiate: $\frac{dy}{dx} = 2x - 2$. Set equal to 0: $2x - 2 = 0 \Rightarrow x = 1$.

Step 3: Substitute: $y = 1 - 2 + 3 = 2$.

Final Answer

The point is (1, 2)

Common Mistake: Confusing "parallel to the x -axis" (horizontal, slope = 0) with "parallel to the y -axis" (vertical, slope undefined) – these require completely different equations to solve.

Alternate Board-Style Example (an ellipse, both horizontal AND vertical)

Source: Exercise 6.2, Q16

Given: Find the points on $\frac{x^2}{4} + \frac{y^2}{25} = 1$ at which the tangents are (i) parallel to the x -axis (ii) parallel to the y -axis

Solution:

Case: (i) Parallel to x -axis: $\frac{dy}{dx} = 0$

Differentiate implicitly: $\frac{2x}{4} + \frac{2y}{25} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-25x}{4y}$. Set equal to 0: this requires $x = 0$ (with $y \neq 0$). Substitute into the curve: $\frac{y^2}{25} = 1 \Rightarrow y = \pm 5$. Points: (0, 5) and (0, -5).

Case: (ii) Parallel to y -axis: $\frac{dx}{dy} = 0$, equivalently $\frac{dy}{dx}$ undefined (denominator zero)

From $\frac{dy}{dx} = \frac{-25x}{4y}$, this is undefined when $y = 0$. Substitute into the curve: $\frac{x^2}{4} = 1 \Rightarrow x = \pm 2$. Points: (2, 0) and (-2, 0).

Final Answer

(i) (0, 5), (0, -5) (ii) (2, 0), (-2, 0)

Quick Recall: Horizontal tangent: set the NUMERATOR of $\frac{dy}{dx}$ to zero (slope = 0); vertical tangent: set the DENOMINATOR of $\frac{dy}{dx}$ to zero (slope undefined) – for implicit curves this is usually the fastest way to spot both cases at once.

Type 16

Finding the point(s) on a curve where the tangent is PARALLEL to a given line – match the tangent's slope to the given line's slope (read off from its equation) and solve.

★ **Board Favorite** – Matching a tangent's slope to a given line is one of the most common LA question formats in this topic, appearing repeatedly across different curve types.

Source: Exercise 6.2, Q17

Given: Find the points on $x^2 + y^2 = 13$ at which the tangent is parallel to the line $2x + 3y = 7$.

Solution:

Step 1: Find the given line's slope: rewrite $2x + 3y = 7$ as $y = -\frac{2}{3}x + \frac{7}{3}$, so its slope is $-\frac{2}{3}$.

Step 2: Differentiate the curve implicitly: $2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{x}{y}$.

Step 3: Set equal to the line's slope: $-\frac{x}{y} = -\frac{2}{3} \Rightarrow 3x = 2y \Rightarrow y = \frac{3x}{2}$.

Step 4: Substitute into the curve equation: $x^2 + \left(\frac{3x}{2}\right)^2 = 13 \Rightarrow x^2 + \frac{9x^2}{4} = 13 \Rightarrow \frac{13x^2}{4} = 13 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$.

Step 5: Find corresponding y : at $x = 2$, $y = 3$; at $x = -2$, $y = -3$.

Final Answer

The points are $(2, 3)$ and $(-2, -3)$

Common Mistake: Reading the slope off the line's equation incorrectly (e.g. using the coefficient of x directly without first solving for y) – always rearrange to $y = mx + c$ form before reading off m .

Alternate Board-Style Example (cubic curve)

Source: Exercise 6.2, Q18

Given: Find the points on $y = x^3 - 3x^2 - 4x$ at which the tangent lines are parallel to the line $4x + y - 3 = 0$.

Solution:

Step 1: Line's slope: $y = -4x + 3$, so slope $= -4$.

Step 2: Differentiate the curve: $\frac{dy}{dx} = 3x^2 - 6x - 4$.

Step 3: Set equal to -4 : $3x^2 - 6x - 4 = -4 \Rightarrow 3x^2 - 6x = 0 \Rightarrow 3x(x - 2) = 0 \Rightarrow x = 0$ or $x = 2$.

Step 4: Find y : at $x = 0$, $y = 0$; at $x = 2$, $y = 8 - 12 - 8 = -12$.

Final Answer

The points are $(0, 0)$ and $(2, -12)$

Quick Recall: Read the given line's slope by rearranging to $y = mx + c$ form, set the curve's $\frac{dy}{dx}$ equal to that same m , solve for x , then substitute back into the ORIGINAL curve to find each corresponding y .

Type 17

Finding the point on a curve where the tangent is PERPENDICULAR to a given line – the tangent's slope must equal the NEGATIVE RECIPROCAL of the given line's slope.

Formula Used: Perpendicular lines: $m_1 \times m_2 = -1$

Source: Exercise 6.2, Q13

Given: Determine the point on $y = 3x^2 - 5$ at which the tangent is perpendicular to a line whose slope is $-\frac{1}{3}$.

Solution:

Step 1: Since the tangent must be perpendicular to a line of slope $-\frac{1}{3}$, the tangent's own slope must be the negative reciprocal: $m = \frac{-1}{-1/3} = 3$.

Step 2: Differentiate the curve: $\frac{dy}{dx} = 6x$. Set equal to 3: $6x = 3 \Rightarrow x = \frac{1}{2}$.

Step 3: Substitute into the curve: $y = 3\left(\frac{1}{2}\right)^2 - 5 = \frac{3}{4} - 5 = -\frac{17}{4}$.

Final Answer

The point is $\left(\frac{1}{2}, -\frac{17}{4}\right)$

Common Mistake: Using the GIVEN line's slope directly as the tangent's slope (as if it were a "parallel" question) instead of taking its negative reciprocal for a "perpendicular" question – always double-check which relationship (parallel or perpendicular) the question actually asks for.

Quick Recall: "Parallel" means match the slope exactly; "perpendicular" means take the negative reciprocal of the given slope FIRST, then match the tangent's $\frac{dy}{dx}$ to that.

Type 18

Proving that the tangents to a curve at two DIFFERENT specified points are parallel to EACH OTHER – compute the slope at both points separately and show they're equal (rather than matching to any external given line).

Source: Exercise 6.2, Q11 (NCERT)

Given: Show that the tangents to the curve $y = 7x^3 + 11$ at the points where $x = 2$ and $x = -2$ are parallel.

Solution:

Step 1: Differentiate: $\frac{dy}{dx} = 21x^2$.

Step 2: Evaluate at $x = 2$: $\frac{dy}{dx} = 21(4) = 84$.

Step 3: Evaluate at $x = -2$: $\frac{dy}{dx} = 21(4) = 84$ – the SAME value, since x^2 is unaffected by the sign of x .

Step 4: Since both tangents have the identical slope 84, they are parallel to each other.

Final Answer

Both slopes equal 84, so the tangents at $x = 2$ and $x = -2$ are parallel

Common Mistake: Forgetting that even-power terms in the derivative (like x^2) automatically give the SAME value at $+a$ and $-a$ – this is often exactly why the question was set up with symmetric x -values in the first place.

Quick Recall: To prove two tangents are parallel, simply compute $\frac{dy}{dx}$ at both specified points independently and show the two numeric results match – no need to compare against any external line.

Type 19

Finding unknown constants in a curve's equation, using BOTH the fact that a given point lies on the curve AND a tangent condition (parallel to a given line) at that point – forms a system of two equations.

Source: Exercise 6.2, Q12

Given: If the tangent to $y = x^3 + ax + b$ at $(1, -6)$ is parallel to the line $2x - 2y + 7 = 0$, find a and b .

Solution:

Step 1: **Equation 1 (point lies on curve):** substitute $(1, -6)$ into $y = x^3 + ax + b$:
 $-6 = 1 + a + b \Rightarrow a + b = -7$.

Step 2: **Equation 2 (tangent condition):** find the given line's slope: $2x - 2y + 7 = 0 \Rightarrow$
 $y = x + \frac{7}{2}$, slope = 1.

Step 3: Differentiate the curve: $\frac{dy}{dx} = 3x^2 + a$. At $x = 1$: $\frac{dy}{dx} = 3 + a$. Set equal to 1:
 $3 + a = 1 \Rightarrow a = -2$.

Step 4: Substitute $a = -2$ into Equation 1: $-2 + b = -7 \Rightarrow b = -5$.

Final Answer

$$a = -2, b = -5$$

Common Mistake: Using only the tangent-slope condition and forgetting that the point must ALSO satisfy the original curve equation – both conditions are needed together to pin down two unknowns.

Quick Recall: Two unknowns need two equations: "point lies on the curve" gives one equation (direct substitution), and the tangent-slope condition (parallel/perpendicular to a given line) gives the second, via differentiation.

Type 20

Finding the slopes of the tangents at MULTIPLE points simultaneously – first solve for where the curve crosses the x -axis (or another specified condition) to locate the points, THEN find the slope at each one.

Source: Exercise 6.2, Q10

Given: Find the slopes of the tangents at the three points where the curve $y = x(x^2 - 3x - 4)$ cuts the x -axis.

Solution:

Step 1: First find where the curve crosses the x -axis, i.e. where $y = 0$: $x(x^2 - 3x - 4) = 0 \Rightarrow x(x - 4)(x + 1) = 0 \Rightarrow x = 0, 4, -1$ – three distinct crossing points.

Step 2: Expand the curve for easier differentiation: $y = x^3 - 3x^2 - 4x$, so $\frac{dy}{dx} = 3x^2 - 6x - 4$.

Step 3: Evaluate the slope at each of the three points: at $x = 0$: $\frac{dy}{dx} = -4$. At $x = 4$:
 $\frac{dy}{dx} = 3(16) - 24 - 4 = 48 - 24 - 4 = 20$. At $x = -1$: $\frac{dy}{dx} = 3(1) + 6 - 4 = 5$.

Final Answer

Slopes are -4 , 20 , and 5 at $x = 0, 4, -1$ respectively

Common Mistake: Trying to find the slope BEFORE locating the actual points – since the question needs the slope AT the crossing points specifically, the roots must be found first (by factoring), then substituted into the already-differentiated formula.

Quick Recall: Two-stage questions like this always go in order: FIRST solve the stated condition (roots, intersections, etc.) to find the relevant x -values, THEN differentiate and substitute each one in turn.

Type 21

Finding the full EQUATIONS of both the tangent AND the normal at a given point – compute the tangent's slope, then use the point-slope form twice (once with the tangent's slope, once with the normal's negative-reciprocal slope).

★ **Board Favorite** – This is the master combined skill of the whole tangent/normal topic – writing full line equations (not just slopes) is the most commonly graded long-answer format, tagged repeatedly to NCERT.

Formula Used: $y - y_1 = m(x - x_1)$

Source: Exercise 6.2, Q19(ii) (NCERT)

Given: Find the tangent and normal to $y = x^4 - 6x^3 + 13x^2 - 10x + 5$ at $(1, 3)$.

Solution:

Step 1: Differentiate: $\frac{dy}{dx} = 4x^3 - 18x^2 + 26x - 10$.

Step 2: Evaluate at $x = 1$: $\frac{dy}{dx} = 4 - 18 + 26 - 10 = 2$ – this is the tangent's slope.

Step 3: **Tangent equation:** using point-slope form with $(1, 3)$ and slope 2: $y - 3 = 2(x - 1) \Rightarrow y = 2x + 1 \Rightarrow 2x - y + 1 = 0$.

Step 4: **Normal's slope:** negative reciprocal of 2 is $-\frac{1}{2}$.

Step 5: **Normal equation:** $y - 3 = -\frac{1}{2}(x - 1) \Rightarrow 2y - 6 = -(x - 1) \Rightarrow x + 2y - 7 = 0$.

Final Answer

Tangent: $2x - y + 1 = 0$; Normal: $x + 2y - 7 = 0$

Common Mistake: Computing the tangent equation correctly but then forgetting to take the **NEGATIVE RECIPROCAL** for the normal's slope – accidentally reusing the tangent's own slope for the normal equation.

Alternate Board-Style Example (trig curve)

Source: Exercise 6.2, Q19(v)

Given: Find the tangent and normal to $y = \sin^2 x$ at $x = \frac{\pi}{2}$.

Solution:

Step 1: First find the y -coordinate: $y = \sin^2\left(\frac{\pi}{2}\right) = 1^2 = 1$, so the point is $\left(\frac{\pi}{2}, 1\right)$.

Step 2: Differentiate: $\frac{dy}{dx} = 2 \sin x \cos x = \sin 2x$. At $x = \frac{\pi}{2}$: $\frac{dy}{dx} = \sin \pi = 0$.

Step 3: **Tangent equation** (slope 0, i.e. horizontal): $y - 1 = 0\left(x - \frac{\pi}{2}\right) \Rightarrow y = 1$, i.e. $y - 1 = 0$.

Step 4: **Normal's slope:** since the tangent's slope is 0, the normal is **VERTICAL** (slope undefined) – a special case where the negative-reciprocal formula breaks down, so reason directly: a normal perpendicular to a horizontal line is simply the vertical line through the point.

Step 5: **Normal equation:** $x = \frac{\pi}{2}$, i.e. $2x - \pi = 0$.

Final Answer

Tangent: $y - 1 = 0$; Normal: $2x - \pi = 0$

Quick Recall: Find the point (computing y if only x is given), find the tangent's slope via differentiation, write the tangent equation using point-slope form, then write the normal equation using the negative-reciprocal slope – **EXCEPT** when the tangent slope is 0 (then the normal is simply the vertical line $x = x_1$) or undefined (then the normal is the horizontal line $y = y_1$).

Type 22

Finding the tangent or normal equation at a **GENERAL** parametric point (expressed with a parameter like m , not a specific number) – the algebra stays symbolic throughout.

Source: Exercise 6.2, Q20

Given: Find the equation of the normal to the curve $ay^2 = x^3$ at the point (am^2, am^3) .

Solution:

Step 1: Differentiate $ay^2 = x^3$ implicitly with respect to x : $2ay \frac{dy}{dx} = 3x^2 \Rightarrow \frac{dy}{dx} = \frac{3x^2}{2ay}$.

Step 2: Substitute the general point (am^2, am^3) : $\frac{dy}{dx} = \frac{3(am^2)^2}{2a(am^3)} = \frac{3a^2m^4}{2a^2m^3} = \frac{3m}{2}$ – the algebra simplifies cleanly since powers of a cancel.

Step 3: Normal's slope: negative reciprocal $= \frac{-2}{3m}$.

Step 4: Write the normal equation using point-slope form with the general point: $y - am^3 = \frac{-2}{3m}(x - am^2)$.

Step 5: Clear the fraction by multiplying through by $3m$: $3m(y - am^3) = -2(x - am^2) \Rightarrow 3my - 3am^4 = -2x + 2am^2 \Rightarrow 2x + 3my - 2am^2 - 3am^4 = 0$, which rearranges to $2x + 3my - am^2(2 + 3m^2) = 0$.

Final Answer

$$2x + 3my - am^2(2 + 3m^2) = 0$$

Common Mistake: Getting intimidated by the symbolic parameter m and trying to substitute a numeric value for it – treat m exactly like any other constant throughout the algebra; the final answer is SUPPOSED to still contain m .

Quick Recall: Parametric-point questions use the exact same tangent/normal method as numeric-point questions – just carry the parameter symbol through the algebra carefully, simplifying powers as they naturally cancel.

Type 23

Finding the tangent equation for a curve given IMPLICITLY (not solved for y), requiring implicit differentiation before the point-slope method can be applied.

★ **Board Favorite** – Implicit curves (especially the "astroid" family $x^{2/3} + y^{2/3} = k$) are a recurring NCERT-tagged pattern testing implicit differentiation combined with tangent/normal equations.

Source: Exercise 6.2, Q21(ii) (NCERT)

Given: Find the tangent and normal to $x^{2/3} + y^{2/3} = 2$ at $(1, 1)$.

Solution:

Step 1: Differentiate implicitly: $\frac{2}{3}x^{-1/3} + \frac{2}{3}y^{-1/3}\frac{dy}{dx} = 0$.

Step 2: Solve for $\frac{dy}{dx}$: $\frac{dy}{dx} = -\frac{x^{-1/3}}{y^{-1/3}} = -\left(\frac{y}{x}\right)^{1/3}$.

Step 3: Substitute $(1, 1)$: $\frac{dy}{dx} = -1$.

Step 4: **Tangent:** $y - 1 = -1(x - 1) \Rightarrow y - 1 = -x + 1 \Rightarrow x + y - 2 = 0$.

Step 5: **Normal:** slope = 1 (negative reciprocal of -1): $y - 1 = 1(x - 1) \Rightarrow y = x \Rightarrow x - y = 0$.

Final Answer

Tangent: $x + y - 2 = 0$; Normal: $x - y = 0$

Common Mistake: Trying to solve the implicit equation for y explicitly before differentiating – with fractional powers like $x^{2/3} + y^{2/3} = 2$, this is unnecessary and often impossible cleanly; implicit differentiation handles it directly.

Alternate Board-Style Example

Source: Exercise 6.2, Q21(i)

Given: Find the tangent to $\sqrt{x} + \sqrt{y} = a$ at the point $\left(\frac{a^2}{4}, \frac{a^2}{4}\right)$.

Solution:

Step 1: Differentiate implicitly: $\frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{y}} \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = -\frac{\sqrt{y}}{\sqrt{x}}$.

Step 2: Substitute the point (where $x = y = \frac{a^2}{4}$, so $\sqrt{x} = \sqrt{y}$): $\frac{dy}{dx} = -1$.

Step 3: Tangent equation: $y - \frac{a^2}{4} = -1\left(x - \frac{a^2}{4}\right) \Rightarrow y - \frac{a^2}{4} = -x + \frac{a^2}{4} \Rightarrow x + y = \frac{a^2}{2} \Rightarrow 2(x + y) = a^2$.

Final Answer

$2(x + y) = a^2$

Quick Recall: For any curve with x and y mixed (or with fractional/radical powers of both), differentiate term-by-term treating y as a function of x , collect the $\frac{dy}{dx}$ terms, THEN substitute the given point into the resulting slope formula.

Type 24

Finding the tangent equation for a PARAMETRIC curve (given as $x = f(t)$, $y = g(t)$) at a specific NUMERIC value of the parameter – find $\frac{dy}{dx}$ via $\frac{dy/dt}{dx/dt}$, then find the actual point by substituting the parameter value into both $x(t)$ and $y(t)$.

★ **Board Favorite** – Parametric tangent questions are frequently tested, tagged directly to NCERT, combining the parametric-differentiation skill with the tangent-equation method in one question.

Source: Exercise 6.2, Q22(i) (NCERT)

Given: Find the tangent to $x = a \sin^3 t$, $y = b \cos^3 t$ at $t = \frac{\pi}{2}$.

Solution:

Step 1: Find the actual point by substituting $t = \frac{\pi}{2}$: $x = a \sin^3 \left(\frac{\pi}{2}\right) = a(1)^3 = a$; $y = b \cos^3 \left(\frac{\pi}{2}\right) = b(0)^3 = 0$. So the point is $(a, 0)$.

Step 2: Differentiate w.r.t. t : $\frac{dx}{dt} = 3a \sin^2 t \cos t$; $\frac{dy}{dt} = -3b \cos^2 t \sin t$.

Step 3: Form $\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-3b \cos^2 t \sin t}{3a \sin^2 t \cos t} = \frac{-b \cos t}{a \sin t}$.

Step 4: Substitute $t = \frac{\pi}{2}$: $\cos \frac{\pi}{2} = 0$, so $\frac{dy}{dx} = \frac{-b(0)}{a(1)} = 0$ – a horizontal tangent.

Step 5: Tangent equation at $(a, 0)$ with slope 0: $y - 0 = 0(x - a) \Rightarrow y = 0$.

Final Answer

$$y = 0$$

Common Mistake: Substituting the parameter value into the SLOPE formula $\frac{dy}{dx}$ but forgetting to ALSO substitute it into $x(t)$ and $y(t)$ separately to find the actual point needed for the point-slope equation.

Alternate Board-Style Example

Source: Exercise 6.2, Q22(ii)

Given: Find the tangent to $x = \sin 3t$, $y = \cos 2t$ at $t = \frac{\pi}{4}$.

Solution:

Step 1: Find the point: $x = \sin \frac{3\pi}{4} = \frac{1}{\sqrt{2}}$; $y = \cos \frac{\pi}{2} = 0$. Point: $\left(\frac{1}{\sqrt{2}}, 0\right)$.

Step 2: $\frac{dx}{dt} = 3 \cos 3t$; $\frac{dy}{dt} = -2 \sin 2t$.

Step 3: $\frac{dy}{dx} = \frac{-2 \sin 2t}{3 \cos 3t}$. At $t = \frac{\pi}{4}$: $\sin \frac{\pi}{2} = 1$, $\cos \frac{3\pi}{4} = \frac{-1}{\sqrt{2}}$, so $\frac{dy}{dx} = \frac{-2(1)}{3(-1/\sqrt{2})} = \frac{-2}{-3/\sqrt{2}} = \frac{2\sqrt{2}}{3}$.

Step 4: Tangent equation: $y - 0 = \frac{2\sqrt{2}}{3} \left(x - \frac{1}{\sqrt{2}} \right) \Rightarrow 3y = 2\sqrt{2}x - 2 \Rightarrow 2\sqrt{2}x - 3y - 2 = 0$.

Final Answer

$$2\sqrt{2}x - 3y - 2 = 0$$

Quick Recall: Parametric tangent questions need TWO separate substitutions of the given parameter value: once into $x(t)$ and $y(t)$ to locate the actual point, and once into the ratio $\frac{dy/dt}{dx/dt}$ to find the slope – don't skip either one.

Type 25

Finding tangent line(s) to a curve with a GIVEN slope, solved by setting up and solving an equation for the point of tangency – including the possibility that NO such tangent exists (the resulting equation has no real solution).

Source: Exercise 6.2, Q23(i) (NCERT)

Given: Find the equations of all lines having slope 2 that are tangent to the curve $y = \frac{1}{x-3}$.

Solution:

Step 1: Differentiate: $\frac{dy}{dx} = \frac{-1}{(x-3)^2}$.

Step 2: Set equal to the given slope 2: $\frac{-1}{(x-3)^2} = 2 \Rightarrow (x-3)^2 = -\frac{1}{2}$.

Step 3: The left side, $(x-3)^2$, is a SQUARE and can never be negative for any real x – but the right side is $-\frac{1}{2}$, a negative number. This is a contradiction.

Step 4: Since no real x satisfies the equation, no such tangent line exists at all.

Final Answer

No tangent to this curve has slope 2 – the curve has no tangent line of this slope

Common Mistake: Assuming every "find the tangent with slope m " question must have a solution – always check whether the resulting equation is actually solvable over the reals before concluding; $\frac{dy}{dx} = \frac{-1}{(x-3)^2}$ is ALWAYS negative here, so it can never equal a positive slope like 2.

Quick Recall: After setting $\frac{dy}{dx}$ equal to the target slope, always sanity-check whether the resulting equation can actually be solved for real x – a squared term equalling a negative number is an immediate sign that no such tangent exists.

Type 26

Finding the full tangent EQUATION (not just the point) that is parallel or perpendicular to a given line, for a curve where the point of tangency itself must first be found algebraically.

★ **Board Favorite** – This combined "find the point via slope-matching, then write the full tangent equation" format is the most common complete long-answer task in the tangent/normal topic, tagged to NCERT.

Source: Exercise 6.2, Q24(ii) (NCERT)

Given: Find the equation of the tangent to $y = x^2 - 2x + 7$ which is perpendicular to the line $5y - 15x = 13$.

Solution:

Step 1: Given line's slope: $5y = 15x + 13 \Rightarrow y = 3x + \frac{13}{5}$, slope = 3.

Step 2: Since the tangent must be PERPENDICULAR, its slope is the negative reciprocal:
 $m = -\frac{1}{3}$.

Step 3: Differentiate the curve: $\frac{dy}{dx} = 2x - 2$. Set equal to $-\frac{1}{3}$: $2x - 2 = -\frac{1}{3} \Rightarrow 2x = \frac{5}{3} \Rightarrow x = \frac{5}{6}$.

Step 4: Find y : $y = \left(\frac{5}{6}\right)^2 - 2\left(\frac{5}{6}\right) + 7 = \frac{25}{36} - \frac{10}{6} + 7$. Convert to a common denominator of 36: $\frac{25}{36} - \frac{60}{36} + \frac{252}{36} = \frac{217}{36}$.

Step 5: Write the tangent equation with point $\left(\frac{5}{6}, \frac{217}{36}\right)$ and slope $-\frac{1}{3}$, then clear fractions to reach the final form: $12x + 36y - 227 = 0$.

Final Answer

$$12x + 36y - 227 = 0$$

Common Mistake: Stopping after finding the point of tangency and forgetting the question asked for the full tangent EQUATION, not just the point – always finish with the point-slope form and simplify.

Alternate Board-Style Example*Source: Exercise 6.2, Q25(i)*

Given: Find the equations of the tangents to $y = 4x^3 - 3x + 5$ which are perpendicular to the line $9y + x + 3 = 0$.

Solution:

Step 1: Given line's slope: $9y = -x - 3 \Rightarrow y = -\frac{1}{9}x - \frac{1}{3}$, slope $= -\frac{1}{9}$.

Step 2: Perpendicular tangent slope: negative reciprocal $= 9$.

Step 3: Differentiate: $\frac{dy}{dx} = 12x^2 - 3$. Set equal to 9: $12x^2 - 3 = 9 \Rightarrow 12x^2 = 12 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$ – TWO points this time, giving two separate tangent lines.

Step 4: At $x = 1$: $y = 4 - 3 + 5 = 6$. Tangent: $y - 6 = 9(x - 1) \Rightarrow 9x - y - 3 = 0$.

Step 5: At $x = -1$: $y = -4 + 3 + 5 = 4$. Tangent: $y - 4 = 9(x + 1) \Rightarrow 9x - y + 13 = 0$.

Final Answer

$9x - y - 3 = 0$ and $9x - y + 13 = 0$

Quick Recall: Convert the given line to $y = mx + c$ to read its slope, apply parallel (same m) or perpendicular (negative reciprocal) as required, solve $\frac{dy}{dx} = m$ for x (watch for MULTIPLE solutions), then write a separate tangent equation for each point found.

Type 27

Finding the full NORMAL equation(s) parallel or perpendicular to a given line – the mirror image of Type 26, but using the normal's own slope ($\frac{-1}{dy/dx}$) matched to the line instead.

★ **Board Favorite** – Normal-equation questions with a parallel/perpendicular condition are just as commonly tested as their tangent counterparts, and the book tags an instance as a Competency Focused Question.

Source: Exercise 6.2, Q27 (Competency Focused Question)

Given: Find the equations of the normals to $3x^2 - y^2 = 8$ which are parallel to the line $x + 3y = 4$.

Solution:

Step 1: Given line's slope: $x + 3y = 4 \Rightarrow y = -\frac{1}{3}x + \frac{4}{3}$, slope $= -\frac{1}{3}$.

Step 2: Since the NORMAL must be parallel to this line, the normal's slope is $-\frac{1}{3}$ – so the TANGENT's slope (negative reciprocal of the normal) is 3.

Step 3: Differentiate the curve implicitly: $6x - 2y\frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{3x}{y}$. Set equal to 3:
 $\frac{3x}{y} = 3 \Rightarrow y = x$.

Step 4: Substitute $y = x$ into the curve: $3x^2 - x^2 = 8 \Rightarrow 2x^2 = 8 \Rightarrow x^2 = 4 \Rightarrow x = \pm 2$, giving points $(2, 2)$ and $(-2, -2)$.

Step 5: Write the normal equation (slope $-\frac{1}{3}$) at each point: at $(2, 2)$: $y - 2 = -\frac{1}{3}(x - 2) \Rightarrow x + 3y - 8 = 0$. At $(-2, -2)$: $y + 2 = -\frac{1}{3}(x + 2) \Rightarrow x + 3y + 8 = 0$.

Final Answer

$$x + 3y - 8 = 0 \text{ and } x + 3y + 8 = 0$$

Common Mistake: Setting $\frac{dy}{dx}$ (the TANGENT's slope) directly equal to the given line's slope, forgetting that the question is about the NORMAL – when the condition is on the normal, first convert to find what the TANGENT's slope must be (via the negative-reciprocal relationship) before differentiating.

Alternate Board-Style Example

Source: Exercise 6.2, Q28

Given: Find the equation of the normal to $y = x + \frac{1}{x}$, $x > 0$, perpendicular to the line $3x - 4y = 7$.

Solution:

Step 1: Given line's slope: $3x - 4y = 7 \Rightarrow y = \frac{3}{4}x - \frac{7}{4}$, slope = $\frac{3}{4}$.

Step 2: The normal must be PERPENDICULAR to this line, so the normal's slope is the negative reciprocal: $-\frac{4}{3}$.

Step 3: The tangent's slope is therefore the negative reciprocal of THAT: $\frac{3}{4}$.

Step 4: Differentiate the curve: $\frac{dy}{dx} = 1 - \frac{1}{x^2}$. Set equal to $\frac{3}{4}$: $1 - \frac{1}{x^2} = \frac{3}{4} \Rightarrow \frac{1}{x^2} = \frac{1}{4} \Rightarrow x^2 = 4 \Rightarrow x = 2$ (taking the positive root, since $x > 0$ is given).

Step 5: Find y : $y = 2 + \frac{1}{2} = \frac{5}{2}$.

Step 6: Normal equation at $\left(2, \frac{5}{2}\right)$ with slope $-\frac{4}{3}$: $y - \frac{5}{2} = -\frac{4}{3}(x - 2) \Rightarrow 3y - \frac{15}{2} = -4x + 8$,

clearing fractions: $6y - 15 = -8x + 16 \Rightarrow 8x + 6y = 31$.

Final Answer

$$8x + 6y = 31$$

Quick Recall: For normal-equation questions, there's a DOUBLE negative-reciprocal relay: given line $\rightarrow$ normal's slope (via parallel/perpendicular condition) $\rightarrow$ tangent's slope (negative reciprocal of the normal) $\rightarrow$ solve $\frac{dy}{dx}$ = that tangent slope for the point.

Type 28

Finding the normal equation that passes through a given EXTERNAL point (a point NOT necessarily on the curve itself) – the point of tangency is unknown and must be solved for using the condition that the normal line passes through the external point.

Source: Exercise 6.2, Q29

Given: Find the equation of the normal to $x^2 = 4y$ which passes through the point $(-1, 4)$.

Solution:

Step 1: Let the (unknown) point of tangency be (x_1, y_1) on the curve, so $x_1^2 = 4y_1$ – this is the curve-membership condition.

Step 2: Differentiate implicitly: $2x = 4\frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{x}{2}$. At (x_1, y_1) : slope of tangent = $\frac{x_1}{2}$, so slope of normal = $-\frac{2}{x_1}$.

Step 3: Write the normal equation at (x_1, y_1) : $y - y_1 = \frac{-2}{x_1}(x - x_1)$.

Step 4: Since this normal must pass through the EXTERNAL point $(-1, 4)$, substitute $x = -1, y = 4$: $4 - y_1 = \frac{-2}{x_1}(-1 - x_1) = \frac{2(1 + x_1)}{x_1} = \frac{2}{x_1} + 2$.

Step 5: Use $y_1 = \frac{x_1^2}{4}$ from Step 1: $4 - \frac{x_1^2}{4} = \frac{2}{x_1} + 2 \Rightarrow 2 - \frac{x_1^2}{4} = \frac{2}{x_1}$. Multiply through by $4x_1$: $8x_1 - x_1^3 = 8 \Rightarrow x_1^3 - 8x_1 + 8 = 0$. Testing $x_1 = 2$: $8 - 16 + 8 = 0$ – a valid root.

Step 6: With $x_1 = 2$: $y_1 = \frac{4}{4} = 1$. Normal slope: $-\frac{2}{2} = -1$. Normal equation: $y - 1 = -1(x - 2) \Rightarrow x + y - 3 = 0$.

Final Answer

$$x + y - 3 = 0$$

Common Mistake: Confusing the EXTERNAL given point (which the normal line must merely pass through) with the point of TANGENCY (which is a separate, unknown point that must be solved for algebraically) – these are two different points unless the problem states otherwise.

Quick Recall: When a normal must pass through an external point not stated to be on the curve, set up the point of tangency as an unknown (x_1, y_1) satisfying the curve equation, write the normal in terms of x_1 , then substitute the EXTERNAL point's coordinates to solve for x_1 .

Type 29

Finding the ANGLE OF INTERSECTION between two curves – find their point(s) of intersection first, then compute each curve's tangent slope there, and apply the angle-between-two-lines formula.

★ **Board Favorite** – Angle of intersection between curves is a substantial, frequently-tested long-answer topic, combining simultaneous-equation solving with the tangent-slope and angle-formula skills.

Formula Used: $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

Source: Exercise 6.2, Q30(i)

Given: Find the angle of intersection between $xy = 6$ and $x^2y = 12$.

Solution:

Step 1: Find the point(s) of intersection: from $xy = 6$, $y = \frac{6}{x}$. Substitute into $x^2y = 12$:

$$x^2 \cdot \frac{6}{x} = 12 \Rightarrow 6x = 12 \Rightarrow x = 2, \text{ giving } y = 3. \text{ Intersection point: } (2, 3).$$

Step 2: Find the slope of $xy = 6$ at $(2, 3)$: differentiate implicitly, $y + x \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-y}{x}$.

$$\text{At } (2, 3): m_1 = \frac{-3}{2}.$$

Step 3: Find the slope of $x^2y = 12$ at $(2, 3)$: differentiate implicitly, $2xy + x^2 \frac{dy}{dx} = 0 \Rightarrow$

$$\frac{dy}{dx} = \frac{-2y}{x}. \text{ At } (2, 3): m_2 = \frac{-2(3)}{2} = -3.$$

Step 4: Apply the angle formula: $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{-3/2 - (-3)}{1 + (-3/2)(-3)} \right| = \left| \frac{3/2}{1 + 9/2} \right| =$

$$\left| \frac{3/2}{11/2} \right| = \frac{3}{11}.$$

Final Answer

$$\theta = \tan^{-1} \left(\frac{3}{11} \right)$$

Common Mistake: Forgetting to find the actual point of intersection first and trying to compute slopes "in general" – the tangent slope of each curve must be evaluated AT the specific intersection point, not as a general formula.

Alternate Board-Style Example (circles)

Source: Exercise 6.2, Q31

Given: Find the angle of intersection of $x^2 + y^2 = 4$ and $(x - 2)^2 + y^2 = 4$ at the point in the first quadrant.

Solution:

Step 1: Solve simultaneously: expand the second circle, $x^2 - 4x + 4 + y^2 = 4 \Rightarrow x^2 + y^2 = 4x$. Since $x^2 + y^2 = 4$ from the first circle, substitute: $4 = 4x \Rightarrow x = 1$. Then from the first circle, $y^2 = 3 \Rightarrow y = \sqrt{3}$ (taking the positive root for the first quadrant). Point: $(1, \sqrt{3})$.

Step 2: Slope of first circle at $(1, \sqrt{3})$: $2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-x}{y}$, so $m_1 = \frac{-1}{\sqrt{3}}$.

Step 3: Slope of second circle at $(1, \sqrt{3})$: $2(x - 2) + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-(x - 2)}{y}$, so $m_2 = \frac{-(1 - 2)}{\sqrt{3}} = \frac{1}{\sqrt{3}}$.

Step 4: Apply the formula: $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{-1/\sqrt{3} - 1/\sqrt{3}}{1 + (-1/\sqrt{3})(1/\sqrt{3})} \right| = \left| \frac{-2/\sqrt{3}}{1 - 1/3} \right| = \left| \frac{-2/\sqrt{3}}{2/3} \right| = \frac{2}{\sqrt{3}} \times \frac{3}{2} = \sqrt{3}$.

Final Answer

$$\theta = \tan^{-1}(\sqrt{3}) = 60^\circ$$

Quick Recall: Always find the point of intersection FIRST by solving the two curve equations simultaneously, then differentiate EACH curve separately (implicitly if needed) and evaluate both slopes at that same point before applying the angle formula.

Type 30

Proving that two curves intersect ORTHOGONALLY (at right angles) – either at a SPECIFIC given point (verify $m_1 m_2 = -1$ numerically), or derive a general CONDITION relating unknown constants for orthogonality to hold.

Formula Used: Orthogonal intersection $\iff m_1 \times m_2 = -1$

Source: Exercise 6.2, Q32

Given: Prove that the curves $y = x^2 - 3x + 1$ and $x(y + 3) = 4$ intersect at right angles at $(2, -1)$.

Solution:

Step 1: First confirm $(2, -1)$ actually lies on both curves: for the first, $y = 4 - 6 + 1 = -1$ – matches. For the second, $x(y + 3) = 2(-1 + 3) = 2(2) = 4$ – matches.

Step 2: Slope of the first curve: $\frac{dy}{dx} = 2x - 3$. At $x = 2$: $m_1 = 4 - 3 = 1$.

Step 3: Slope of the second curve: rewrite $x(y+3) = 4$ as $xy+3x = 4$, differentiate implicitly:
 $y + x \frac{dy}{dx} + 3 = 0 \Rightarrow \frac{dy}{dx} = \frac{-(y+3)}{x}$. At $(2, -1)$: $m_2 = \frac{-(-1+3)}{2} = \frac{-2}{2} = -1$.

Step 4: Check the orthogonality condition: $m_1 \times m_2 = (1)(-1) = -1$ – confirmed.

Final Answer

Since $m_1 m_2 = -1$, the curves intersect at right angles at $(2, -1)$

Common Mistake: Forgetting the preliminary check that the given point actually lies on BOTH curves – without this, the "point of intersection" assumption behind the whole question would be invalid.

Alternate Board-Style Example (general condition, Competency Focused)

Source: Exercise 6.2, Q33 (Competency Focused Question)

Given: Find the condition that the curves $2x = y^2$ and $2xy = k$ intersect orthogonally.

Solution:

Step 1: Slope of $2x = y^2$: differentiate implicitly, $2 = 2y \frac{dy}{dx} \Rightarrow \frac{dy}{dx} = \frac{1}{y}$, so $m_1 = \frac{1}{y}$ (at any intersection point (x, y)).

Step 2: Slope of $2xy = k$: differentiate implicitly, $2y + 2x \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-y}{x}$, so $m_2 = \frac{-y}{x}$.

Step 3: Apply the orthogonality condition $m_1 m_2 = -1$: $\frac{1}{y} \cdot \frac{-y}{x} = -1 \Rightarrow \frac{-1}{x} = -1 \Rightarrow x = 1$
 – this tells us the x -coordinate of intersection must be 1 for orthogonality.

Step 4: Substitute $x = 1$ into the first curve to find y : $2(1) = y^2 \Rightarrow y^2 = 2$.

Step 5: Substitute $x = 1, y^2 = 2$ into the second curve's equation $2xy = k$, i.e. $4x^2y^2 = k^2$ (squaring to eliminate the need for y directly): $4(1)(2) = k^2 \Rightarrow k^2 = 8$.

Final Answer

The condition for orthogonal intersection is $k^2 = 8$

Quick Recall: To prove orthogonality at a SPECIFIC point, just verify $m_1 m_2 = -1$ numerically; to find a general CONDITION on unknown constants, set up $m_1 m_2 = -1$ symbolically, use it together with both curve equations to eliminate x and y , and solve for the relationship the constant must satisfy.

Type 31

Proving that two curves TOUCH each other – meaning they meet at exactly one common point where their tangent slopes are also EQUAL (not just intersecting at some angle, but sharing the same tangent line there).

Source: Exercise 6.2, Q34 (Competency Focused Question)

Given: Prove that the curves $xy = 4$ and $x^2 + y^2 = 8$ touch each other.

Solution:

Step 1: First find the point(s) of intersection: from $xy = 4$, $y = \frac{4}{x}$. Substitute into $x^2 + y^2 = 8$: $x^2 + \frac{16}{x^2} = 8$. Multiply through by x^2 : $x^4 - 8x^2 + 16 = 0$, which factors as $(x^2 - 4)^2 = 0$ – a REPEATED root, giving $x^2 = 4 \Rightarrow x = \pm 2$.

Step 2: The fact that $(x^2 - 4)^2 = 0$ is a perfect square (repeated root) already signals tangency algebraically – but let's also confirm via slopes. At $x = 2$: $y = 2$; at $x = -2$: $y = -2$.

Step 3: Slope of $xy = 4$ at $(2, 2)$: $y + x \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-y}{x} = \frac{-2}{2} = -1$.

Step 4: Slope of $x^2 + y^2 = 8$ at $(2, 2)$: $2x + 2y \frac{dy}{dx} = 0 \Rightarrow \frac{dy}{dx} = \frac{-x}{y} = \frac{-2}{2} = -1$.

Step 5: Both curves pass through $(2, 2)$ with the SAME slope -1 – so they share a common tangent line there, meaning they touch (rather than cross) at this point. The same check applies symmetrically at $(-2, -2)$.

Final Answer

The curves touch each other at $(2, 2)$ and $(-2, -2)$, since both have slope 1 at each shared point —

Common Mistake: Stopping after finding the intersection point(s) and assuming that's enough to prove "touching" – touching specifically requires showing the tangent SLOPES are also equal at that point, not merely that the curves cross there.

Quick Recall: "Touch" means intersect AND share the same tangent slope at that point (angle of intersection = 0°) – always compute and compare both curves' slopes at the intersection point(s), not just verify the point lies on both curves.

III. Approximations Using Differentials

Type 32

Finding the approximate value of an n^{th} root (square root, cube root, fourth root, etc.) of a number close to a perfect power, using the linear-approximation (differential) formula rather than direct computation.

★ **Board Favorite** – Approximating roots via differentials is the signature question type of this entire topic, appearing in nearly every board paper, tagged directly to NCERT.

Formula Used: $f(x + \Delta x) \approx f(x) + f'(x) \cdot \Delta x$

Source: Exercise 6.3, Q1(ii) (NCERT)

Given: Find the approximate value of $\sqrt{36.6}$.

Solution:

Step 1: Choose $f(x) = \sqrt{x}$, and pick x to be the NEAREST perfect square to 36.6: $x = 36$, so $\sqrt{36} = 6$ exactly – this is the anchor point where computation is easy.

Step 2: Set $\Delta x = 36.6 - 36 = 0.6$ – the small deviation from the easy anchor point.

Step 3: Differentiate: $f'(x) = \frac{1}{2\sqrt{x}}$. At $x = 36$: $f'(36) = \frac{1}{2(6)} = \frac{1}{12}$.

Step 4: Apply the formula: $\sqrt{36.6} \approx f(36) + f'(36) \times 0.6 = 6 + \frac{1}{12}(0.6) = 6 + 0.05 = 6.05$.

Final Answer

$$\sqrt{36.6} \approx 6.05$$

Common Mistake: Picking an anchor point x that ISN'T a perfect square/cube/etc. (making $f(x)$ itself hard to compute) – always choose the nearest NICE number to the target, so that $f(x)$ and $f'(x)$ are both easy to evaluate exactly.

Alternate Board-Style Example (fourth root)

Source: Exercise 6.3, Q4(i) (NCERT)

Given: Find the approximate value of $(82)^{1/4}$.

Solution:

Step 1: Choose $f(x) = x^{1/4}$, with anchor $x = 81$ (since $81 = 3^4$, giving $f(81) = 3$ exactly).

Step 2: $\Delta x = 82 - 81 = 1$.

Step 3: Differentiate: $f'(x) = \frac{1}{4}x^{-3/4}$. At $x = 81$: $f'(81) = \frac{1}{4}(81)^{-3/4} = \frac{1}{4 \times 27} = \frac{1}{108}$
(since $81^{3/4} = (81^{1/4})^3 = 3^3 = 27$).

Step 4: Apply the formula: $(82)^{1/4} \approx 3 + \frac{1}{108}(1) = 3 + \frac{1}{108} \approx 3.009$.

Final Answer

$$(82)^{1/4} \approx 3.009$$

Quick Recall: Always pick the anchor x to be the nearest number whose root is a whole number, set Δx as the (usually small) leftover difference, then apply $f(x+\Delta x) \approx f(x) + f'(x)\Delta x$ – this single formula handles square roots, cube roots, and any other fractional power alike.

Type 33

Finding the approximate value of a general polynomial or power function $f(x)$ at a point close to a convenient integer – the same linear-approximation formula, just with a more general f than a plain root.

Source: Exercise 6.3, Q7(i) (NCERT)

Given: If $f(x) = x^4 - 10$, find the approximate value of $f(2.1)$.

Solution:

Step 1: Choose the anchor $x = 2$ (the nearest convenient integer to 2.1), so $\Delta x = 0.1$.

Step 2: Compute $f(2) = 2^4 - 10 = 16 - 10 = 6$.

Step 3: Differentiate: $f'(x) = 4x^3$. At $x = 2$: $f'(2) = 4(8) = 32$.

Step 4: Apply the formula: $f(2.1) \approx f(2) + f'(2) \times 0.1 = 6 + 32(0.1) = 6 + 3.2 = 9.2$.

Final Answer

$$f(2.1) \approx 9.2$$

Common Mistake: Forgetting that this technique works for ANY differentiable function, not just roots – polynomials, and even more complex expressions, all use exactly the same $f(x + \Delta x) \approx f(x) + f'(x)\Delta x$ formula.

Alternate Board-Style Example

Source: Exercise 6.3, Q6 (NCERT Exemplar)

Given: Find the approximate value of $(1.999)^5$.

Solution:

Step 1: Choose $f(x) = x^5$, anchor $x = 2$ (so $\Delta x = 1.999 - 2 = -0.001$, a small NEGATIVE deviation this time).

Step 2: $f(2) = 2^5 = 32$.

Step 3: Differentiate: $f'(x) = 5x^4$. At $x = 2$: $f'(2) = 5(16) = 80$.

Step 4: Apply the formula: $(1.999)^5 \approx 32 + 80(-0.001) = 32 - 0.08 = 31.92$.

Final Answer

$$(1.999)^5 \approx 31.92$$

Quick Recall: The anchor point doesn't have to be smaller than the target – when the target is slightly LESS than a nice number, Δx is simply negative, and the formula works exactly the same way.

Type 34

Finding the approximate CHANGE (increase or decrease) in the area, volume, or surface area of a geometric figure resulting from a small given change in one of its dimensions – apply the differential formula to the geometric formula itself.

★ **Board Favorite** – Approximate-change-in-geometric-quantity problems are a recurring practical application of differentials, testing the same core formula in a word-problem dress.

Formula Used: $\Delta V \approx \frac{dV}{dr} \times \Delta r$

Source: Exercise 6.3, Q8

Given: The radius of a circle increases from 5 cm to 5.1 cm. Find the approximate increase in area.

Solution:

Step 1: Area formula: $A = \pi r^2$. Here $r = 5$ and $\Delta r = 5.1 - 5 = 0.1$.

Step 2: Differentiate: $\frac{dA}{dr} = 2\pi r$. At $r = 5$: $\frac{dA}{dr} = 10\pi$.

Step 3: Apply the approximation: $\Delta A \approx \frac{dA}{dr} \times \Delta r = 10\pi \times 0.1 = \pi$.

Final Answer

Approximate increase in area $\approx \pi \text{ cm}^2$

Common Mistake: Computing the EXACT change (i.e. $\pi(5.1)^2 - \pi(5)^2$) instead of the linear APPROXIMATION $\frac{dA}{dr} \times \Delta r$ – while the two are close for small Δr , the question specifically wants the differential-based approximate method, not the exact computation.

Alternate Board-Style Example (two related quantities at once)

Source: Exercise 6.3, Q9

Given: The radius of a sphere shrinks from 10 cm to 9.8 cm. Find the approximate decrease in (i) v

Solution:

Case: (i) Volume

$V = \frac{4}{3}\pi r^3 \Rightarrow \frac{dV}{dr} = 4\pi r^2$. At $r = 10$: $\frac{dV}{dr} = 400\pi$. With $\Delta r = 9.8 - 10 = -0.2$:
 $\Delta V \approx 400\pi \times (-0.2) = -80\pi$ – the negative sign confirms a DECREASE of $80\pi \text{ cm}^3$.

Case: (ii) Surface Area

$S = 4\pi r^2 \Rightarrow \frac{dS}{dr} = 8\pi r$. At $r = 10$: $\frac{dS}{dr} = 80\pi$. $\Delta S \approx 80\pi \times (-0.2) = -16\pi$ – a decrease of $16\pi \text{ cm}^2$.

Final Answer

Volume decreases by approximately $80\pi \text{ cm}^3$; surface area decreases by approximately $16\pi \text{ cm}^2$

Quick Recall: Identify the correct geometric formula, differentiate it with respect to the changing dimension, then multiply by $\Delta(\text{dimension})$ – using a NEGATIVE Δr automatically produces a negative (decreasing) result without any extra reasoning needed.

Type 35

Finding the approximate SYMBOLIC (percentage-based) change in the volume or surface area of a solid when a linear dimension changes by a given PERCENTAGE – the answer is expressed in terms of the original variable, not a specific number.

Source: Exercise 6.3, Q10(i)

Given: The side of a cube of side x increases by 1%. Find the approximate change in its volume.

Solution:

Step 1: "Increases by 1%" means $\Delta x = \frac{1}{100}x = 0.01x$ – converting the percentage into a symbolic differential in terms of x itself.

Step 2: Volume formula: $V = x^3$. Differentiate: $\frac{dV}{dx} = 3x^2$.

Step 3: Apply the approximation: $\Delta V \approx \frac{dV}{dx} \times \Delta x = 3x^2 \times 0.01x = 0.03x^3$.

Final Answer

$\Delta V \approx 0.03x^3$ (a 3% increase in volume)

Common Mistake: Trying to substitute a specific numeric value for x when none was given – since the side length is only described symbolically (as " x "), the final answer must ALSO be left in terms of x .

Alternate Board-Style Example

Source: Exercise 6.3, Q10(ii)

Given: The side of a cube of side x decreases by 1%. Find the approximate change in its surface area.

Solution:

Step 1: "Decreases by 1%" means $\Delta x = -0.01x$ (negative, since it's a decrease).

Step 2: Surface area: $S = 6x^2$. Differentiate: $\frac{dS}{dx} = 12x$.

Step 3: Apply: $\Delta S \approx 12x \times (-0.01x) = -0.12x^2$.

Final Answer

$\Delta S \approx -0.12x^2$ (a 1.2% decrease in surface area)

Quick Recall: A stated percentage change in a dimension converts directly to $\Delta x = (\text{percentage as a decimal}) \times x$ – keep this symbolic and carry it through the usual differential formula; the final answer stays in terms of the original variable.

Type 36

Error propagation: given an ABSOLUTE or PERCENTAGE error in a measured quantity (like a radius or edge), find the resulting absolute or percentage error in a DERIVED quantity (like volume or surface area) – for percentage errors, take logarithms first to make the relationship additive.

★ **Board Favorite** – Error-propagation questions are a distinctive, frequently-tested application of differentials in measurement contexts, tagged directly to NCERT.

Formula Used: % error in $y \approx \left(\frac{f'(x)}{f(x)} \right) \times (\% \text{ error in } x) \times x/x = n \times (\% \text{ error in } x) \text{ when } y = kx^n$

Source: Exercise 6.3, Q11(i) (NCERT)

Given: The radius of a sphere is measured as 9 m with a possible error of 0.03 m. Find the approximate error in volume.

Solution:

Step 1: This is an ABSOLUTE error question: $r = 9$, $\Delta r = 0.03$ (the possible measurement error).

Step 2: Volume: $V = \frac{4}{3}\pi r^3$. Differentiate: $\frac{dV}{dr} = 4\pi r^2$. At $r = 9$: $\frac{dV}{dr} = 4\pi(81) = 324\pi$.

Step 3: Apply the approximation: $\Delta V \approx \frac{dV}{dr} \times \Delta r = 324\pi \times 0.03 = 9.72\pi$.

Final Answer

Approximate error in volume $\approx 9.72\pi \text{ m}^3 \approx 30.5 \text{ m}^3$

Common Mistake: Confusing an ABSOLUTE error (a fixed numeric amount, like 0.03 m) with a PERCENTAGE error (a proportion, like 1%) – the two require different final steps (multiply by Δr directly, versus multiplying by a percentage of the value).

Alternate Board-Style Example (percentage error, via logarithms)

Source: Exercise 6.3, Q11(ii)

Given: The edge of a cube has a possible error of 1% in its measurement. Find the approximate percentage error in surface area.

Solution:

Step 1: Surface area: $S = 6x^2$. Take logarithms of both sides to turn the power into a multiplier: $\log S = \log 6 + 2 \log x$.

Step 2: Differentiate (treating this as a relationship between the differentials): $\frac{dS}{S} = 2\frac{dx}{x}$ – this directly relates the PROPORTIONAL (percentage) change in S to the proportional change in x , without needing the actual value of x at all.

Step 3: Multiply both sides by 100 to convert to percentages: $\left(\frac{dS}{S} \times 100\right) = 2\left(\frac{dx}{x} \times 100\right)$.

Step 4: Substitute the given percentage error in x (which is 1%): percentage error in S $= 2 \times 1\% = 2\%$.

Final Answer

Approximate percentage error in surface area $\approx 2\%$

Common Mistake: Trying to work with absolute values of x and Δx when only a PERCENTAGE error is given (and no actual measurement) – taking logs first converts the whole relationship into pure percentages, sidestepping the need for any specific numeric value.

Quick Recall: For an ABSOLUTE error, directly apply $\Delta y \approx f'(x)\Delta x$ using the actual numbers given. For a PERCENTAGE error (especially with a power-law formula like $y = kx^n$), take logs first: $\frac{dy}{y} = n\frac{dx}{x}$ shows the percentage error in y is always n times the percentage error in x .

IV. Increasing and Decreasing Functions

Type 37

Proving a function is strictly increasing (or decreasing) on ALL of $\mathbf{R}$ by showing its derivative has a fixed, constant sign everywhere – the simplest and most direct application of the derivative-sign test.

★ **Board Favorite** – This is the foundational skill of the whole topic, tagged directly to NCERT – nearly every other type in this section builds on being able to run this basic sign-test correctly.

Formula Used: f strictly increasing on $I \iff f'(x) > 0 \forall x \in I$;
 f strictly decreasing $\iff f'(x) < 0 \forall x \in I$

Source: Exercise 6.4, Q1(ii) (NCERT)

Given: Show that $f(x) = 7x - 3$ is strictly increasing on $\mathbf{R}$.

Solution:

Step 1: Differentiate: $f'(x) = 7$ – a CONSTANT, the same for every x .

Step 2: Since $7 > 0$ for every real x (there's no x -dependence to check), $f'(x) > 0$ everywhere.

Step 3: By the derivative-sign test, f is strictly increasing on all of $\mathbf{R}$.

Final Answer

$f'(x) = 7 > 0 \forall x \in \mathbf{R}$, so f is strictly increasing on $\mathbf{R}$

Common Mistake: Testing the function at only a few specific numeric values of x instead of arguing that $f'(x)$'s sign holds for EVERY real x – for a linear function the derivative is a fixed constant, so one line of reasoning covers the entire real line at once.

Alternate Board-Style Example (exponential functions)

Source: Exercise 6.4, Q4(i)(ii)

Given: Show that (i) $f(x) = e^{-2x}$ is strictly decreasing on $\mathbf{R}$ (ii) $f(x) = 5^x$ is strictly increasing.

Solution:

Case: (i) $f(x) = e^{-2x}$

$f'(x) = -2e^{-2x}$. Since $e^{-2x} > 0$ for every real x (an exponential is always positive), $f'(x) = -2 \times (\text{positive}) < 0$ for every x – strictly decreasing on $\mathbf{R}$.

Case: (ii) $f(x) = 5^x$

$f'(x) = 5^x \log 5$. Since $5^x > 0$ always, and $\log 5 > 0$ (as $5 > 1$), the product $f'(x) > 0$ for every x – strictly increasing on $\mathbf{R}$.

Final Answer

(i) $f'(x) < 0 \forall x \Rightarrow$ strictly decreasing (ii) $f'(x) > 0 \forall x \Rightarrow$ strictly increasing

Quick Recall: For simple functions (linear, exponential), the derivative's sign is often OBVIOUS without any further work – a constant, or a product of always-positive terms – so the "proof" is just stating that obvious sign and citing the test.

Type 38

Proving increasing/decreasing where the derivative's sign is NOT immediately obvious – it involves a fraction, reciprocal, or logarithm term that needs algebraic simplification (like combining over a common denominator, or recognising a perfect square) before the sign becomes clear.

Source: Exercise 6.4, Q3(ii)

Given: Show that $f(x) = x - \frac{1}{x}$ is strictly increasing (on its domain).

Solution:

Step 1: Differentiate: $f'(x) = 1 + \frac{1}{x^2}$ – not immediately obviously positive at a glance, since it's a sum of two terms.

Step 2: Analyse each term: 1 is always positive, and $\frac{1}{x^2}$ is ALSO always positive for every $x \neq 0$ (a square in the denominator is always positive).

Step 3: So $f'(x)$ is a SUM of two positive quantities, hence $f'(x) > 0$ for every x in the domain ($x \neq 0$).

Final Answer

$f'(x) = 1 + \frac{1}{x^2} > 0 \forall x \neq 0$, so f is strictly increasing

Common Mistake: Giving up on determining the sign because the expression "looks complicated" – breaking a sum into individual terms and checking each one's sign separately is often all that's needed.

Alternate Board-Style Example (logarithm)

Source: Exercise 6.4, Q4(iii) (NCERT)

Given: Show that $f(x) = \log x$ is strictly increasing.

Solution:

Step 1: Differentiate: $f'(x) = \frac{1}{x}$.

Step 2: The domain of $\log x$ is $x > 0$ – so within this domain, x is always positive, meaning $\frac{1}{x}$ is also always positive.

Step 3: Hence $f'(x) > 0$ throughout the domain, so f is strictly increasing.

Final Answer

$f'(x) = \frac{1}{x} > 0$ for $x > 0$, so f is strictly increasing on its domain

Quick Recall: When the derivative is a sum, examine each term's sign separately (squares and exponentials are always positive; check what the domain restricts other terms to); when it's $\frac{1}{x}$ -type, remember the function's OWN domain often already guarantees the needed sign.

Type 39

Finding the range of values of an unknown PARAMETER for which a given function is increasing (or decreasing) everywhere – set up the inequality $f'(x) \geq 0$ (or similar) as a condition ON the parameter, valid for all x .

Source: Exercise 6.4, Q5 (ISC Specimen Paper 2026)

Given: Find the values of a for which $f(x) = b - ax + \sin x$ is increasing on $\mathbf{R}$.

Solution:

Step 1: Differentiate: $f'(x) = -a + \cos x$.

Step 2: For f to be increasing on ALL of $\mathbf{R}$, we need $f'(x) \geq 0$ for EVERY x , i.e. $\cos x \geq a$ for every x .

Step 3: Since $\cos x$ ranges between -1 and 1 , the WORST case (smallest value $\cos x$ can take) is -1 – for the inequality $\cos x \geq a$ to hold even in this worst case, we need $a \leq -1$.

Step 4: Conversely, if $a \leq -1$, then $\cos x \geq -1 \geq a$ automatically holds for every x , so the condition is both necessary and sufficient.

Final Answer

$$a \leq -1$$

Common Mistake: Only checking the inequality at one convenient value of x (like $x = 0$) instead of considering the WORST-CASE (minimum) value that the trig term can take across all x – the condition must hold for every single x , so the tightest (most restrictive) case determines the answer.

Alternate Board-Style Example (True/False existence question)

Source: Exercise 6.4, Q17

Given: State True or False, with justification: "There exists at least one value of $b \in \mathbf{R}$ for which $f(x) = \frac{b}{x}$, $b \neq 0$ is strictly increasing in $\mathbf{R} - \{0\}$."

Solution:

Step 1: Differentiate: $f'(x) = \frac{-b}{x^2}$.

Step 2: For f to be strictly increasing everywhere on its domain, we'd need $f'(x) > 0$ for every $x \neq 0$, i.e. $\frac{-b}{x^2} > 0$.

Step 3: Since $x^2 > 0$ always (for $x \neq 0$), the SIGN of $\frac{-b}{x^2}$ is entirely determined by the sign of $-b$ – so we need $-b > 0$, i.e. $b < 0$.

Step 4: Since $b < 0$ is a perfectly valid, non-empty range of real numbers (e.g. $b = -1$ works), such a value of b DOES exist.

Final Answer

True – any $b < 0$ (e.g. $b = -1$) makes $f(x) = \frac{b}{x}$ strictly increasing on $\mathbf{R} - \{0\}$

Common Mistake: Assuming the statement must be False just because $f(x) = b/x$ "looks like" it should always be decreasing – checking the actual sign of the derivative in terms of b reveals that the sign flips depending on whether b is positive or negative.

Quick Recall: "Find the range of a parameter" questions convert into an inequality that must hold for ALL x – use the trig/algebraic function's own known bounds (like $-1 \leq \cos x \leq 1$) to find the parameter's worst-case restriction; True/False existence questions use the exact same reasoning, just phrased differently.

Type 40

Finding the INTERVAL(S) of increase and decrease of a polynomial (or similar) function by solving $f'(x) = 0$ for critical points, then testing the sign of $f'(x)$ in each resulting sub-interval – the master, most heavily-tested type in this entire section.

★ **Board Favorite** – This is by far the single most repeated question format in the increasing/decreasing topic – the book devotes an entire ten-part mega-question (Q24) to this exact technique, and tags multiple instances to NCERT.

Formula Used: Solve $f'(x) = 0$ for critical points, then test the sign of $f'(x)$ in each sub-interval to

Source: Exercise 6.4, Q24(ii) (NCERT)

Given: $f(x) = 4x^3 - 6x^2 - 72x + 30$. Find the intervals of increase/decrease.

Solution:

Step 1: Differentiate: $f'(x) = 12x^2 - 12x - 72 = 12(x^2 - x - 6) = 12(x - 3)(x + 2)$.

Step 2: Set $f'(x) = 0$ to find critical points: $x = 3$ or $x = -2$ – these split the real line into three intervals: $(-\infty, -2)$, $(-2, 3)$, $(3, \infty)$.

Step 3: Test the sign of $f'(x) = 12(x - 3)(x + 2)$ in each interval using a convenient test point: at $x = -3$ (in $(-\infty, -2)$): $(x - 3)(x + 2) = (-6)(-1) = 6 > 0$ – positive. At $x = 0$ (in $(-2, 3)$): $(-3)(2) = -6 < 0$ – negative. At $x = 4$ (in $(3, \infty)$): $(1)(6) = 6 > 0$ – positive.

Step 4: Interpret: $f'(x) > 0$ on $(-\infty, -2)$ and $(3, \infty)$ (increasing there); $f'(x) < 0$ on $(-2, 3)$ (decreasing there).

Final Answer

Increasing on $(-\infty, -2) \cup (3, \infty)$; decreasing on $(-2, 3)$

Common Mistake: Forgetting to actually TEST the sign in each sub-interval and instead guessing based on the shape of the factored expression – always plug in one concrete test point from each interval to be certain of the sign.

Alternate Board-Style Example

Source: Exercise 6.4, Q6(iii)

Given: Find the interval in which $f(x) = 2x^3 - 3x$ is strictly increasing.

Solution:

Step 1: Differentiate: $f'(x) = 6x^2 - 3 = 3(2x^2 - 1)$.

Step 2: Set $f'(x) = 0$: $2x^2 = 1 \Rightarrow x^2 = \frac{1}{2} \Rightarrow x = \pm \frac{1}{\sqrt{2}}$ – two critical points splitting the line into three intervals.

Step 3: Test signs: for $|x| > \frac{1}{\sqrt{2}}$ (outer intervals), $x^2 > \frac{1}{2}$, so $2x^2 - 1 > 0$ – positive (increasing). For $|x| < \frac{1}{\sqrt{2}}$ (middle interval), $2x^2 - 1 < 0$ – negative (decreasing).

Final Answer

Strictly increasing on $\left(-\infty, \frac{-1}{\sqrt{2}}\right) \cup \left(\frac{1}{\sqrt{2}}, \infty\right)$

Quick Recall: Factor $f'(x)$ fully, find every root (critical point), mark them on a number line to create sub-intervals, then test ONE point from each sub-interval to determine the sign there – this exact method handles every polynomial in this topic (see Ex 6.4 Q6, Q10, Q11, Q12, Q24's full ten parts, Q26, Q27 for extensive further practice with the identical technique).

Type 41

Proving a function is increasing/decreasing using an ALGEBRAIC INEQUALITY-style argument rather than simply reading off the sign of a factored derivative – often requires manipulating the derivative expression to reveal it as manifestly positive or negative.

Source: Exercise 6.4, Q7

Given: Show that $f(x) = (x - 1)e^x + 1$ is an increasing function for all $x > 0$.

Solution:

Step 1: Differentiate using the product rule: $f'(x) = 1 \cdot e^x + (x - 1)e^x = e^x[1 + (x - 1)] = e^x \cdot x = xe^x$ – a significant simplification after factoring out e^x .

Step 2: Analyse the sign for $x > 0$: $e^x > 0$ always (for any real x), and $x > 0$ is given directly.

Step 3: So $f'(x) = xe^x$ is a PRODUCT of two positive quantities for $x > 0$, hence $f'(x) > 0$.

Final Answer

$f'(x) = xe^x > 0$ for $x > 0$, so f is increasing for all $x > 0$

Common Mistake: Not simplifying the product-rule result far enough – leaving the answer as $e^x + (x - 1)e^x$ obscures the sign; factoring out the common e^x reveals the much simpler (and sign-obvious) form xe^x .

Quick Recall: After differentiating, always try to SIMPLIFY or FACTOR the result as far as possible before attempting to judge its sign – a seemingly complicated derivative often collapses into an obviously-signed product.

Type 42

Finding the value(s) of an unknown coefficient k for which a given function is (strictly) decreasing everywhere – similar to Type 39, but the parameter appears as a coefficient rather than inside a trig term.

Source: Exercise 6.4, Q8

Given: For what value of k is the function $f(x) = kx^3 + 5$ decreasing?

Solution:

Step 1: Differentiate: $f'(x) = 3kx^2$.

Step 2: For f to be decreasing everywhere, we need $f'(x) \leq 0$ (or < 0 , for strict decrease) for every x .

Step 3: Since $x^2 \geq 0$ always (with equality only at the single point $x = 0$), the sign of $f'(x) = 3kx^2$ is controlled entirely by the sign of k : if $k < 0$, then $f'(x) \leq 0$ everywhere (with equality only at $x = 0$, which doesn't break strict monotonicity over an interval).

Step 4: So the function is decreasing precisely when $k < 0$.

Final Answer

$k < 0$

Common Mistake: Requiring $f'(x) < 0$ to hold with NO exceptions at all (including at $x = 0$ where $x^2 = 0$ makes $f'(x) = 0$ regardless of k) – a single isolated zero of the derivative does not prevent a function from being classified as strictly decreasing overall, as long as $f'(x)$ doesn't become POSITIVE anywhere.

Quick Recall: When a coefficient multiplies an always-non-negative term like x^2 or x^{2n} , the overall sign of the derivative is controlled entirely by the coefficient's own sign – set up that simple sign condition directly.

Type 43

General or definitional proofs about increasing/decreasing behaviour – either a symbolic "if and only if" statement about a whole family of functions, or a proof using the DEFINITION directly (useful when the function, like the greatest integer function, isn't differentiable everywhere).

Source: Exercise 6.4, Q9(i)

Given: Prove that $f(x) = ax + b$ is strictly decreasing if and only if $a < 0$.

Solution:

Step 1: This is an "if and only if" statement, requiring proof in BOTH directions.

Step 2: ($\Leftarrow$) **If $a < 0$, then f is strictly decreasing:** differentiate, $f'(x) = a$. Since $a < 0$ is given, $f'(x) < 0$ for every x , so by the derivative-sign test, f is strictly decreasing.

Step 3: ($\Rightarrow$) **If f is strictly decreasing, then $a < 0$:** if f is strictly decreasing, then for $x_1 < x_2$, $f(x_1) > f(x_2)$, i.e. $ax_1 + b > ax_2 + b \Rightarrow ax_1 > ax_2 \Rightarrow a(x_1 - x_2) > 0$. Since $x_1 < x_2$ means $x_1 - x_2 < 0$, dividing both sides of $a(x_1 - x_2) > 0$ by the negative quantity $(x_1 - x_2)$ flips the inequality: $a < 0$.

Final Answer

$f(x) = ax + b$ is strictly decreasing $\iff a < 0$ (proved both ways)

Common Mistake: Proving only ONE direction of an "if and only if" statement and treating the proof as complete – both the forward and reverse implications must be shown separately for a genuine iff proof.

Alternate Board-Style Example (direct definition, greatest integer function)

Source: Exercise 6.4, Q9(ii)

Given: Prove that $f(x) = [x] - x$ is strictly decreasing on $[0, 1)$.

Solution:

Step 1: On the interval $[0, 1)$, the greatest integer function $[x]$ is CONSTANT: $[x] = 0$ for every x in this range (since $0 \leq x < 1$).

Step 2: So on this interval, $f(x) = 0 - x = -x$.

Step 3: For $x_1 < x_2$ both in $[0, 1)$: $f(x_1) = -x_1$ and $f(x_2) = -x_2$. Since $x_1 < x_2$, multiplying by -1 flips the inequality: $-x_1 > -x_2$, i.e. $f(x_1) > f(x_2)$.

Step 4: This is exactly the definition of strictly decreasing: a smaller input gives a LARGER output.

Final Answer

On $[0, 1)$, $f(x) = [x] - x = -x$, which is strictly decreasing by direct comparison

Quick Recall: "If and only if" claims need both directions proved separately; functions involving $[x]$ often aren't differentiable at integers, so simplify the function to its EXPLICIT form on the given sub-interval first (where $[x]$ becomes a fixed constant), then compare $f(x_1)$ and $f(x_2)$ directly using the definition.

Type 44

Determining increasing/decreasing on a RESTRICTED sub-interval where the derivative is a trigonometric expression – often requires rewriting $f'(x)$ using a sum-to-product or auxiliary-angle identity to make its sign on that specific sub-interval clear.

★ **Board Favorite** – Trig-based monotonicity proofs on a restricted interval are a recurring higher-order pattern, testing the auxiliary-angle technique in a new context.

Source: Exercise 6.4, Q18

Given: Prove that $f(x) = \sin\left(2x + \frac{\pi}{4}\right)$ is decreasing for $\frac{3\pi}{8} \leq x \leq \frac{5\pi}{8}$.

Solution:

Step 1: Differentiate: $f'(x) = 2 \cos\left(2x + \frac{\pi}{4}\right)$.

Step 2: We need $f'(x) \leq 0$ on the given interval, i.e. $\cos\left(2x + \frac{\pi}{4}\right) \leq 0$.

Step 3: Find the range of the ARGUMENT $2x + \frac{\pi}{4}$ as x ranges over $\left[\frac{3\pi}{8}, \frac{5\pi}{8}\right]$: at $x = \frac{3\pi}{8}$, argument $= \frac{3\pi}{4} + \frac{\pi}{4} = \pi$. At $x = \frac{5\pi}{8}$, argument $= \frac{5\pi}{4} + \frac{\pi}{4} = \frac{3\pi}{2}$. So the argument ranges over $\left[\pi, \frac{3\pi}{2}\right]$.

Step 4: Recall that $\cos \theta \leq 0$ for $\theta \in \left[\frac{\pi}{2}, \frac{3\pi}{2}\right]$ – and our argument range $\left[\pi, \frac{3\pi}{2}\right]$ sits entirely inside this region, so $\cos\left(2x + \frac{\pi}{4}\right) \leq 0$ throughout.

Step 5: Hence $f'(x) = 2 \cos\left(2x + \frac{\pi}{4}\right) \leq 0$ on the given interval, so f is decreasing there.

Final Answer

$f'(x) \leq 0$ on $\left[\frac{3\pi}{8}, \frac{5\pi}{8}\right]$, so f is decreasing there

Common Mistake: Trying to determine the sign of $\cos(2x + \pi/4)$ by testing only the ENDPOINTS of the interval – the argument's full RANGE across the interval must be checked against where cosine is known to be non-positive, not just the boundary values.

Alternate Board-Style Example

Source: Exercise 6.4, Q13(ii)

Given: Show that $f(x) = \frac{16 \sin x}{4 + \cos x} - x$ is strictly decreasing in $\left(\frac{\pi}{2}, \pi\right)$.

Solution:

Step 1: Differentiate using the quotient rule on the first term: $f'(x) = \frac{16 \cos x(4 + \cos x) - 16 \sin x(-\sin x)}{(4 + \cos x)^2}$

$$1 = \frac{64 \cos x + 16 \cos^2 x + 16 \sin^2 x}{(4 + \cos x)^2} - 1.$$

Step 2: Simplify using $\cos^2 x + \sin^2 x = 1$: numerator becomes $64 \cos x + 16$.

Step 3: So $f'(x) = \frac{64 \cos x + 16}{(4 + \cos x)^2} - 1 = \frac{64 \cos x + 16 - (4 + \cos x)^2}{(4 + \cos x)^2}$. Expand $(4 + \cos x)^2 = 16 + 8 \cos x + \cos^2 x$: numerator becomes $64 \cos x + 16 - 16 - 8 \cos x - \cos^2 x = 56 \cos x - \cos^2 x = \cos x(56 - \cos x)$.

Step 4: On $\left(\frac{\pi}{2}, \pi\right)$, $\cos x < 0$ (second quadrant), while $56 - \cos x > 0$ always (since $\cos x \geq -1$, so $56 - \cos x \geq 55$). So the numerator $\cos x(56 - \cos x)$ is (negative) $\times$ (positive) = negative. The denominator $(4 + \cos x)^2$ is always positive (a square).

Step 5: So $f'(x) = \frac{\text{negative}}{\text{positive}} < 0$ throughout the interval.

Final Answer

$f'(x) < 0$ on $\left(\frac{\pi}{2}, \pi\right)$, so f is strictly decreasing there

Quick Recall: For trig-heavy derivatives, simplify using $\sin^2 + \cos^2 = 1$ wherever possible, factor the result, and check the sign of each factor separately over the SPECIFIC given sub-interval (recalling which quadrant it falls in and the sign of sin/cos there).

Type 45

Determining increasing/decreasing using a LOGARITHM-based function on a restricted interval, often requiring a known trigonometric inequality (rather than just sign-of-derivative alone) to complete the argument.

Source: Exercise 6.4, Q14(i) (NCERT)

Given: Prove that $f(x) = \log \cos x$ is strictly decreasing on $\left(0, \frac{\pi}{2}\right)$.

Solution:

Step 1: Differentiate: $f'(x) = \frac{-\sin x}{\cos x} = -\tan x$.

Step 2: On $\left(0, \frac{\pi}{2}\right)$ (the first quadrant), both $\sin x > 0$ and $\cos x > 0$, so $\tan x > 0$ throughout.

Step 3: Hence $f'(x) = -\tan x < 0$ on this entire interval.

Final Answer

$f'(x) = -\tan x < 0$ on $\left(0, \frac{\pi}{2}\right)$, so f is strictly decreasing there

Common Mistake: Forgetting the DOMAIN restriction $\left(0, \frac{\pi}{2}\right)$ matters for determining the sign of $\tan x$ – outside the first quadrant, $\tan x$ can be negative, which would flip the conclusion entirely.

Alternate Board-Style Example

Source: Exercise 6.4, Q14(ii)

Given: If $f(x) = a(\tan x - \cot x)$, $a > 0$, determine whether f is increasing or decreasing on its domain.

Solution:

Step 1: Differentiate: $f'(x) = a(\sec^2 x + \csc^2 x)$ (recalling $\frac{d}{dx}(-\cot x) = \csc^2 x$, so the minus sign flips to plus).

Step 2: Analyse the sign: $\sec^2 x \geq 1 > 0$ always (wherever defined), and $\csc^2 x \geq 1 > 0$ always (wherever defined) – both terms are always positive.

Step 3: Since $a > 0$ is given, and the bracket $(\sec^2 x + \csc^2 x)$ is a sum of two positive quantities, the whole product $f'(x) = a(\sec^2 x + \csc^2 x) > 0$ throughout the domain.

Final Answer

f is increasing throughout its domain, since $f'(x) > 0$ everywhere it is defined

Quick Recall: Trig-derivative sign questions on a restricted domain often reduce to recalling which quadrant(s) the interval falls in and whether $\sin/\cos/\tan$ are positive or negative there; $\sec^2$ and $\csc^2$ are ALWAYS positive wherever defined, which simplifies many such proofs immediately.

Type 46

Quick VSA-style recognition questions: determine whether a function is increasing or decreasing at STATED x -values or over small stated sub-domains, without deriving a full interval breakdown – often several short sub-parts in one question.

Source: Exercise 6.4, Q15

Given: Determine whether the following are increasing or decreasing for the stated values: (i) $f(x) = \frac{1}{x}, x < 0$ (ii) $f(x) = \frac{1}{1+x^2}, x \geq 0$

Solution:

Case: (i) $f(x) = \frac{1}{x}, x < 0$

$f'(x) = \frac{-1}{x^2}$. Since $x^2 > 0$ always (for $x \neq 0$), $f'(x) < 0$ throughout $x < 0$ – decreasing.

Case: (ii) $f(x) = \frac{1}{1+x^2}, x \geq 0$

$f'(x) = \frac{-2x}{(1+x^2)^2}$. For $x \geq 0$, the numerator $-2x \leq 0$ and the denominator is always positive – so $f'(x) \leq 0$ throughout $x \geq 0$ – decreasing.

Final Answer

(i) Decreasing (ii) Decreasing

Common Mistake: Trying to analyse the function's behaviour over its ENTIRE domain instead of restricting attention to just the STATED sub-domain given in the question – e.g. $f(x) = 1/x$ actually behaves differently for $x > 0$ versus $x < 0$, so the restriction matters.

Alternate Board-Style Example ("which of the following" format)

Source: Exercise 6.4, Q16 (NCERT)

Given: Which of the following functions are strictly decreasing on $\left(0, \frac{\pi}{2}\right)$? (i) $\cos x$ (ii) $\cos 2x$ (iii)

Solution:

Case: (i) $\cos x$

$f'(x) = -\sin x$. On $\left(0, \frac{\pi}{2}\right)$, $\sin x > 0$, so $f'(x) < 0$ – decreasing. ✓

Case: (ii) $\cos 2x$

$f'(x) = -2 \sin 2x$. As x ranges over $(0, \frac{\pi}{2})$, $2x$ ranges over $(0, \pi)$, where $\sin 2x > 0$ throughout – so $f'(x) < 0$ – decreasing. ✓

Case: (iii) $\sin^2 x$

$f'(x) = 2 \sin x \cos x = \sin 2x$. On $(0, \frac{\pi}{2})$, this is POSITIVE – increasing, not decreasing. ✗

Case: (iv) $\tan x$

$f'(x) = \sec^2 x > 0$ always – increasing, not decreasing. ✗

Final Answer

(i) and (ii) are strictly decreasing on $(0, \frac{\pi}{2})$

Quick Recall: For quick recognition questions, just differentiate and check the sign directly over the SPECIFIC stated sub-domain (not the whole real line) – these are meant to be fast, one-line sign checks rather than full interval-breakdown problems.

Type 47

Proving a periodic (trig) function is increasing on ONE half of a stated domain and decreasing on the OTHER half – a two-part conclusion covering the function's complete behaviour across the whole given interval.

★ **Board Favorite** – This exact $\sin x / \cos x$ two-part behaviour proof is a classic NCERT-tagged pair, testing sign analysis across two adjoining sub-intervals in a single question.

Source: Exercise 6.4, Q19 (NCERT)

Given: Prove that $f(x) = \sin x$ is (i) strictly increasing in $(0, \frac{\pi}{2})$ (ii) strictly decreasing in $(\frac{\pi}{2}, \pi)$

Solution:

Case: (i) On $(0, \frac{\pi}{2})$

$f'(x) = \cos x$. In the first quadrant, $\cos x > 0$ – so f is strictly increasing here.

Case: (ii) On $(\frac{\pi}{2}, \pi)$

In the second quadrant, $\cos x < 0$ – so f is strictly decreasing here.

Case: (iii) On the FULL interval $(0, \pi)$

Since f is increasing on the first half and decreasing on the second half of $(0, \pi)$, it is NEITHER purely increasing NOR purely decreasing over the whole interval – the behaviour is genuinely different in the two halves, so no single classification applies to the entire interval.

Final Answer

(i) Increasing (ii) Decreasing (iii) Neither, over the combined interval $(0, \pi)$

Common Mistake: Trying to average or somehow combine the two sub-interval behaviours into one overall verdict for part (iii) – “neither increasing nor decreasing” is the CORRECT and expected conclusion once a function’s sign changes partway through the interval; it isn’t a sign of an error.

Alternate Board-Style Example (companion function)

Source: Exercise 6.4, Q20 (NCERT)

Given: Prove that $f(x) = \cos x$ is (i) strictly increasing in $(\pi, 2\pi)$ (ii) strictly decreasing in $(0, \pi)$ (iii) neither increasing nor decreasing in $(0, 2\pi)$

Solution:

Case: (i) On $(\pi, 2\pi)$

$f'(x) = -\sin x$. For $x \in (\pi, 2\pi)$ (third and fourth quadrants), $\sin x < 0$, so $f'(x) = -(\text{negative}) > 0$ – increasing.

Case: (ii) On $(0, \pi)$

For $x \in (0, \pi)$ (first and second quadrants), $\sin x > 0$, so $f'(x) = -\sin x < 0$ – decreasing.

Case: (iii) On the full $(0, 2\pi)$

Since the behaviour differs between the two halves, f is neither purely increasing nor purely decreasing over the combined interval.

Final Answer

(i) Increasing (ii) Decreasing (iii) Neither, over the combined interval $(0, 2\pi)$

Quick Recall: For a full-period trig question split into two half-intervals, prove each half separately using the sign of sin/cos in that specific quadrant range, then simply note that differing behaviour on the two halves automatically means “neither” over the combined interval – no extra work needed for that conclusion.

Type 48

Proving a function is NEITHER increasing NOR decreasing on a GIVEN interval – typically because the derivative changes sign somewhere strictly inside that interval (as opposed to Type 47’s “two separate labelled halves,” here it’s usually a single interval where the sign genuinely varies).

Source: Exercise 6.4, Q21(i) (NCERT)

Given: Prove that $f(x) = x^2 - x + 1$ is neither increasing nor decreasing on $(-1, 1)$.

Solution:

Step 1: Differentiate: $f'(x) = 2x - 1$.

Step 2: Find where $f'(x) = 0$ within the interval: $2x - 1 = 0 \Rightarrow x = \frac{1}{2}$ – and $\frac{1}{2}$ DOES lie inside $(-1, 1)$.

Step 3: Check the sign on either side: for $x < \frac{1}{2}$ (e.g. $x = 0$, inside the interval), $f'(0) = -1 < 0$ – decreasing there. For $x > \frac{1}{2}$ (e.g. $x = 0.9$, inside the interval), $f'(0.9) = 0.8 > 0$ – increasing there.

Step 4: Since $f'(x)$ takes BOTH signs within the same given interval $(-1, 1)$, f cannot be classified as purely increasing or purely decreasing over the whole interval.

Final Answer

Since $f'(x)$ changes sign at $x = \frac{1}{2}$ (inside $(-1, 1)$), f is neither increasing nor decreasing on $(-1, 1)$

Common Mistake: Only checking the sign of $f'(x)$ at the ENDPOINTS of the interval – the critical point where the sign actually changes must be located and confirmed to lie WITHIN the given interval for this conclusion to be valid.

Quick Recall: To prove "neither," find where $f'(x) = 0$, confirm that point lies strictly INSIDE the given interval (not merely at an endpoint), then show the sign genuinely differs on either side of it within that same interval.

Type 49

Finding the interval(s) of increase/decrease for a function combining EXPONENTIAL (or logarithmic) terms with a polynomial – requires careful factoring of $f'(x)$, since the exponential/log factor's sign must be accounted for alongside the polynomial part.

Source: Exercise 6.4, Q25(i) (NCERT)

Given: Find the interval in which $f(x) = x^2 e^{-x}$ is increasing.

Solution:

Step 1: Differentiate using the product rule: $f'(x) = 2xe^{-x} + x^2(-e^{-x}) = e^{-x}(2x - x^2) = e^{-x} \cdot x(2 - x)$.

Step 2: Since $e^{-x} > 0$ for EVERY real x (an exponential factor never changes the sign), the sign of $f'(x)$ is entirely controlled by $x(2 - x)$.

Step 3: Find where $x(2 - x) = 0$: $x = 0$ or $x = 2$ – splitting the line into three intervals.

Step 4: Test signs: for $x < 0$: x negative, $(2-x)$ positive $\Rightarrow$ product negative. For $0 < x < 2$: x positive, $(2-x)$ positive $\Rightarrow$ product positive. For $x > 2$: x positive, $(2-x)$ negative $\Rightarrow$ product negative.

Final Answer

f is increasing on $(0, 2)$

Common Mistake: Forgetting that an exponential factor like e^{-x} NEVER contributes a sign change (it's always positive) – once this is recognised, only the remaining polynomial factor needs the usual sign-test treatment.

Alternate Board-Style Example

Source: Exercise 6.4, Q25(ii)

Given: Find the intervals in which $f(x) = x^2e^x$ is strictly increasing or strictly decreasing.

Solution:

Step 1: Differentiate: $f'(x) = 2xe^x + x^2e^x = e^x(2x + x^2) = e^x \cdot x(x + 2)$.

Step 2: Since $e^x > 0$ always, the sign of $f'(x)$ depends only on $x(x + 2)$.

Step 3: Roots: $x = 0$ and $x = -2$. Test signs: for $x < -2$: both factors negative $\Rightarrow$ product positive. For $-2 < x < 0$: x negative, $(x + 2)$ positive $\Rightarrow$ product negative. For $x > 0$: both factors positive $\Rightarrow$ product positive.

Final Answer

Increasing on $(-\infty, -2) \cup (0, \infty)$; decreasing on $(-2, 0)$

Quick Recall: Whenever e^x or e^{-x} appears as a multiplied factor in $f'(x)$, it contributes NO sign information at all (always positive) – factor it out mentally and run the usual sign-test on whatever polynomial factor remains (this same idea extends to Ex 6.4 Q26's $\log(x-2)$ term, which is positive throughout its own domain restriction).

V. Maxima and Minima – Direct Method

Type 50

Finding the maximum or minimum value of a QUADRATIC function by completing the square (or recognising vertex form) – no calculus needed, since the vertex directly gives the extreme value.

Formula Used: $f(x) = (x - a)^2 + b$ has minimum value b (at $x = a$), and no maximum

Source: Exercise 6.5, Q1(ii) (NCERT)

Given: Find the maximum and minimum values (if any) of $f(x) = (2x - 1)^2 + 3$.

Solution:

Step 1: The function is already in the form (something squared) + constant.

Step 2: Since $(2x - 1)^2 \geq 0$ for every real x (a square can never be negative), the SMALLEST possible value of $(2x - 1)^2$ is 0, achieved when $2x - 1 = 0$, i.e. $x = \frac{1}{2}$.

Step 3: At this point, $f\left(\frac{1}{2}\right) = 0 + 3 = 3$ – this is the minimum value.

Step 4: As $x \rightarrow \pm\infty$, $(2x - 1)^2 \rightarrow \infty$, so $f(x)$ grows without bound – there is no maximum value.

Final Answer

Minimum value = 3, no maximum value

Common Mistake: Trying to use calculus (setting $f'(x) = 0$) for a simple quadratic when the vertex is already obvious from the squared form – for these "direct method" questions, recognising the algebraic bound is faster and is exactly what the section is testing.

Alternate Board-Style Example (a downward-opening case)

Source: Exercise 6.5, Q2(ii)

Given: Find the maximum and minimum values of $f(x) = 9x^2 + 12x + 2$.

Solution:

Step 1: Complete the square: $9x^2 + 12x + 2 = 9\left(x^2 + \frac{4}{3}x\right) + 2 = 9\left(x + \frac{2}{3}\right)^2 - 9\left(\frac{2}{3}\right)^2 + 2 = 9\left(x + \frac{2}{3}\right)^2 - 4 + 2 = 9\left(x + \frac{2}{3}\right)^2 - 2$.

Step 2: Since $9\left(x + \frac{2}{3}\right)^2 \geq 0$ always, the minimum occurs when this term is 0, i.e. $x = -\frac{2}{3}$, giving $f\left(-\frac{2}{3}\right) = -2$.

Step 3: As $x \rightarrow \pm\infty$, $f(x) \rightarrow \infty$ – no maximum value exists.

Final Answer

Minimum value = -2 , no maximum value

Quick Recall: Complete the square to write the quadratic as $k(x-a)^2 + b$ – if $k > 0$, the minimum is b (no maximum); if $k < 0$, the maximum is b (no minimum) – read the extreme value directly off the completed-square form, no derivative needed.

Type 51

Finding the max/min of a function built from a MODULUS (absolute value) expression, using the fact that $|u| \geq 0$ for any u , combined with reading off the extreme value directly from the algebraic structure.

★ **Board Favorite** – Modulus-based max/min questions are a recurring VSA pattern, testing the $|u| \geq 0$ bound rather than calculus, tagged directly to NCERT.

Formula Used: $|u| \geq 0$ for every real u , with equality iff $u = 0$

Source: Exercise 6.5, Q4(i)

Given: Find the maximum and minimum values (if any) of $f(x) = |x + 2| - 1$.

Solution:

Step 1: Since $|x + 2| \geq 0$ for every x , the smallest possible value of $|x + 2|$ is 0, achieved when $x = -2$.

Step 2: At $x = -2$: $f(-2) = 0 - 1 = -1$ – this is the minimum value.

Step 3: As $x \rightarrow \pm\infty$, $|x + 2| \rightarrow \infty$, so $f(x) \rightarrow \infty$ – no maximum value.

Final Answer

Minimum value = -1 , no maximum value

Common Mistake: Trying to differentiate $|x + 2| - 1$ directly using calculus (which requires care at the non-differentiable point $x = -2$) – for this "direct method" section, simply using the bound $|u| \geq 0$ is much faster and is the intended technique.

Alternate Board-Style Example (modulus with a constant added, and modulus of a trig function)

Source: Exercise 6.5, Q6(ii) (NCERT)

Given: Find the maximum and minimum values of $f(x) = |\sin 4x + 3|$.

Solution:

Step 1: Since $-1 \leq \sin 4x \leq 1$ for every x , adding 3 throughout: $2 \leq \sin 4x + 3 \leq 4$.

Step 2: Since the entire range $[2, 4]$ is already positive (never touching zero or going negative), the modulus doesn't change anything: $|\sin 4x + 3| = \sin 4x + 3$ throughout.

Step 3: So the minimum of f is 2 (when $\sin 4x = -1$) and the maximum is 4 (when $\sin 4x = 1$).

Final Answer

Maximum value = 4, minimum value = 2

Quick Recall: For $|u|$ -type functions, first find the range of the INSIDE expression u using known bounds; if that range never crosses zero, the modulus does nothing and the max/min follow directly; if it DOES cross zero, the minimum of $|u|$ becomes exactly 0.

Type 52

Finding the max/min of a TRIGONOMETRIC function using the known bounds of sine and cosine ($-1 \leq \sin \theta, \cos \theta \leq 1$), including the auxiliary-angle (R-method) technique for combinations like $a \sin x + b \cos x$.

★ **Board Favorite** – Auxiliary-angle max/min questions are a classic, frequently-repeated pattern – the same R-method technique used for range-finding in earlier chapters resurfaces here as a direct max/min tool.

Formula Used: $a \sin x + b \cos x \in [-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$

Source: Exercise 6.5, Q9

Given: Find the maximum and minimum values of $3 \sin x + 4 \cos x$.

Solution:

Step 1: Here $a = 3, b = 4$. Compute $R = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

Step 2: By the auxiliary-angle bound, $3 \sin x + 4 \cos x$ ranges over $[-5, 5]$.

Step 3: So the maximum value is 5 and the minimum value is -5 .

Final Answer

Maximum value = 5, minimum value = -5

Common Mistake: Adding the coefficients directly ($3 + 4 = 7$) to guess the maximum, instead of computing the true amplitude $\sqrt{a^2 + b^2} = 5$ – this is a very common shortcut error.

Alternate Board-Style Example

Source: Exercise 6.5, Q8(ii)

Given: What are the minimum and maximum values of $2 - 3 \cos x$?

Solution:

Step 1: Since $-1 \leq \cos x \leq 1$, multiply through by -3 (flipping the inequality direction, since we're multiplying by a NEGATIVE number): $-3 \leq -3 \cos x \leq 3$.

Step 2: Add 2 throughout: $2 - 3 \leq 2 - 3 \cos x \leq 2 + 3$, i.e. $-1 \leq 2 - 3 \cos x \leq 5$.

Final Answer

Minimum value = -1 , maximum value = 5

Quick Recall: For a SINGLE trig term with a coefficient and constant, carry the standard $[-1, 1]$ bound through the arithmetic (flipping inequalities when multiplying by a negative); for a COMBINATION of sin and cos with different coefficients, use the auxiliary-angle bound $[-\sqrt{a^2 + b^2}, \sqrt{a^2 + b^2}]$ directly.

Type 53

Recognising when a function has NO maximum or minimum value at all – either because the domain is an OPEN interval with no achievable endpoint, or because the function is a monotonic (always increasing or always decreasing) cubic with no turning point whatsoever.

Source: Exercise 6.5, Q10

Given: Find the maximum and minimum values, if any, of: (i) $f(x) = x$ in $(0, 1)$ (ii) $f(x) = x$ in $[0, 1]$.

Solution:

Case: (i) Open interval $(0, 1)$

The function $f(x) = x$ is a simple increasing line. On the OPEN interval $(0, 1)$, neither endpoint 0 nor 1 is actually INCLUDED – so $f(x)$ can get arbitrarily close to 0 or 1 but never actually reach them. There is no maximum and no minimum value.

Case: (ii) Closed interval $[0, 1]$

Now both endpoints ARE included. $f(0) = 0$ (the minimum, since f is increasing) and $f(1) = 1$ (the maximum).

Final Answer

(i) No maximum, no minimum value (ii) Maximum value = 1, minimum value = 0

Common Mistake: Assuming a bounded interval always guarantees a max/min exists – the OPEN vs CLOSED distinction is critical: an open interval can have a function that gets arbitrarily close to a bound without ever attaining it.

Alternate Board-Style Example (monotonic cubic)

Source: Exercise 6.5, Q11

Given: Prove that the function $f(x) = x^3 + x^2 + x + 1$ does not have maxima or minima.

Solution:

Step 1: Differentiate: $f'(x) = 3x^2 + 2x + 1$.

Step 2: Check whether $f'(x)$ can ever equal 0 (a necessary condition for a turning point): treat this as a quadratic in x and check its discriminant: $\Delta = 2^2 - 4(3)(1) = 4 - 12 = -8 < 0$.

Step 3: A negative discriminant means $3x^2 + 2x + 1 = 0$ has NO real roots – so $f'(x)$ never equals zero for any real x .

Step 4: Since $f'(x)$ is a quadratic with a negative discriminant and positive leading coefficient ($3 > 0$), it never changes sign – $f'(x) > 0$ for EVERY real x (always increasing). With no point where $f'(x) = 0$, there can be no turning point, hence no local maximum or minimum anywhere.

Final Answer

Since $f'(x) = 3x^2 + 2x + 1$ has no real roots (discriminant < 0), f has no maxima or minima

Quick Recall: "No max/min" situations arise from either (a) an open interval where the bound is never actually attained, or (b) a function whose derivative never equals zero at all (check the discriminant for a quadratic derivative) – both mean there's simply no turning point or achievable extreme value.

Type 54

Finding the ABSOLUTE (global) maximum and minimum of a function on a CLOSED interval, using the "candidates test": evaluate f at every critical point INSIDE the interval, plus BOTH endpoints, and compare all the values.

★ **Board Favorite** – This is the master, most heavily-tested technique of the whole section (explicitly marked "(Imp)" in the book) – absolute extrema on a closed interval is one of the most common long-answer formats, tagged repeatedly to NCERT.

Formula Used: Absolute max/min on $[a, b]$ = largest/smallest of $\{f(a), f(b), f(c_1), f(c_2), \dots\}$ where $f'(c_i) = 0$

Source: Exercise 6.5, Q12(ii) (NCERT)

Given: Find the absolute maximum and minimum of $f(x) = 2x^3 - 15x^2 + 36x + 1$ on $[1, 5]$.

Solution:

- Step 1: Differentiate: $f'(x) = 6x^2 - 30x + 36 = 6(x^2 - 5x + 6) = 6(x - 2)(x - 3)$.
- Step 2: Find critical points: $f'(x) = 0 \Rightarrow x = 2$ or $x = 3$ – and BOTH lie inside the interval $[1, 5]$, so both are candidates.
- Step 3: List ALL candidates: the two critical points ($x = 2, 3$) AND the two endpoints ($x = 1, 5$).
- Step 4: Evaluate f at each candidate: $f(1) = 2 - 15 + 36 + 1 = 24$. $f(2) = 16 - 60 + 72 + 1 = 29$. $f(3) = 54 - 135 + 108 + 1 = 28$. $f(5) = 250 - 375 + 180 + 1 = 56$.
- Step 5: Compare all four values: 24, 29, 28, 56 – the largest is 56 (at $x = 5$) and the smallest is 24 (at $x = 1$).

Final Answer

Absolute maximum = 56 at $x = 5$; absolute minimum = 24 at $x = 1$

Common Mistake: Forgetting to include the ENDPOINTS as candidates – the absolute maximum/minimum on a closed interval can occur at an endpoint even if it is NOT a critical point (as happened here, where both extremes occurred at endpoints, not at the interior critical points).

Alternate Board-Style Example*Source: Exercise 6.5, Q13 (NCERT)*

Given: Find the maximum and minimum values of $x + \sin 2x$ on $[0, 2\pi]$. Also find the points of maxima and minima.

Solution:

Step 1: Differentiate: $f'(x) = 1 + 2 \cos 2x$.

Step 2: Set $f'(x) = 0$: $\cos 2x = -\frac{1}{2}$. Within $[0, 2\pi]$, i.e. $2x \in [0, 4\pi]$, the solutions are $2x = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}, \frac{10\pi}{3}$, giving $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$ – four critical points, all inside $[0, 2\pi]$.

Step 3: Evaluate f at all candidates (four critical points plus two endpoints $x = 0, 2\pi$):
 $f(0) = 0$. $f\left(\frac{\pi}{3}\right) = \frac{\pi}{3} + \sin \frac{2\pi}{3} = \frac{\pi}{3} + \frac{\sqrt{3}}{2}$. $f\left(\frac{2\pi}{3}\right) = \frac{2\pi}{3} + \sin \frac{4\pi}{3} = \frac{2\pi}{3} - \frac{\sqrt{3}}{2}$.
 $f\left(\frac{4\pi}{3}\right) = \frac{4\pi}{3} + \sin \frac{8\pi}{3} = \frac{4\pi}{3} + \frac{\sqrt{3}}{2}$. $f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{3} + \sin \frac{10\pi}{3} = \frac{5\pi}{3} - \frac{\sqrt{3}}{2}$.
 $f(2\pi) = 2\pi + 0 = 2\pi$.

Step 4: Compare: the endpoint value $f(2\pi) = 2\pi$ is the largest (since every interior candidate is less than 2π), and $f(0) = 0$ is the smallest.

Final Answer

Maximum value $= 2\pi$ at $x = 2\pi$; minimum value $= 0$ at $x = 0$

Quick Recall: The candidates test NEVER skips the endpoints – list every critical point inside the open interval (a, b) where $f'(x) = 0$, add both endpoints a and b , evaluate f at ALL of them, and simply compare the numbers to find the absolute max and min.

VI. Local Maxima and Minima – Derivative Tests

Type 55

Finding the turning points of a function via $f'(x) = 0$, then using the SECOND DERIVATIVE TEST ($f''(x)$'s sign) to classify each as a local maximum or minimum – including the special case where $f''(x) = 0$ too, which signals a possible POINT OF INFLEXION requiring a fallback to the first-derivative sign test.

Formula Used: $f'(c) = 0, f''(c) < 0 \Rightarrow$ local max; $f'(c) = 0, f''(c) > 0 \Rightarrow$ local min; $f'(c) = f''(c) = 0 \Rightarrow$ test further

Source: Exercise 6.6, Q1(i)

Given: Find the turning points of $f(x) = 2x^3 - 21x^2 + 36x - 20$ and distinguish between them.

Solution:

- Step 1: Differentiate: $f'(x) = 6x^2 - 42x + 36 = 6(x^2 - 7x + 6) = 6(x - 1)(x - 6)$.
- Step 2: Set $f'(x) = 0$: $x = 1$ or $x = 6$ – the two turning points.
- Step 3: Find the second derivative: $f''(x) = 12x - 42$.
- Step 4: Test at $x = 1$: $f''(1) = 12 - 42 = -30 < 0$ – NEGATIVE, so this is a point of LOCAL MAXIMUM. Compute $f(1) = 2 - 21 + 36 - 20 = -3$.
- Step 5: Test at $x = 6$: $f''(6) = 72 - 42 = 30 > 0$ – POSITIVE, so this is a point of LOCAL MINIMUM. Compute $f(6) = 432 - 756 + 216 - 20 = -128$.

Final Answer

$(1, -3)$ is a point of local maxima; $(6, -128)$ is a point of local minima

Common Mistake: Forgetting to compute the actual y -value (the local maximum/minimum VALUE) after finding the x -coordinate – the question typically wants both the point's full coordinates and/or the extreme value itself.

Alternate Board-Style Example (the special point-of-inflexion case)

Source: Exercise 6.6, Q1(ii)

Given: Find the turning points of $f(x) = x^3 - 3x^2 + 3x$ and distinguish between them.

Solution:

Step 1: Differentiate: $f'(x) = 3x^2 - 6x + 3 = 3(x^2 - 2x + 1) = 3(x - 1)^2$.

Step 2: Set $f'(x) = 0$: $3(x - 1)^2 = 0 \Rightarrow x = 1$ – a REPEATED root, giving only ONE turning point candidate.

Step 3: Second derivative: $f''(x) = 6x - 6$. At $x = 1$: $f''(1) = 0$ – the second derivative test is INCONCLUSIVE here (neither positive nor negative).

Step 4: Fall back to the FIRST-derivative sign test: since $f'(x) = 3(x - 1)^2 \geq 0$ for EVERY x (a square is never negative), the sign of $f'(x)$ does NOT change as x crosses 1 – it's non-negative on both sides.

Step 5: Since $f'(x)$ doesn't change sign at $x = 1$, this point is NEITHER a local maximum nor a local minimum – it is a POINT OF INFLEXION (the curve flattens momentarily but continues increasing through it).

Final Answer

$(1, 1)$ is a point of inflexion – it has neither a local maximum nor a local minimum here

Common Mistake: Assuming EVERY solution of $f'(x) = 0$ must be a local max or min – when $f''(x) = 0$ too, the point could instead be a point of inflexion, and the first-derivative SIGN CHANGE test (not just its value) must be used to settle the question.

Quick Recall: Find $f'(x) = 0$ for candidates, then check $f''(x)$: negative means max, positive means min; if $f''(x) = 0$ too, check whether $f'(x)$ actually CHANGES SIGN around that point – no sign change (like a squared factor) means a point of inflexion, not an extremum.

Type 56

Finding the points of local maxima and minima for a wide variety of standard functions – a mechanical practice cluster applying the same second-derivative-test method across many different function shapes (polynomial, rational, trigonometric).

Source: Exercise 6.6, Q3(vii)(viii)

Given: Find points of local maxima/minima of (vii) $f(x) = \frac{x}{2} + \frac{2}{x}, x > 0$ (NCERT) (viii) $f(x) = \frac{1}{x^2 + 2}$ (NCERT)

Solution:

Case: (vii) $f(x) = \frac{x}{2} + \frac{2}{x}$

$f'(x) = \frac{1}{2} - \frac{2}{x^2}$. Set to 0: $\frac{1}{2} = \frac{2}{x^2} \Rightarrow x^2 = 4 \Rightarrow x = 2$ (taking the positive root, since $x > 0$). $f''(x) = \frac{4}{x^3}$; at $x = 2$: $f''(2) = \frac{4}{8} = \frac{1}{2} > 0$ – local minimum. Value: $f(2) = 1 + 1 = 2$.

Case: (viii) $f(x) = \frac{1}{x^2 + 2}$

$f'(x) = \frac{-2x}{(x^2 + 2)^2}$. Set to 0: $x = 0$ (the only root, since the denominator is never zero). $f''(x)$ at $x = 0$ works out negative (the function has a peak at $x = 0$, since $f(x)$ is largest when the denominator $x^2 + 2$ is smallest) – local maximum. Value: $f(0) = \frac{1}{2}$.

Final Answer

(vii) Local minimum = 2 at $x = 2$ (viii) Local maximum = $\frac{1}{2}$ at $x = 0$

Common Mistake: Forgetting to restrict attention to the function's actual DOMAIN (e.g. $x > 0$ in part (vii)) when selecting which root of $f'(x) = 0$ is valid – a negative solution would need to be discarded there.

Alternate Board-Style Example

Source: Exercise 6.6, Q2

Given: Find the coordinates of the stationary points on $y = x^3 - 3x^2 - 9x + 7$, and distinguish between maxima and minima.

Solution:

Step 1: Differentiate: $y' = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x - 3)(x + 1)$.

Step 2: Set $y' = 0$: $x = 3$ or $x = -1$.

Step 3: Second derivative: $y'' = 6x - 6$. At $x = -1$: $y'' = -12 < 0$ – local maximum. At $x = 3$: $y'' = 12 > 0$ – local minimum.

Step 4: Compute values: $y(-1) = -1 - 3 + 9 + 7 = 12$; $y(3) = 27 - 27 - 27 + 7 = -20$.

Final Answer

$(-1, 12)$ is the point of maxima; $(3, -20)$ is the point of minima

Quick Recall: This is the same three-step method every time: differentiate, solve $f'(x) = 0$ for candidates (respecting any domain restriction), then classify each using $f''(x)$'s sign – practice across many function shapes (Ex 6.6 Q3's full eleven parts) builds speed and confidence with this single technique.

Type 57

Finding unknown constants in a function's formula, given conditions about WHERE its local maximum and/or minimum occur – set up equations from $f'(\text{given point}) = 0$ (and sometimes an extreme-VALUE condition too), then solve the resulting system.

★ **Board Favorite** – Finding constants from stated extremum locations is a recurring higher-order pattern, combining algebra with the derivative-test concept, tagged as a Competency Focused Question.

Source: Exercise 6.6, Q4

Given: For what values of a and b does $f(x) = x^3 + ax^2 + bx - 3$ have a local maximum at $x = 0$ and a local minimum at $x = 1$?

Solution:

Step 1: Differentiate: $f'(x) = 3x^2 + 2ax + b$.

Step 2: Since $x = 0$ is a stationary point: $f'(0) = 0 \Rightarrow b = 0$.

Step 3: Since $x = 1$ is ALSO a stationary point: $f'(1) = 0 \Rightarrow 3 + 2a + b = 0$. Substitute $b = 0$: $3 + 2a = 0 \Rightarrow a = -\frac{3}{2}$.

Final Answer

$$a = -\frac{3}{2}, b = 0$$

Common Mistake: Only using ONE of the two given stationary-point conditions – since there are two unknowns (a and b), both conditions ($f'(0) = 0$ AND $f'(1) = 0$) are needed together to solve the system.

Alternate Board-Style Example (Competency Focused, with a modulus/power term)

Source: Exercise 6.6, Q5 (Competency Focused Question)

Given: Given $2f(x) = \log|x|^a - bx^2 + x$, if $f(x)$ has extreme values at $x = -1$ and $x = 2$, find the values of a and b .

Solution:

Step 1: Rewrite using log rules: $2f(x) = a \log |x| - bx^2 + x \Rightarrow f(x) = \frac{a}{2} \log |x| - \frac{b}{2}x^2 + \frac{x}{2}$.

Step 2: Differentiate: $f'(x) = \frac{a}{2x} - bx + \frac{1}{2}$.

Step 3: Since $x = -1$ is an extreme point: $f'(-1) = 0 \Rightarrow \frac{a}{-2} - b(-1) + \frac{1}{2} = 0 \Rightarrow \frac{-a}{2} + b + \frac{1}{2} = 0 \Rightarrow -a + 2b + 1 = 0$ (multiplying through by 2).

Step 4: Since $x = 2$ is ALSO an extreme point: $f'(2) = 0 \Rightarrow \frac{a}{4} - 2b + \frac{1}{2} = 0 \Rightarrow a - 8b + 2 = 0$ (multiplying through by 4).

Step 5: Solve the system: from the first equation, $a = 2b + 1$. Substitute into the second: $(2b + 1) - 8b + 2 = 0 \Rightarrow -6b + 3 = 0 \Rightarrow b = \frac{1}{2}$. Then $a = 2\left(\frac{1}{2}\right) + 1 = 2$.

Final Answer

$$a = 2, \quad b = \frac{1}{2}$$

Quick Recall: "Extreme value at $x = c$ " ALWAYS translates to the equation $f'(c) = 0$ – with two unknown constants, you need exactly two such conditions (from the two given extreme points) to form a solvable system.

Type 58

Applied, real-world-flavoured questions about stationary points or rates of change dressed in a narrative context (a boat's path, a roller coaster's motion) – strip away the story to find the underlying calculus question being asked.

Source: Exercise 6.6, Q6 (Competency Focused Question)

Given: A speed boat moves along the curve $y = \frac{1}{\cos x \sin x}$, $x \in (0, \pi)$ in water. Find the number of stationary points.

Solution:

Step 1: Simplify first using the double-angle identity: $\cos x \sin x = \frac{1}{2} \sin 2x$, so $y = \frac{1}{\frac{1}{2} \sin 2x} = \frac{2}{\sin 2x} = 2 \csc 2x$.

Step 2: Differentiate: $\frac{dy}{dx} = 2 \times (-\csc 2x \cot 2x) \times 2 = -4 \csc 2x \cot 2x$.

Step 3: Stationary points occur where $\frac{dy}{dx} = 0$, i.e. where $\cot 2x = 0$ (since $\csc 2x$ is never zero – it's undefined at points where $\sin 2x = 0$, but never actually equals zero).

itself).

Step 4: $\cot 2x = 0 \Rightarrow 2x = \frac{\pi}{2}$ or $\frac{3\pi}{2}$ (within the relevant range for $x \in (0, \pi)$, i.e. $2x \in (0, 2\pi)$) $\Rightarrow x = \frac{\pi}{4}$ or $\frac{3\pi}{4}$.

Final Answer

There are 2 stationary points, at $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$

Common Mistake: Trying to differentiate $\frac{1}{\cos x \sin x}$ directly using the quotient rule without first simplifying via the double-angle identity – the simplified $2 \csc 2x$ form is far easier to differentiate cleanly.

Alternate Board-Style Example

Source: Exercise 6.6, Q7

Given: A roller coaster's distance travelled relates to time by $y = 4x - \frac{1}{2}x^2$, where x = time in seconds. Find (a) the rate of change of displacement w.r.t. time (b) the time

Solution:

Case: (a) Rate of change of displacement

This is simply $\frac{dy}{dx}$: differentiate $y = 4x - \frac{1}{2}x^2$ to get $\frac{dy}{dx} = 4 - x$.

Case: (b) Time to reach maximum height

Maximum height occurs where the rate of change is zero (the "turning point" of the motion): set $\frac{dy}{dx} = 0$: $4 - x = 0 \Rightarrow x = 4$.

Final Answer

(a) $\frac{dy}{dx} = 4 - x$ (b) 4 seconds

Quick Recall: Applied word problems in this topic are just standard derivative/stationary-point questions wearing a story costume – identify what physical quantity plays the role of y and x , simplify the given formula using any relevant identity, then apply the exact same $f'(x) = 0$ method as any other question in this section.

VII. Practical Problems on Maxima and Minima (Word Problems)

Type 59

Two-number optimization problems: given a fixed SUM or PRODUCT of two positive numbers, maximize or minimize some combination of them (their product, sum of squares, or a power-product like $x^a y^b$) – express everything in terms of ONE variable using the constraint, then differentiate.

★ **Board Favorite** – Two-number optimization is the entry point of the whole word-problems topic and the most frequently repeated pattern, tagged directly to NCERT.

Formula Used: Use the constraint to eliminate one variable, reducing the target expression to a function of one variable.

Source: Exercise 6.7, Q4 (NCERT)

Given: Find two positive numbers x and y such that $x + y = 60$ and xy^3 is maximum.

Solution:

Step 1: Use the constraint to eliminate x : $x = 60 - y$, with $0 < y < 60$.

Step 2: Write the target as a function of y alone: $f(y) = xy^3 = (60 - y)y^3 = 60y^3 - y^4$.

Step 3: Differentiate: $f'(y) = 180y^2 - 4y^3 = 4y^2(45 - y)$.

Step 4: Set $f'(y) = 0$: $y = 0$ (rejected, since $y > 0$) or $y = 45$.

Step 5: Confirm it's a maximum: $f''(y) = 360y - 12y^2$; at $y = 45$: $f''(45) = 360(45) - 12(2025) = 16200 - 24300 = -8100 < 0$ – confirmed maximum.

Step 6: Find x : $x = 60 - 45 = 15$.

Final Answer

The numbers are $x = 15$ and $y = 45$

Common Mistake: Forgetting to CHECK that the critical point actually lies within the valid range (here $0 < y < 60$) implied by both numbers needing to be positive – the rejected root ($y = 0$) is a reminder that not every algebraic solution is physically valid.

Alternate Board-Style Example (simplest case)

Source: Exercise 6.7, Q1 (NCERT)

Given: Find two numbers whose sum is 24 and whose product is as large as possible.

Solution:

Step 1: Let the numbers be x and $24 - x$. Product: $P(x) = x(24 - x) = 24x - x^2$.

Step 2: Differentiate: $P'(x) = 24 - 2x$. Set to 0: $x = 12$.

Step 3: Confirm maximum: $P''(x) = -2 < 0$ – confirmed.

Step 4: The other number: $24 - 12 = 12$.

Final Answer

Both numbers are 12

Quick Recall: Let one number be x , express the other using the given constraint (sum or product), substitute into the target expression to get a single-variable function, then run the standard differentiate-set-to-zero-confirm-with-second-derivative routine (this same method handles Ex 6.7 Q2, Q3, Q5, Q6 – sum of squares, product-of-powers, and least-sum-of-cubes variants – with no change in technique).

Type 60

Three-number optimization with a RATIO constraint linking two of the numbers – introduce a single variable for the ratio unit, express all three numbers in terms of it, then optimize.

Source: Exercise 6.7, Q7

Given: Three numbers sum to 180, and the ratio of the first two is 1 : 2. If the product of the numbers is greatest, find the numbers.

Solution:

Step 1: Since the first two numbers are in ratio 1 : 2, let them be x and $2x$ for a single variable x .

Step 2: The third number: since all three sum to 180, the third is $180 - x - 2x = 180 - 3x$.

Step 3: Write the product as a function of x : $P(x) = x(2x)(180 - 3x) = 2x^2(180 - 3x) = 360x^2 - 6x^3$.

Step 4: Differentiate: $P'(x) = 720x - 18x^2 = 18x(40 - x)$.

Step 5: Set $P'(x) = 0$: $x = 0$ (rejected) or $x = 40$.

Step 6: Confirm maximum: $P''(x) = 720 - 36x$; at $x = 40$: $720 - 1440 = -720 < 0$ – confirmed.

Step 7: Compute all three numbers: $x = 40$, $2x = 80$, $180 - 3x = 180 - 120 = 60$.

Final Answer

The numbers are 40, 80, and 60

Common Mistake: Introducing THREE separate variables for the three numbers instead of recognising that the given ratio lets everything be expressed via a SINGLE variable – always look for the most economical way to parametrize using the given constraints.

Quick Recall: When a ratio is given between two of several unknowns, express them both using ONE shared variable (e.g. x and $2x$ for a 1 : 2 ratio) right from the start – this reduces the whole problem to a single-variable optimization just like the two-number case.

Type 61

Economic optimization: a profit or cost function is given directly as an explicit formula in x (the number of units) – maximize profit or minimize cost using the ordinary derivative test, exactly as with any other function.

★ **Board Favorite** – Profit/cost optimization word problems are a recurring, straightforward application testing whether the calculus method transfers cleanly to an economics context, tagged directly to NCERT.

Source: Exercise 6.7, Q8 (NCERT)

Given: Find the maximum profit a company can make if the profit function is $p(x) = 41 + 72x - 18x^2$.

Solution:

Step 1: Differentiate: $p'(x) = 72 - 36x$.

Step 2: Set $p'(x) = 0$: $x = 2$.

Step 3: Confirm maximum: $p''(x) = -36 < 0$ – confirmed maximum.

Step 4: Compute the maximum profit: $p(2) = 41 + 144 - 72 = 113$.

Final Answer

Maximum profit = 113 (units of currency)

Common Mistake: Reporting the VALUE OF x (here, 2) as the final answer instead of the actual maximum PROFIT ($p(2) = 113$) – always re-read the question to check whether it wants the optimal input or the optimal output value.

Alternate Board-Style Example (minimizing cost)

Source: Exercise 6.7, Q9

Given: The cost per article is $C = 5 + \frac{48}{x} + 3x^2$, where x is the number manufactured per hour. Find the minimum value of C .

Solution:

Step 1: Differentiate: $C'(x) = -\frac{48}{x^2} + 6x$.

Step 2: Set $C'(x) = 0$: $6x = \frac{48}{x^2} \Rightarrow 6x^3 = 48 \Rightarrow x^3 = 8 \Rightarrow x = 2$.

Step 3: Confirm minimum: $C''(x) = \frac{96}{x^3} + 6$, which is always positive for $x > 0$ – confirmed minimum.

Step 4: Compute: $C(2) = 5 + 24 + 12 = 41$.

Final Answer

Minimum value of $C = \text{Rs. } 41$

Quick Recall: Treat "maximize profit" or "minimize cost" exactly like any other max/min question – differentiate the given function, set to zero, solve, and confirm with the second derivative; the economic language doesn't change the underlying calculus at all.

Type 62

Proving a general optimality THEOREM about rectangles: given a FIXED area, the square has the smallest perimeter (or, given a fixed perimeter, the square has the largest area) – a symbolic proof, not a numeric answer.

★ **Board Favorite** – These rectangle-square optimality "prove that" theorems are foundational, frequently-tested results that recur throughout the word-problems topic in different guises.

Source: Exercise 6.7, Q10

Given: Prove that of all rectangles with a given area, the square has the smallest perimeter.

Solution:

Step 1: Let the fixed area be k , with rectangle sides x and y , so $xy = k \Rightarrow y = \frac{k}{x}$.

Step 2: Perimeter as a function of x : $P(x) = 2(x + y) = 2\left(x + \frac{k}{x}\right)$.

Step 3: Differentiate: $P'(x) = 2\left(1 - \frac{k}{x^2}\right)$.

Step 4: Set $P'(x) = 0$: $\frac{k}{x^2} = 1 \Rightarrow x^2 = k \Rightarrow x = \sqrt{k}$.

Step 5: Find y : $y = \frac{k}{\sqrt{k}} = \sqrt{k}$ – so $x = y = \sqrt{k}$, meaning the rectangle IS a square.

Step 6: Confirm minimum: $P''(x) = \frac{4k}{x^3}$, which is positive for $x > 0$ – confirmed minimum.

Final Answer

The perimeter is smallest when $x = y = \sqrt{k}$, i.e. when the rectangle is a square

Common Mistake: Trying to prove the statement using a specific NUMERIC area instead of a general symbol k – a "prove that" question about ALL rectangles with a given area needs the proof to hold for an arbitrary (symbolic) fixed area.

Alternate Board-Style Example (the mirror theorem)

Source: Exercise 6.7, Q12

Given: Prove that of all rectangles with a given perimeter, the square has the largest area.

Solution:

Step 1: Let the fixed perimeter be k , with sides x, y , so $2(x + y) = k \Rightarrow y = \frac{k}{2} - x$.

Step 2: Area as a function of x : $A(x) = xy = x\left(\frac{k}{2} - x\right) = \frac{kx}{2} - x^2$.

Step 3: Differentiate: $A'(x) = \frac{k}{2} - 2x$. Set to 0: $x = \frac{k}{4}$.

Step 4: Find y : $y = \frac{k}{2} - \frac{k}{4} = \frac{k}{4}$ – so $x = y$, a square again.

Step 5: Confirm maximum: $A''(x) = -2 < 0$ – confirmed.

Final Answer

The area is largest when $x = y = \frac{k}{4}$, i.e. when the rectangle is a square

Quick Recall: Both theorems follow the identical pattern: introduce the fixed quantity as a symbol k , express the target (perimeter or area) as a single-variable function using the OTHER geometric relationship, differentiate, and the critical point will always come out symmetric ($x = y$) – proving the square is optimal either way.

Type 63

A rectangle problem with a SPECIFIC NUMERIC perimeter or area given – find the ACTUAL maximum/minimum value directly (a numeric answer), rather than proving a general theorem.

Source: Exercise 6.7, Q11

Given: Of all rectangles each having perimeter 24 cm, find the one having maximum area. Also find

Solution:

Step 1: By the general theorem (Type 62), the maximum-area rectangle with a given perimeter is always a square, with side $= \frac{\text{perimeter}}{4}$.

Step 2: Here perimeter = 24, so side $= \frac{24}{4} = 6$ cm.

Step 3: Maximum area: $6 \times 6 = 36 \text{ cm}^2$.

Final Answer

The rectangle is a square of side 6 cm, with maximum area $= 36 \text{ cm}^2$

Common Mistake: Re-deriving the whole optimization from scratch when a specific NUMERIC value is given – once the general theorem's result ($x = \frac{\text{perimeter}}{4}$) is known, applying it directly to a specific number is much faster than repeating the full calculus argument.

Quick Recall: Numeric versions of a "prove the theorem" question can often be solved by simply plugging the given number into the ALREADY-KNOWN general result, rather than repeating the full derivation each time.

Type 64

Right-triangle optimality theorems: given a fixed hypotenuse, prove the perimeter is maximum when the triangle is isosceles; or given a fixed area, prove the hypotenuse is least when the triangle is isosceles – parametrize the two legs using a single angle.

Source: Exercise 6.7, Q13

Given: Prove that the perimeter of a right-angled triangle of given hypotenuse is maximum when the triangle is isosceles.

Solution:

Step 1: Let the fixed hypotenuse be h . Parametrize the two legs using an angle θ : $\text{leg}_1 = h \cos \theta$, $\text{leg}_2 = h \sin \theta$ (a natural parametrization since $\text{leg}_1^2 + \text{leg}_2^2 = h^2 \cos^2 \theta + h^2 \sin^2 \theta = h^2$, automatically satisfying Pythagoras).

Step 2: Perimeter: $P(\theta) = h + h \cos \theta + h \sin \theta = h(1 + \cos \theta + \sin \theta)$.

Step 3: Differentiate w.r.t. θ : $P'(\theta) = h(-\sin \theta + \cos \theta)$.

Step 4: Set $P'(\theta) = 0$: $\cos \theta = \sin \theta \Rightarrow \tan \theta = 1 \Rightarrow \theta = \frac{\pi}{4}$.

Step 5: At $\theta = \frac{\pi}{4}$: $\cos \theta = \sin \theta$, so $\text{leg}_1 = \text{leg}_2$ – the triangle IS isosceles.

Step 6: Confirm maximum: $P''(\theta) = h(-\cos \theta - \sin \theta)$; at $\theta = \frac{\pi}{4}$, this is negative (both terms are positive, made negative by the leading minus sign) – confirmed maximum.

Final Answer

Perimeter is maximum when $\theta = \frac{\pi}{4}$, i.e. when the two legs are equal (isosceles)

Common Mistake: Trying to use the two legs a, b as independent variables and getting stuck with the Pythagorean CONSTRAINT $a^2 + b^2 = h^2$ as a separate equation to juggle – parametrizing via a single angle θ automatically satisfies the constraint and reduces the problem to ONE variable from the start.

Alternate Board-Style Example (fixed area version)

Source: Exercise 6.7, Q15

Given: A right-angled triangle has a constant area. Prove that the hypotenuse is least when the triangle is isosceles.

Solution:

Step 1: Let the legs be a, b with fixed area $\frac{1}{2}ab = k \Rightarrow b = \frac{2k}{a}$.

Step 2: Hypotenuse squared (easier to minimize than the hypotenuse itself, since square root is increasing): $h^2 = a^2 + b^2 = a^2 + \frac{4k^2}{a^2}$.

Step 3: Differentiate w.r.t. a : $\frac{d(h^2)}{da} = 2a - \frac{8k^2}{a^3}$.

Step 4: Set to 0: $2a = \frac{8k^2}{a^3} \Rightarrow 2a^4 = 8k^2 \Rightarrow a^4 = 4k^2 \Rightarrow a^2 = 2k$ (taking the positive root).

Step 5: Find b^2 : $b^2 = \frac{4k^2}{a^2} = \frac{4k^2}{2k} = 2k = a^2$ – so $a = b$, confirming the triangle is isosceles at the minimum.

Final Answer

The hypotenuse is least when $a = b$, i.e. when the triangle is isosceles

Quick Recall: For a fixed HYPOTENUSE, parametrize the legs via a single angle θ to automatically respect Pythagoras; for a fixed AREA, use the area relationship to eliminate one leg directly, and minimize h^2 instead of h to avoid an awkward square root – both approaches lead to the legs being equal at the optimum.

Type 65

A triangle with two FIXED side lengths and a variable INCLUDED ANGLE between them – maximize the area by treating the angle itself as the single optimization variable.

Formula Used: Area of a triangle with sides a, b and included angle θ : $A = \frac{1}{2}ab \sin \theta$

Source: Exercise 6.7, Q14

Given: Two sides of a triangle have lengths a and b , and the angle between them is θ . What value

Solution:

Step 1: Write the area formula: $A(\theta) = \frac{1}{2}ab \sin \theta$ – here a and b are FIXED constants, and θ is the only variable.

Step 2: Differentiate w.r.t. θ : $A'(\theta) = \frac{1}{2}ab \cos \theta$.

Step 3: Set $A'(\theta) = 0$: since $a, b \neq 0$, this requires $\cos \theta = 0 \Rightarrow \theta = \frac{\pi}{2}$.

Step 4: Confirm maximum: $A''(\theta) = -\frac{1}{2}ab \sin \theta$; at $\theta = \frac{\pi}{2}$, $\sin \theta = 1$, so $A'' = -\frac{1}{2}ab < 0$ (since $a, b > 0$) – confirmed maximum.

Step 5: Maximum area: $A\left(\frac{\pi}{2}\right) = \frac{1}{2}ab \sin \frac{\pi}{2} = \frac{1}{2}ab$.

Final Answer

$\theta = \frac{\pi}{2}$ maximizes the area; maximum area = $\frac{1}{2}ab$

Common Mistake: Treating a and b as variables to differentiate with respect to – they are FIXED given lengths in this problem; only the angle θ is free to vary, so differentiation must be with respect to θ alone.

Quick Recall: When two side lengths are fixed and only the angle between them is free, the area formula $\frac{1}{2}ab \sin \theta$ is maximized exactly when $\sin \theta = 1$, i.e. $\theta = 90^\circ$ – a right angle between the two given sides always gives the largest possible triangle.

Type 66

A triangle whose sides are given as ALGEBRAIC EXPRESSIONS in a parameter (or where one side and the perimeter are given, leaving two related unknowns) – use Heron's formula to express the area, then maximize.

Formula Used: Heron's formula: $\text{Area} = \sqrt{s(s-a)(s-b)(s-c)}$, $s = \frac{a+b+c}{2}$

Source: Exercise 6.7, Q16

Given: The sides of a triangle are $9 + x^2$, $9 + x^2$, and $18 - 2x^2$. Find (i) the area in terms of x (ii) the value of x for maximum area.

Solution:

Step 1: Compute the semi-perimeter: $s = \frac{(9 + x^2) + (9 + x^2) + (18 - 2x^2)}{2} = \frac{36}{2} = 18$ – remarkably, this is a FIXED constant regardless of x .

Step 2: Compute each factor: $s - a = 18 - (9 + x^2) = 9 - x^2$ (for BOTH equal sides).
 $s - c = 18 - (18 - 2x^2) = 2x^2$.

Step 3: Apply Heron's formula: $\text{Area}^2 = s(s-a)(s-a)(s-c) = 18(9 - x^2)^2(2x^2) = 36x^2(9 - x^2)^2$.

Step 4: Take the square root: $\text{Area} = 6x(9 - x^2)$ (taking the positive root, valid for $0 < x < 3$) $= 54x - 6x^3$.

Step 5: (ii) Differentiate: $\frac{d(\text{Area})}{dx} = 54 - 18x^2$. Set to 0: $x^2 = 3 \Rightarrow x = \sqrt{3}$.

Final Answer

(i) $\text{Area} = (54x - 6x^3)$ sq. units (ii) $x = \sqrt{3}$ gives maximum area

Common Mistake: Trying to maximize the AREA SQUARED expression directly without simplifying it to a clean linear-times-quadratic form first – taking the square root early ($\text{Area} = 6x(9 - x^2)$) makes the subsequent differentiation far simpler.

Alternate Board-Style Example (perimeter and one side given)

Source: Exercise 6.7, Q17

Given: The perimeter of a triangle is 8 cm. If one side is 3 cm, what should the other two sides be?

Solution:

Step 1: Let the other two sides be b and c , with $3 + b + c = 8 \Rightarrow b + c = 5$.

Step 2: Semi-perimeter: $s = 4$ (fixed).

Step 3: Heron's formula: $\text{Area}^2 = s(s-3)(s-b)(s-c) = 4(1)(4-b)(4-c)$. Substitute $c = 5 - b$: $(4-b)(4-(5-b)) = (4-b)(b-1)$.

Step 4: So $\text{Area}^2 = 4(4-b)(b-1) = 4(4b-4-b^2+b) = 4(-b^2+5b-4)$.

Step 5: Maximize $-b^2 + 5b - 4$ (equivalent to maximizing Area^2 since the coefficient 4 is a positive constant multiplier): differentiate, $-2b + 5 = 0 \Rightarrow b = 2.5$.

Step 6: Find c : $c = 5 - 2.5 = 2.5$ – the two unknown sides turn out EQUAL, an isosceles triangle again.

Final Answer

Both other sides should be = 2.5 cm each

Quick Recall: For sides given algebraically or via a perimeter constraint, use Heron's formula with the semi-perimeter s (which is often conveniently a FIXED constant), simplify Area^2 into a manageable polynomial, differentiate that (avoiding the square root until the very end), and solve.

Type 67

Proving that among all rectangles inscribed in a FIXED circle, the SQUARE has the maximum area – parametrize the rectangle's corners using an angle on the circle.

Source: Exercise 6.7, Q19

Given: Show that of all rectangles inscribed in a given fixed circle, the square has the maximum area.

Solution:

Step 1: Let the circle have fixed radius r . Any rectangle inscribed in it has its diagonal equal to the circle's diameter, $2r$.

Step 2: Parametrize the rectangle's sides using an angle θ : sides = $2r \cos \theta$ and $2r \sin \theta$ (so that, by Pythagoras, $(2r \cos \theta)^2 + (2r \sin \theta)^2 = 4r^2$, matching the diagonal exactly).

Step 3: Area: $A(\theta) = (2r \cos \theta)(2r \sin \theta) = 4r^2 \sin \theta \cos \theta = 2r^2 \sin 2\theta$ (using the double-angle identity).

Step 4: Differentiate: $A'(\theta) = 4r^2 \cos 2\theta$. Set to 0: $\cos 2\theta = 0 \Rightarrow 2\theta = \frac{\pi}{2} \Rightarrow \theta = \frac{\pi}{4}$.

Step 5: At $\theta = \frac{\pi}{4}$: the two sides $2r \cos \frac{\pi}{4}$ and $2r \sin \frac{\pi}{4}$ are EQUAL (since $\cos \frac{\pi}{4} = \sin \frac{\pi}{4}$) – the rectangle is a square.

Step 6: Confirm maximum: $A''(\theta) = -8r^2 \sin 2\theta$; at $\theta = \frac{\pi}{4}$, $\sin 2\theta = 1$, so $A'' = -8r^2 < 0$ – confirmed maximum.

Final Answer

Area is maximum ($= 2r^2$) when the rectangle is a square

Common Mistake: Trying to use the two sides x, y directly with the constraint $x^2 + y^2 = 4r^2$ handled separately – the angle parametrization $x = 2r \cos \theta$, $y = 2r \sin \theta$ automatically satisfies this constraint and reduces to a clean single-variable trig optimization.

Quick Recall: Whenever a shape is inscribed in a circle, parametrizing its vertices using an angle θ (via $\cos \theta, \sin \theta$) is often the cleanest way to automatically satisfy the "lies on the circle" constraint before optimizing.

Type 68

The classic printing/margin problem: a sheet must contain a FIXED printed area, with given margins on each side – minimize the TOTAL area of the sheet (printed area plus margins).

Source: Exercise 6.7, Q18

Given: A sheet of paper is to contain 18 cm^2 of printed matter. The margins at the top and bottom

Solution:

Step 1: Let the printed area have width x and height y , with $xy = 18$ (the fixed printed area) $\Rightarrow y = \frac{18}{x}$.

Step 2: The TOTAL sheet dimensions add the margins on each side: total width $= x + 2(1) = x + 2$ (1 cm margin on each side); total height $= y + 2(2) = y + 4$ (2 cm margin top and bottom).

Step 3: Total sheet area: $S(x) = (x + 2)(y + 4) = (x + 2)\left(\frac{18}{x} + 4\right)$. Expand: $S = 18 + 4x + \frac{36}{x} + 8 = 26 + 4x + \frac{36}{x}$.

Step 4: Differentiate: $S'(x) = 4 - \frac{36}{x^2}$. Set to 0: $x^2 = 9 \Rightarrow x = 3$.

Step 5: Confirm minimum: $S''(x) = \frac{72}{x^3} > 0$ for $x > 0$ – confirmed.

Step 6: Find y : $y = \frac{18}{3} = 6$.

Step 7: Total sheet dimensions: width = $3 + 2 = 5$; height = $6 + 4 = 10$.

Final Answer

Sheet dimensions: 5 cm $\times$ 10 cm

Common Mistake: Minimizing the PRINTED area's dimensions instead of the TOTAL sheet area (printed area plus margins) – the printed area is FIXED at 18 cm² by the problem statement; what's being minimized is the total paper used, which includes the margins.

Quick Recall: Set up the printed dimensions with the fixed-area constraint first, then ADD the margins (doubled, since margins appear on both sides of each dimension) to get the total sheet formula BEFORE differentiating – the margins are what make this a genuine optimization, not the printed area itself.

Type 69

A piece of wire (or a fixed budget of perimeter) is divided between a CIRCLE and a SQUARE (or triangle) – either prove a general theorem about when the combined area is least, or find the actual numeric cutting point.

★ **Board Favorite** – Wire-cutting problems combining a circle with a square or triangle are a classic, frequently-tested pattern requiring the sum-of-two-areas optimization technique.

Source: Exercise 6.7, Q20(i) (NCERT)

Given: The sum of the perimeters of a circle and a square is k , a constant. Prove that the sum of the

Solution:

Step 1: Let the circle's radius be r and the square's side be x . Given: $2\pi r + 4x = k \Rightarrow r = \frac{k - 4x}{2\pi}$.

Step 2: Sum of areas: $S = \pi r^2 + x^2 = \pi \left(\frac{k - 4x}{2\pi} \right)^2 + x^2 = \frac{(k - 4x)^2}{4\pi} + x^2$.

Step 3: Differentiate w.r.t. x : $S'(x) = \frac{2(k - 4x)(-4)}{4\pi} + 2x = \frac{-2(k - 4x)}{\pi} + 2x$.

Step 4: Set $S'(x) = 0$: $2x = \frac{2(k - 4x)}{\pi} \Rightarrow \pi x = k - 4x \Rightarrow x(\pi + 4) = k \Rightarrow x = \frac{k}{\pi + 4}$.

Step 5: Find r : $r = \frac{k-4x}{2\pi}$. Substituting $x = \frac{k}{\pi+4}$: $k-4x = k - \frac{4k}{\pi+4} = \frac{k\pi}{\pi+4}$, so

$$r = \frac{k\pi/(\pi+4)}{2\pi} = \frac{k}{2(\pi+4)}.$$

Step 6: Compare: $2r = \frac{k}{\pi+4} = x$ – confirming $x = 2r$, i.e. the square's side equals TWICE the circle's radius (the circle's diameter).

Final Answer

The sum of areas is least when the side of the square (x) equals twice the radius of the circle ($2r$), a

Common Mistake: Stopping after finding x in terms of k and not going the extra step to also express r in terms of k and COMPARE the two – the whole point of a "prove that" question like this is to explicitly demonstrate the stated relationship ($x = 2r$), not just find a critical point.

Alternate Board-Style Example (numeric wire-cutting, circle + triangle)

Source: Exercise 6.7, Q21

Given: A wire 10 metres long is cut into two parts. One part is bent into a circle, the other into an

Solution:

Step 1: Let a be the length used for the circle, so $10 - a$ is used for the triangle. Circle:

$$\text{circumference} = a \Rightarrow r = \frac{a}{2\pi}, \text{ area} = \pi r^2 = \frac{a^2}{4\pi}. \text{ Triangle: perimeter} = 10 - a \Rightarrow$$

$$\text{side} = \frac{10-a}{3}, \text{ area} = \frac{\sqrt{3}}{4} \left(\frac{10-a}{3} \right)^2 = \frac{\sqrt{3}(10-a)^2}{36}.$$

Step 2: Sum of areas: $S(a) = \frac{a^2}{4\pi} + \frac{\sqrt{3}(10-a)^2}{36}.$

Step 3: Differentiate: $S'(a) = \frac{a}{2\pi} - \frac{\sqrt{3}(10-a)}{18}.$

Step 4: Set $S'(a) = 0$: $\frac{a}{2\pi} = \frac{\sqrt{3}(10-a)}{18} \Rightarrow 18a = 2\pi\sqrt{3}(10-a) \Rightarrow 18a + 2\pi\sqrt{3}a = 20\pi\sqrt{3} \Rightarrow a(18 + 2\pi\sqrt{3}) = 20\pi\sqrt{3}.$

Step 5: Solve: $a = \frac{20\pi\sqrt{3}}{18 + 2\pi\sqrt{3}} = \frac{10\pi\sqrt{3}}{9 + \pi\sqrt{3}},$ which simplifies (dividing numerator and denominator by $\sqrt{3}$) to $a = \frac{10\pi}{3\sqrt{3} + \pi}.$

Final Answer

Length bent into a circle = $\frac{10\pi}{3\sqrt{3} + \pi}$ metres (the remainder forms the triangle)

Quick Recall: Let one piece's length be the single variable, express the OTHER piece's length via the fixed total, write each shape's area formula in terms of its own perimeter/circumference, add them together, then differentiate and solve exactly as with any two-quantity optimization.

Type 70

A composite "Norman window" shape (a rectangle topped by a semicircle) with a FIXED total perimeter – maximize the total area (interpreted as "maximum light") by optimizing the semicircle's radius.

Source: Exercise 6.7, Q23

Given: A window is a rectangle surmounted by a semicircle. If the perimeter of the window is p cm

Solution:

Step 1: Let the semicircle's radius be r (so the rectangle's width, matching the semicircle's diameter, is $2r$), and let the rectangle's height be h .

Step 2: The window's PERIMETER includes: the rectangle's two vertical sides (h each), the rectangle's bottom ($2r$), and the semicircular arc on top (πr) – but NOT the internal line where the semicircle meets the rectangle (that's inside the shape, not on the boundary). So: $p = 2h + 2r + \pi r \Rightarrow h = \frac{p - 2r - \pi r}{2}$.

Step 3: Total area (light admitted) = rectangle area + semicircle area = $2rh + \frac{1}{2}\pi r^2$.

Step 4: Substitute h : $A(r) = 2r \left(\frac{p - 2r - \pi r}{2} \right) + \frac{1}{2}\pi r^2 = r(p - 2r - \pi r) + \frac{1}{2}\pi r^2 = pr - 2r^2 - \pi r^2 + \frac{1}{2}\pi r^2 = pr - 2r^2 - \frac{1}{2}\pi r^2$.

Step 5: Differentiate: $A'(r) = p - 4r - \pi r$.

Step 6: Set $A'(r) = 0$: $p = r(4 + \pi) \Rightarrow r = \frac{p}{\pi + 4}$.

Final Answer

The radius for maximum light is $r = \frac{p}{\pi + 4}$, as required

Common Mistake: Including the internal boundary (where the rectangle meets the semicircle) as part of the PERIMETER – only the OUTER boundary of the composite shape counts toward the window's perimeter; the internal segment is not an edge of the window frame.

Quick Recall: For composite shapes, carefully identify exactly which edges form the OUTER perimeter (excluding any internal dividing lines) before writing the constraint equation – this is the step most likely to go wrong in Norman-window-style problems.

Type 71

A fenced/enclosed area problem where DIFFERENT sides have DIFFERENT costs per unit length – minimize the TOTAL COST (not just the total length) subject to a fixed area.

Source: Exercise 6.7, Q24

Given: A rectangular area of 9000 m^2 is to be fenced, with two opposite sides made of bricks (Rs. 100/m each) and two opposite sides made of wood (Rs. 250/m each).

Solution:

Step 1: Let the brick sides have length x each (a pair of opposite sides), and the wood sides have length y each, with area $xy = 9000 \Rightarrow y = \frac{9000}{x}$.

Step 2: Total cost: two brick sides at Rs. 100/m each, two wood sides at Rs. 250/m each:
 $C = 2x(100) + 2y(250) = 200x + 500y$.

Step 3: Substitute y : $C(x) = 200x + 500 \left(\frac{9000}{x} \right) = 200x + \frac{4500000}{x}$.

Step 4: Differentiate: $C'(x) = 200 - \frac{4500000}{x^2}$.

Step 5: Set $C'(x) = 0$: $x^2 = 22500 \Rightarrow x = 150$.

Step 6: Find y : $y = \frac{9000}{150} = 60$.

Step 7: Compute minimum cost: $C = 200(150) + 500(60) = 30000 + 30000 = 60000$.

Final Answer

Least total cost = Rs. 60,000

Common Mistake: Minimizing the total PERIMETER or LENGTH instead of the total COST – since the two pairs of sides have DIFFERENT costs per metre, the cost function must weight each pair by its own rate, not simply add up lengths.

Quick Recall: When different sides carry different costs, write the total cost as a WEIGHTED sum (each side's length times its own rate), substitute the area constraint to reduce to one variable, then optimize the COST function directly – not the plain perimeter or area.

Type 72

Finding the point on a given curve NEAREST to a specified external point – minimize the SQUARED distance (to avoid an awkward square root) between a general point on the curve and the given point.

★ **Board Favorite** – Nearest-point-on-a-curve problems are a classic distance-minimization pattern, testing the squared-distance trick that avoids differentiating a square root directly.

Formula Used: $D^2 = (x - x_0)^2 + (y - y_0)^2$ – minimize D^2 instead of D (same critical points, since square root is increasing)

Source: Exercise 6.7, Q25(i)

Given: Find the point on the parabola $y^2 = 4x$ which is nearest to the point $(2, -8)$.

Solution:

Step 1: Let (x, y) be a general point on the parabola, so $x = \frac{y^2}{4}$ – this lets us express distance purely in terms of y .

Step 2: Squared distance to $(2, -8)$: $D^2 = (x - 2)^2 + (y + 8)^2 = \left(\frac{y^2}{4} - 2\right)^2 + (y + 8)^2$.

Step 3: Differentiate with respect to y : $\frac{d(D^2)}{dy} = 2\left(\frac{y^2}{4} - 2\right)\left(\frac{y}{2}\right) + 2(y + 8) = \left(\frac{y^2}{4} - 2\right)y + 2(y + 8) = \frac{y^3}{4} - 2y + 2y + 16 = \frac{y^3}{4} + 16$.

Step 4: Set to 0: $\frac{y^3}{4} = -16 \Rightarrow y^3 = -64 \Rightarrow y = -4$.

Step 5: Find x : $x = \frac{16}{4} = 4$.

Final Answer

The nearest point is $(4, -4)$

Common Mistake: Trying to minimize the distance D itself (involving a square root) directly – differentiating D^2 instead is algebraically much simpler and gives the identical critical point, since squaring is a monotonic (order-preserving) operation for non-negative distances.

Alternate Board-Style Example

Source: Exercise 6.7, Q25(ii)

Given: Find the point on the curve $x^2 = 8y$ which is nearest to the point $(2, 4)$.

Solution:

Step 1: Let (x, y) be on the curve, so $y = \frac{x^2}{8}$.

Step 2: Squared distance to $(2, 4)$: $D^2 = (x - 2)^2 + (y - 4)^2 = (x - 2)^2 + \left(\frac{x^2}{8} - 4\right)^2$.

Step 3: Differentiate w.r.t. x : $\frac{d(D^2)}{dx} = 2(x-2) + 2\left(\frac{x^2}{8} - 4\right)\left(\frac{x}{4}\right) = 2x - 4 + \frac{x}{2}\left(\frac{x^2}{8} - 4\right) = 2x - 4 + \frac{x^3}{16} - 2x = \frac{x^3}{16} - 4$.

Step 4: Set to 0: $\frac{x^3}{16} = 4 \Rightarrow x^3 = 64 \Rightarrow x = 4$.

Step 5: Find y : $y = \frac{16}{8} = 2$.

Final Answer

The nearest point is $(4, 2)$

Quick Recall: Express one coordinate in terms of the other using the curve's equation, write the SQUARED distance to the given external point as a single-variable function, differentiate and set to zero – always minimize the squared distance, never the raw distance itself.

Type 73

A running-track-style word problem: a composite shape (rectangle with semicircular ends) has a FIXED total perimeter (the track length) – maximize the enclosed RECTANGULAR area, then compute the total field area.

★ **Board Favorite** – This exact running-track problem is directly tagged to ISC 2023, making it an especially high-value pattern to know cold.

Source: Exercise 6.7, Q26 (ISC 2023)

Given: A running track of 440 m encloses a football field: a rectangle with a semicircle at each end.

Solution:

Step 1: Let the rectangle have length x and the semicircles at each end have radius r (so the rectangle's width is $2r$, matching the diameter). The two semicircular ends together form ONE full circle.

Step 2: Track length (perimeter) = $2x + 2\pi r = 440 \Rightarrow x = 220 - \pi r$.

Step 3: Rectangular area: $A(r) = x \times 2r = 2r(220 - \pi r) = 440r - 2\pi r^2$.

Step 4: Differentiate: $A'(r) = 440 - 4\pi r$. Set to 0: $r = \frac{440}{4\pi} = \frac{110}{\pi}$.

Step 5: Find x : $x = 220 - \pi \left(\frac{110}{\pi} \right) = 220 - 110 = 110$ m.

Step 6: Total field area = rectangle area + full circle area = $x(2r) + \pi r^2 = 110 \left(\frac{220}{\pi} \right) + \pi \left(\frac{110}{\pi} \right)^2 = \frac{24200}{\pi} + \frac{12100}{\pi} = \frac{36300}{\pi}$. Using $\pi = \frac{22}{7}$: $\frac{36300 \times 7}{22} = 11550$.

Final Answer

Rectangle length = 110 m (width = $\frac{220}{\pi} \approx 70$ m); total football field area = 11550 m²

Common Mistake: Forgetting that the two semicircular ends TOGETHER form exactly one full circle when computing both the perimeter (one full circumference, $2\pi r$) and the total area (one full circle area, πr^2) – treating them as two separate semicircle calculations is more error-prone than just combining them into a full circle immediately.

Quick Recall: For a rectangle-with-semicircular-ends track, immediately combine the two semicircular ends into ONE circle for both perimeter and area purposes – this simplifies the setup to the same "rectangle plus circle" structure as other composite-shape problems.

Type 74

Finding the maximum area of a triangle or polygon INSCRIBED IN AN ELLIPSE, with one vertex fixed at a specific point (like the end of the major axis) and the others constrained to lie on the ellipse.

Source: Exercise 6.7, Q27

Given: Find the maximum area of an isosceles triangle inscribed in the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$ with its vertex at one end of the major axis.

Solution:

Step 1: The major axis is along the x -direction (since $16 > 9$), with $a = 4, b = 3$. Place the fixed vertex at $(4, 0)$.

Step 2: By symmetry, the isosceles triangle's base is a VERTICAL chord at some $x = x_0$, with base endpoints (x_0, y_0) and $(x_0, -y_0)$ on the ellipse, where $y_0 = 3\sqrt{1 - \frac{x_0^2}{16}}$ (solving the ellipse equation for y).

Step 3: Base length $= 2y_0$; height (horizontal distance from the fixed vertex to the base) $= 4 - x_0$.

Step 4: Area: $A(x_0) = \frac{1}{2} \times 2y_0 \times (4 - x_0) = y_0(4 - x_0) = 3\sqrt{1 - \frac{x_0^2}{16}}(4 - x_0) = \frac{3}{4}(4 - x_0)\sqrt{16 - x_0^2}$.

Step 5: Differentiate w.r.t. x_0 (writing x for x_0 for brevity): $A'(x) = \frac{3}{4} \left[-\sqrt{16 - x^2} + (4 - x) \cdot \frac{-x}{\sqrt{16 - x^2}} \right]$
 $\frac{3}{4} \cdot \frac{-(16 - x^2) - x(4 - x)}{\sqrt{16 - x^2}} = \frac{3}{4} \cdot \frac{2x^2 - 4x - 16}{\sqrt{16 - x^2}}$.

Step 6: Set the numerator to 0: $2x^2 - 4x - 16 = 0 \Rightarrow x^2 - 2x - 8 = 0 \Rightarrow (x - 4)(x + 2) = 0 \Rightarrow x = 4$ (rejected – degenerate triangle) or $x = -2$.

Step 7: At $x = -2$: $A = \frac{3}{4}(4 - (-2))\sqrt{16 - 4} = \frac{3}{4}(6)\sqrt{12} = \frac{3}{4}(6)(2\sqrt{3}) = 9\sqrt{3}$.

Final Answer

Maximum area $= 9\sqrt{3}$ sq. units

Common Mistake: Failing to REJECT the degenerate solution $x = 4$ (which coincides with the fixed vertex itself, giving a triangle with zero area/height) – always check that a critical point corresponds to an actual, non-degenerate configuration.

Quick Recall: For a triangle inscribed in an ellipse with one fixed vertex, use the ellipse equation to express the OTHER two vertices' y -coordinate in terms of a single x -parameter, form the area using base $\times$ height/2 with the fixed vertex providing the height, then differentiate – always check for and reject any degenerate (zero-area) critical points.

Type 75

Proving general optimality THEOREMS for a box or cuboid – given a fixed volume, prove the minimum surface area occurs for a specific shape (often a cube), or given a fixed sheet area, prove the maximum volume takes a specific form.

★ **Board Favorite** – Box optimality theorems are among the most frequently tested "prove that" results in this topic, with the book tagging a direct instance to ISC 2017.

Source: Exercise 6.7, Q28 (ISC 2017)

Given: Show that the surface area of a closed cuboid with a square base and given volume is minimum

Solution:

Step 1: Let the square base have side x and height h , with fixed volume $V = x^2h \Rightarrow h = \frac{V}{x^2}$.

Step 2: Surface area (closed box, square base): $S = 2x^2 + 4xh = 2x^2 + 4x \left(\frac{V}{x^2} \right) = 2x^2 + \frac{4V}{x}$.

Step 3: Differentiate: $S'(x) = 4x - \frac{4V}{x^2}$.

Step 4: Set $S'(x) = 0$: $4x = \frac{4V}{x^2} \Rightarrow x^3 = V \Rightarrow x = V^{1/3}$.

Step 5: Find h : $h = \frac{V}{x^2} = \frac{V}{V^{2/3}} = V^{1/3} = x$ – so $h = x$, confirming a CUBE.

Step 6: Confirm minimum: $S''(x) = 4 + \frac{8V}{x^3} > 0$ for $x > 0$ – confirmed minimum.

Final Answer

Surface area is minimum when $h = x$, i.e. when the cuboid is a cube

Common Mistake: Introducing THREE independent side variables for a general cuboid when the problem specifically states a SQUARE base – this halves the number of unknowns from the start and is essential for keeping the problem tractable with basic single-variable calculus.

Alternate Board-Style Example (open tank, depth-width relationship)

Source: Exercise 6.7, Q31

Given: An open tank with a square base and vertical sides is to hold a given volume of water. Show

Solution:

Step 1: Let the square base have side x (the "width") and depth h , with fixed volume $V = x^2h \Rightarrow h = \frac{V}{x^2}$.

Step 2: Since the tank is OPEN at the top, the material used is: base + four sides (NO top): $S = x^2 + 4xh = x^2 + 4x \left(\frac{V}{x^2} \right) = x^2 + \frac{4V}{x}$.

Step 3: Differentiate: $S'(x) = 2x - \frac{4V}{x^2}$. Set to 0: $2x^3 = 4V \Rightarrow x^3 = 2V \Rightarrow x = (2V)^{1/3}$.

Step 4: Find h : $h = \frac{V}{x^2} = \frac{V}{(2V)^{2/3}}$. Simplify by comparing to $\frac{x}{2} = \frac{(2V)^{1/3}}{2}$: writing both with the same base, $h = V^{1/3} \cdot 2^{-2/3}$ and $\frac{x}{2} = V^{1/3} \cdot 2^{1/3-1} = V^{1/3} \cdot 2^{-2/3}$ – these MATCH exactly.

Step 5: So $h = \frac{x}{2}$, i.e. the depth is exactly half the width.

Final Answer

Cost (material) is least when the depth $= \frac{1}{2} \times$ width, as required

Quick Recall: An OPEN box/tank drops the "top" term from the surface-area formula compared to a closed one – this single difference is what shifts the optimal ratio from "height equals side" (closed cube case) to "depth equals half the width" (open tank case); always double check whether the container is open or closed before writing the surface area formula.

Type 76

A NUMERIC box optimization problem with an extra practical twist – either a DIFFERENT cost per unit area for different faces (requiring a weighted surface-area/cost function), or a corner-cutting construction (an open box folded from a flat rectangular sheet).

★ **Board Favorite** – Corner-cutting box problems are a hallmark practical application, frequently tested for their distinctive volume-as-a-cubic-in-one-variable setup.

Source: Exercise 6.7, Q33

Given: A rectangular sheet of tin $45 \text{ cm} \times 24 \text{ cm}$ is made into a box without a top by cutting equal squares from each corner.

Solution:

Step 1: Let the side of each cut-out square be x (so $0 < x < 12$, since the shorter side is 24). After folding, the box has base dimensions $(45 - 2x) \times (24 - 2x)$ and height x .

Step 2: Volume: $V(x) = x(45 - 2x)(24 - 2x)$. Expand the product of the two linear factors first: $(45 - 2x)(24 - 2x) = 1080 - 90x - 48x + 4x^2 = 1080 - 138x + 4x^2$.

Step 3: So $V(x) = x(1080 - 138x + 4x^2) = 1080x - 138x^2 + 4x^3$.

Step 4: Differentiate: $V'(x) = 1080 - 276x + 12x^2$.

Step 5: Set $V'(x) = 0$: divide through by 12: $x^2 - 23x + 90 = 0$. Solve using the quadratic formula: discriminant $= 23^2 - 4(90) = 529 - 360 = 169 = 13^2$, so $x = \frac{23 \pm 13}{2} = 18$ or 5.

Step 6: Since $0 < x < 12$ is required (the cut can't exceed half the shorter side), reject $x = 18$; keep $x = 5$.

Final Answer

Side of the cut-out square = 5 cm (for maximum volume)

Common Mistake: Accepting BOTH roots of the quadratic without checking the physical constraint $0 < x < 12$ – a cut-out square larger than half the shorter side of the sheet is geometrically impossible, so such a root must always be discarded.

Alternate Board-Style Example (weighted cost, closed box)

Source: Exercise 6.7, Q30 (ISC 2019)

Given: A closed metal box with a square base has volume 4096 cm^3 . Polishing the outer surface costs Rs. 4 per cm^2 .

Solution:

Step 1: Since the cost rate (Rs. 4 per cm^2) is the SAME for every face, minimizing cost is equivalent to minimizing the total surface area directly – no weighting between different faces is needed here.

Step 2: This is exactly the closed-cube optimality result from Type 75: for a square-base closed box with fixed volume V , the minimum-surface-area shape is a cube with side $x = V^{1/3}$.

Step 3: Here $V = 4096$. Since $16^3 = 4096$, we get $x = 16$.

Final Answer

Dimensions: $16 \text{ cm} \times 16 \text{ cm} \times 16 \text{ cm}$ (a cube), minimizing the polishing cost

Quick Recall: Corner-cutting problems always reduce to a CUBIC in the cut-size x – factor or use the quadratic formula on $V'(x) = 0$, and always discard roots that violate the physical constraint (cut size less than half the shorter dimension); a uniform cost-per-area box problem is secretly just the ordinary minimum-surface-area problem in disguise.

Type 77

Cone optimization: given a FIXED slant height, find the semi-vertical angle that maximizes the volume – parametrize the cone's radius and height via the slant height and the angle.

Source: Exercise 6.7, Q34

Given: Show that the semi-vertical angle of a cone of maximum volume, with given (fixed) slant height l , is $\theta = \sin^{-1}(\frac{1}{2})$.

Solution:

Step 1: Let the fixed slant height be l , and let θ be the semi-vertical angle. Then $r = l \sin \theta$ and $h = l \cos \theta$ (standard right-triangle decomposition of the cone's cross section).

Step 2: Volume: $V(\theta) = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(l \sin \theta)^2(l \cos \theta) = \frac{1}{3}\pi l^3 \sin^2 \theta \cos \theta$.

Step 3: Differentiate w.r.t. θ (product rule on $\sin^2 \theta \cos \theta$): $V'(\theta) = \frac{1}{3}\pi l^3 [2 \sin \theta \cos \theta \cdot \cos \theta + \sin^2 \theta \cdot (-\sin \theta)]$
 $\frac{1}{3}\pi l^3 \sin \theta [2 \cos^2 \theta - \sin^2 \theta]$.

Step 4: Set $V'(\theta) = 0$: since $\sin \theta \neq 0$ (a genuine cone needs $\theta > 0$), we need $2 \cos^2 \theta - \sin^2 \theta = 0 \Rightarrow 2 \cos^2 \theta = \sin^2 \theta = 1 - \cos^2 \theta \Rightarrow 3 \cos^2 \theta = 1 \Rightarrow \cos^2 \theta = \frac{1}{3} \Rightarrow \cos \theta = \frac{1}{\sqrt{3}}$
 (taking the positive root, since θ is an acute angle).

Step 5: So $\theta = \cos^{-1} \frac{1}{\sqrt{3}}$.

Final Answer

$\theta = \cos^{-1} \frac{1}{\sqrt{3}}$, as required

Common Mistake: Trying to use r and h as two independent variables linked by the Pythagorean constraint $r^2 + h^2 = l^2$ handled separately – parametrizing directly via the angle θ (using $r = l \sin \theta, h = l \cos \theta$) automatically satisfies this constraint and reduces the problem to ONE trig variable.

Quick Recall: For "fixed slant height, optimize over the semi-vertical angle" cone problems, always parametrize with $r = l \sin \theta, h = l \cos \theta$ – this converts the volume into a pure trig function of θ alone, ready for ordinary differentiation.

Type 78

Cylinder optimization: given a FIXED volume, prove the minimum surface area occurs at a specific radius-height ratio; or given a FIXED surface area, prove the maximum volume occurs at a specific ratio – eliminate one variable using the given constraint.

★ **Board Favorite** – Cylinder optimality theorems (both "given volume, minimize area" and "given area, maximize volume" directions) are extensively tested, with the book tagging a direct instance to ISC 2020.

Source: Exercise 6.7, Q37 (ISC 2020)

Given: Show that the radius of a closed right circular cylinder of given surface area and maximum volume is equal to the height.

Solution:

Step 1: Let the fixed surface area be $S = 2\pi r^2 + 2\pi r h \Rightarrow h = \frac{S - 2\pi r^2}{2\pi r} = \frac{S}{2\pi r} - r$.

Step 2: Volume: $V(r) = \pi r^2 h = \pi r^2 \left(\frac{S}{2\pi r} - r \right) = \frac{rS}{2} - \pi r^3$.

Step 3: Differentiate: $V'(r) = \frac{S}{2} - 3\pi r^2$.

Step 4: Set $V'(r) = 0$: $r^2 = \frac{S}{6\pi} \Rightarrow S = 6\pi r^2$.

Step 5: Find h using this value of S : $h = \frac{S}{2\pi r} - r = \frac{6\pi r^2}{2\pi r} - r = 3r - r = 2r$.

Step 6: So $h = 2r$, i.e. $r = \frac{h}{2}$ – the radius is exactly half the height.

Final Answer

$$r = \frac{h}{2}, \text{ as required}$$

Common Mistake: Solving for r in terms of S but forgetting to substitute BACK to find h and explicitly compare the two – a "show that $r = h/2$ " question needs both quantities expressed and compared, not just the critical radius found.

Alternate Board-Style Example (given volume, minimize surface area)

Source: Exercise 6.7, Q36(i)

Given: Show that a closed cylindrical vessel of given volume has the least total surface area when it

Solution:

Step 1: Let the fixed volume be $V = \pi r^2 h \Rightarrow h = \frac{V}{\pi r^2}$.

Step 2: Surface area: $S(r) = 2\pi r^2 + 2\pi r h = 2\pi r^2 + 2\pi r \left(\frac{V}{\pi r^2} \right) = 2\pi r^2 + \frac{2V}{r}$.

Step 3: Differentiate: $S'(r) = 4\pi r - \frac{2V}{r^2}$. Set to 0: $4\pi r^3 = 2V \Rightarrow r^3 = \frac{V}{2\pi}$.

Step 4: Find h : $h = \frac{V}{\pi r^2}$. Using $V = 2\pi r^3$ (from the critical-point equation): $h = \frac{2\pi r^3}{\pi r^2} = 2r$.

Final Answer

$$h = 2r, \text{ i.e. the height is twice the radius, as required}$$

Quick Recall: "Given volume, minimize area" and "given area, maximize volume" are TWO SIDES of the same relationship for a cylinder – both lead to the same optimal ratio between h and r (height = twice radius for a closed cylinder), just approached from opposite starting constraints; always finish by explicitly comparing h and r to state the required ratio.

Type 79

A solid (cylinder or cone) is INSCRIBED IN A SPHERE of given radius – express the solid's dimensions in terms of the sphere's radius and one free variable (using the Pythagorean relationship from the sphere's cross-section), then maximize the solid's volume or curved surface area.

★ **Board Favorite** – Solids inscribed in a sphere are a classic, geometrically rich optimization family, with the book tagging a direct instance to ISC 2018.

Formula Used: Cone inscribed in a sphere of radius R , height h : $r^2 = 2Rh - h^2$

Source: Exercise 6.7, Q39 (ISC 2018)

Given: A cone is inscribed in a sphere of radius 12 cm. If the volume of the cone is maximum, find

Solution:

Step 1: Let the sphere have radius $R = 12$, and let the cone (with its vertex on the sphere's surface, base a chord-circle inside the sphere) have height h and base radius r . A standard geometric relationship for a cone inscribed this way gives $r^2 = 2Rh - h^2$ (derived from the right triangle formed by the sphere's center, the base circle's edge, and the axis).

Step 2: Volume: $V(h) = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(2Rh - h^2)h = \frac{1}{3}\pi(2Rh^2 - h^3)$.

Step 3: Differentiate w.r.t. h : $V'(h) = \frac{1}{3}\pi(4Rh - 3h^2)$.

Step 4: Set $V'(h) = 0$: $h(4R - 3h) = 0 \Rightarrow h = 0$ (rejected, degenerate) or $h = \frac{4R}{3}$.

Step 5: Substitute $R = 12$: $h = \frac{4(12)}{3} = 16$.

Final Answer

Height for maximum volume = 16 cm

Common Mistake: Not knowing (or mis-deriving) the key relationship $r^2 = 2Rh - h^2$ linking the inscribed cone's dimensions to the sphere's radius – this geometric fact is essential background for this entire problem family and is worth memorizing directly.

Alternate Board-Style Example (cylinder inscribed in a sphere)

Source: Exercise 6.7, Q38

Given: Find the maximum volume of a cylinder that can be inscribed in a sphere of radius $3\sqrt{3}$ cm

Solution:

Step 1: Let the sphere have radius $R = 3\sqrt{3}$, and let the inscribed cylinder have height h and radius r . By the Pythagorean relationship in the sphere's cross-section (radius, half-height, and cylinder radius form a right triangle): $r^2 + \left(\frac{h}{2}\right)^2 = R^2 \Rightarrow r^2 = R^2 - \frac{h^2}{4}$.

Step 2: Volume: $V(h) = \pi r^2 h = \pi \left(R^2 - \frac{h^2}{4}\right) h = \pi R^2 h - \frac{\pi h^3}{4}$.

Step 3: Differentiate: $V'(h) = \pi R^2 - \frac{3\pi h^2}{4}$. Set to 0: $h^2 = \frac{4R^2}{3} \Rightarrow h = \frac{2R}{\sqrt{3}}$.

Step 4: Find r^2 : $r^2 = R^2 - \frac{h^2}{4} = R^2 - \frac{R^2}{3} = \frac{2R^2}{3}$.

Step 5: Compute the maximum volume: $V = \pi r^2 h = \pi \left(\frac{2R^2}{3}\right) \left(\frac{2R}{\sqrt{3}}\right) = \frac{4\pi R^3}{3\sqrt{3}}$. Substitute $R = 3\sqrt{3}$, so $R^3 = (3\sqrt{3})^3 = 27 \times 3\sqrt{3} = 81\sqrt{3}$: $V = \frac{4\pi(81\sqrt{3})}{3\sqrt{3}} = \frac{4\pi(81)}{3} = 4\pi(27) = 108\pi$.

Final Answer

Maximum volume = $108\pi \text{ cm}^3$

Quick Recall: For a solid inscribed in a sphere, always start by identifying the RIGHT TRIANGLE that the sphere's radius forms with the solid's own dimensions (half-height and radius, for a cylinder; height and radius via $r^2 = 2Rh - h^2$, for a cone touching the surface) – this relationship is what eliminates one variable before optimizing.

Type 80

A solid inscribed in ANOTHER solid (a cylinder inside a cone, rather than a sphere), or a COMPOSITE-solid optimization combining multiple shapes (like a parallelepiped and a sphere) under one shared constraint – use similar triangles or a shared-constant relationship to link the dimensions before optimizing.

★ **Board Favorite** – Cylinder-inscribed-in-a-cone is a distinctive, high-value pattern (the ratio "cylinder height is one-third of the cone's height" is a famous, frequently-tested result), and the book tags it as a Competency Focused Question.

Source: Exercise 6.7, Q41 (Competency Focused Question)

Given: Show that the height of a right circular cylinder of greatest volume inscribed in a right circular cone is one-third the height of the cone.

Solution:

Step 1: Let the inscribed cylinder have height y and radius x . By SIMILAR TRIANGLES (comparing the cone's full profile to the smaller cone ABOVE the cylinder): $\frac{h-y}{h} = \frac{x}{r} \Rightarrow x = r \left(1 - \frac{y}{h}\right) = r - \frac{ry}{h}$.

Step 2: Cylinder volume: $V(y) = \pi x^2 y = \pi r^2 \left(1 - \frac{y}{h}\right)^2 y$.

Step 3: Differentiate using the product rule: $V'(y) = \pi r^2 \left[\left(1 - \frac{y}{h}\right)^2 + y \cdot 2 \left(1 - \frac{y}{h}\right) \left(-\frac{1}{h}\right) \right] = \pi r^2 \left(1 - \frac{y}{h}\right) \left[\left(1 - \frac{y}{h}\right) - \frac{2y}{h} \right] = \pi r^2 \left(1 - \frac{y}{h}\right) \left(1 - \frac{3y}{h}\right)$.

Step 4: Set $V'(y) = 0$: $y = h$ (rejected, degenerate – zero cylinder radius) or $1 - \frac{3y}{h} = 0 \Rightarrow y = \frac{h}{3}$.

Step 5: Find x at $y = \frac{h}{3}$: $x = r \left(1 - \frac{1}{3}\right) = \frac{2r}{3}$.

Step 6: Compute the maximum cylinder volume: $V = \pi \left(\frac{2r}{3}\right)^2 \left(\frac{h}{3}\right) = \pi \cdot \frac{4r^2}{9} \cdot \frac{h}{3} = \frac{4\pi r^2 h}{27}$.

Step 7: Compare to the cone's own volume $V_{\text{cone}} = \frac{1}{3}\pi r^2 h$: the ratio is $\frac{4\pi r^2 h/27}{\frac{1}{3}\pi r^2 h} = \frac{4}{27} \times 3 = \frac{4}{9}$.

Final Answer

Cylinder height = $\frac{h}{3}$; maximum cylinder volume = $\frac{4}{9} \times \text{cone's volume}$

Common Mistake: Setting up the similar-triangles ratio backwards (comparing the WRONG pair of triangles, e.g. the cylinder's own triangle to itself instead of to the REMAINING smaller cone above it) – always sketch the cross-section and carefully identify which two triangles are actually similar before writing the proportion.

Alternate Board-Style Example (economical manufacturing, differentiated cost)

Source: Exercise 6.7, Q43

Given: A manufacturer plans a cylindrical can to hold 1 m^3 of oil. If the cost of constructing the top

Solution:

Step 1: Let the radius be r , height h , with fixed volume $\pi r^2 h = 1 \Rightarrow h = \frac{1}{\pi r^2}$.

Step 2: Let the cost rate for the side be c per unit area, so the cost rate for the top and bottom (combined) is $2c$ per unit area.

Step 3: Total cost: top+bottom area is $2\pi r^2$ (costing $2c$ per unit area), side area is $2\pi rh$ (costing c per unit area): $C = (2c)(2\pi r^2) + (c)(2\pi rh) = 4c\pi r^2 + 2c\pi rh$.

Step 4: Substitute h : $C(r) = 4c\pi r^2 + 2c\pi r \left(\frac{1}{\pi r^2} \right) = 4c\pi r^2 + \frac{2c}{r}$.

Step 5: Differentiate: $C'(r) = 8c\pi r - \frac{2c}{r^2}$. Set to 0: $8\pi r = \frac{2}{r^2} \Rightarrow r^3 = \frac{2}{8\pi} = \frac{1}{4\pi} \Rightarrow r = \left(\frac{1}{4\pi} \right)^{1/3}$.

Step 6: Find h using $h = \frac{1}{\pi r^2}$ and simplify to match the standard form $h = \left(\frac{16}{\pi} \right)^{1/3}$ (obtained by substituting $r^3 = \frac{1}{4\pi}$ through the algebra).

Final Answer

Radius = $\left(\frac{1}{4\pi} \right)^{1/3}$ metre; height = $\left(\frac{16}{\pi} \right)^{1/3}$ metre

Quick Recall: Whenever different faces of a solid have DIFFERENT cost rates, build the cost function as a WEIGHTED sum of each face's area times its own rate (exactly like Type 71's fenced-area problem) – for a solid inscribed in another solid, always find the SIMILAR-TRIANGLES relationship linking their dimensions first, before writing the volume formula.